Potable Water & Treated Sewage Effluent (TSE) Design Guidelines

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i Document Authorisation

This document is authorised for issue.

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ii Revision History

The following is a brief summary of the most recent revisions to this document. Details of all revisions prior to these are held on file by the issuing department.

Revision No.	Date	Author	Scope / Remarks
Rev. 01	March 2026	Seureca, NWS Asset Engineering	First issue after merger - Integrate the following documents into one document TET-DG-5000, TET-DG-5001 & AM-ENG-WMD-03

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1. Introduction

Oman Water and Wastewater Services Company (OWWSC), operating under the brand name Nama Water Services (NWS), was established by Royal Decree No. 131/2020, issued on 9 December 2020. Nama Water Services (NWS) is a member of the Nama Group and holds the relevant licenses responsible for providing potable water to customers and the management and handling of wastewater and treated effluent in accordance with international standards. The potable water and treated effluent (TSE) guidelines are prepared in compliance with NWS' general policy to provide safe production, storage and distribution systems to ensure best and efficient water utilities performance, at highest quality and environmental (local and international) standards.

1.1. Objective

The objective of this document is to standardise and reduce the variability in designs undertaken on behalf of NWS. The designer is to consider this document to be applicable in the majority of situations and shall read in conjunction with other NWS' standard specifications and drawings or any other document published to replace whole or parts of this document. The intended design lifetime shall be 25 years for all kinds of assets, unless otherwise stated.

In principle, the designer to consider the water and treated effluent design guidelines objectives to cover the following criteria:

- Ensure fitness for purpose;
- Ensure ease of constructability;
- Ensure functional and durable system;
- Ensure ease of maintenance;
- Ensure reliable operation;
- Ensure water quality is not degraded;
- Minimise adverse environmental and community impact;
- Comply with environmental requirements;
- Comply with operational health and safety requirements;
- Minimise energy consumption by efficient operation;
- Achieve extended service life with minimal maintenance and least whole of life cost;
- Provide adequate weather protection and storm water management.
- Provide sufficient vehicular and personnel access for maintenance.

1.2. Scope

This Guidelines volume is part and shall be considered with the other volumes of the Integrated Engineering Standards package including:

- Design Guidelines:
    - PAM-GUD-201 General design guidelines
    - PAM-GUD-202 Water and Treated Sewerage Effluent TSE Design Guidelines
    - PAM-GUD-203 Wastewater Design Guidelines
- Standard Specifications:
    - PAM-SPC-2xx Pipes and Fittings
    - PAM-SPC-3xx Valves and Accessories
    - PAM-SPC-4xx Civil Works
    - PAM-SPC-5xx Metal Works
    - PAM-SPC-6xx Mechanical
    - PAM-SPC-7xx Electrical
    - PAM-SPC-8xx Instrumentation, Control and Automation (ICA)
- Standard Drawings
    - PAM-STD-1xx General
    - PAM-STD-2xx Pipes
    - PAM-STD-3xx Valves & Accessories
    - PAM-STD-4xx Civil Works
    - PAM-STD-5xx Metal Works
    - PAM-STD-6xx Mechanical
    - PAM-STD-7xx Electrical
    - PAM-STD-8xx ICA & OT
    - PAM-STD-9xx Indicative STP, Process & Building
- Procedures
    - PAM-PRO-201 Managing Asset Change Requirements
    - PAM-PRO-202 Asset Failure Investigation Procedure
    - PAM-PRO-203 Technical Audit Procedure
    - PAM-PRO-204 Lessons Learned Procedure
    - PAM-PRO-205 Commissioning Procedure
    - PAM-PRO-206 Asset Handover Procedure
    - PAM-PRO-207 Decommissioning Procedure

- This volume presents general and common guidelines applicable to potable water systems, and treated sewage effluents systems.

1.3. Distribution & Target Audience

i. This document is issued for use by concerned NWS employees and stakeholders, Consultants, Owner Representatives/Engineers, and Contractors & Suppliers.

1.4. Review & Improvement

i. This Document is intended to be formally reviewed every 3 years, in line with the control of documented information procedure "QHSSE-PRO-001". Additional reviews shall be performed, as necessary. Despite the time duration between reviews, Consultants and Contractors shall adhere to the latest version of the Specification at the time a project design commences.

1.5. Abbreviations

Table 1 List of Acronyms

Acronym	Definition
APSR	**Authority of Public Services Regulation**
ASP	**Activated Sludge Process**
AWWA	**American Water Works Association**
BS	**British Standards**
CAA	**Civil Aviation Authority**
CAPEX	**Capital Expenditure**
CDAA	**Public Authority for Civil Defense and Ambulance**
Cp	**Peak coefficient**
CPA	**Critical Parts Analysis**
CSGRE	**Carbon Steel Glass Reinforced Epoxy**
DCF	**Discounted Cash Flow**
DI	**Ductile Iron**
EA	**Extended Aeration**
EIA	**Environmental Impact Assessment**
EMC	**Electromagnetic Compatibility**
EN	**European Norms**
ERI	**Electrical Resistivity Imaging**
FMEA	**Failure Modes and Effects Analysis**
GPR	**Ground-Penetrating Radar**

Acronym	Definition
GRP	Glass Reinforced Plastic
HAZOP	Hazard and Operability Study
HIS	Historical Information System
HPDE	High-Density Polyethylene
HSE	Health Safety and Environment
HW	Hardware
HVAC	Heating, Ventilation, and Air Conditioning
ICA	Instrumentation, Control and Automation
ICPE	"Installations Classées pour la Protection de l'Environnement" or "Installations Classified for the Protection of the Environment"
IEC	International Electrotechnical Commission
IFAS	Integrated Fixed-film Activated Sludge
IS&R	Information Storage and Retrieval
ISO	International Organization for Standardization
L/d/km	liters per day per kilometer
LCC	Life-Cycle Cost
LCCA	Lifecycle Cost Analysis
LOS	Level of Service
LPCD	Litres per capita per day
LRA	Lightning Risk Assessment
MBBR	Moving Bed Biofilm Reactor
MBR	Membrane Bioreactor
MCDA	Multi-Criteria Decision Analysis
MD	Ministerial Decisions
MEP	Mechanical, Electrical and Plumbing
MoAFWR	Ministry of Agriculture, Fisheries and Water Resources

Acronym	Definition
MoT	Ministry of Transport
NCC	National Control Centre
NCRs	Non-Conformance Reports
NCSI	National Centre for Statistics and Information
NFPA	National Fire Protection Association
NPV	Net Present Value
NRW	Non-Revenue Water
NWS	Nama Water Services
OFC	Buildings Fire Prevention and Protection Requirements–Part One
OHMG	Occupational Health Management Guideline
OPEX	Operational Expenditure
OWWSC	Oman Water Supply and Wastewater Company
PAM-GUD-201	General Design Guidelines (Document Number)
PE100	Polyethylene 100
PEAZ	Potentially Explosive Atmosphere Zoning
Pf	Peaking Factor
PfWW	Peak Factor for wastewater flows
P&IDs	Piping and Instrumentation Diagrams
QADF	Average Daily Flow
Qadf	Average daily flow
QA/QC	Quality Assurance / Quality Control
Qm	average flow
QPDF	Peak Flow
Qpdf	Peak daily flow
RBC	Rotating Biological Contactor
RCC	Regional Control Centres / Reinforced Cement Concrete

Acronym	Definition
RO	Reverse Osmosis
ROI	Return on Investment
ROP	Royal Oman Police
SBR	Sequencing Batch Reactors
SCADA	Supervisory Control and Data Acquisition
SDI	Silt Density Index
SDR11	Standard Dimension Ratio 11
SW	Software
STP	Sewerage Treatment Plant
TDS	Total Dissolved Solids
TE	Trade Effluent
TPM	Total Productive Maintenance
TSE	Treated Sewerage Effluent
UL	Underwriters Laboratories
UV	Ultraviolet
VE	Value Engineering
WHO	World Health Organisation
WSZ	Water Supply Zone

The present Guidelines cover the design of potable water and treated effluent systems.

2. Terminology, Numbering & Standards Names

2.1. Terminology

The standard terminology used in these guidelines are present in Appendix I of the General Design Guidelines PAM-GUD-201.

2.2. Roles & Responsibilities

a) NWS - PAM
   i) Discusses the requirements with the line manager.
   ii) Writes the document using the right template.
   iii) Requests document number from IMS Section (new document).
   iv) Follows the agreed upon quality assurance (QA) procedure and keep records accordingly.
   v) Issues document for review.
   vi) Incorporates comments (as appropriate) and issues document for use.
   vii) Conducts a formal review of the document and sends comments to author.
   viii) Checks that comments from other reviewers are incorporated.
   ix) Checks for grammatical and spelling mistakes.
   x) Ensures that control of documents is followed.
   xi) Accepts final document.

b) User – NWS, Consultants, Designer & Contractors
   i) Adheres to the document control procedure established by NWS
   ii) Uses and refers to documents with current revision status only

2.3. Standard Names

The standard names used in these guidelines are present in Section 2.3 of the General Design Guidelines PAM-GUD-201.

3. Wells

3.1. Water Sources

Though ground water is one of the major sources of water in the Sultanate of Oman, due to the poor quality and quantity of available ground water in certain areas, sea water desalination is carried out to meet the potable demands of the citizens. But as a reserve, ground water is used as and when needed in emergencies and hence this exercises to standardize the abstraction techniques of the ground water.

Ground water is abstracted by the following conventional techniques:

a. Open Wells
b. Bore Wells

Of the above, open wells are not much suitable for bulk abstraction due to the limitation in depth. Instead, deep drilled bore wells are widely used by water supply agencies and hence this document focuses mainly on the bore well construction and its related techniques.

3.2. Groundwater Identification

Being it open well or bore well; the well location is a prime factor which would very much influence the performance of the well. Selection of the aquifer/well location depends upon the following aspects:

1. Hydrogeology
2. Surface geophysics
3. Subsurface geophysics

3.2.1. Hydrogeology

In groundwater resource identification, the objective is to locate deposits of ''water bearing sub-strata'' with relatively high permeability. Such a water bearing sub-strata is called ''aquifer''. The potential of an aquifer is mainly determined by:

- its geometry - extent, thickness etc.,
- its hydrodynamic characteristics - transmissivity, stored volume of water,
- its influencing boundaries - presence of surface water bodies (rivers, lakes, wadi courses) or tight boundaries (rock, clay as per existing geological setup),
- proximity of sea-line,
- its natural recharge - effective rainfall, extent of the catchment area.

In order to classify them as potential aquifers such deposits should meet the following requirements.

- Occurrence in sufficient thickness
- Enough areal extent to provide large storage volumes
- Enough annual recharge (rains)

A thorough analysis of the underground stratification, buried topography, inflow and outflow, rainfall, evaporation etc. of the related area needs to be carried out to determine the quality and quantity of water that can be abstracted for a considerably a long time from any aquifer. The performance and the available data from the nearby wells in operation could also provide information on the aquifer under study.

3.2.2. Surface geophysics

Surface geophysical surveys are of major importance to assess the status of underlying water aquifers. The survey is carried out on the ground surface prior to reconnaissance drilling in order to define areas with greatest potential and to find the best locations for water abstraction. Two types of geophysical surveys are widely in use:

- Electric resistivity method.
- Electromagnetic methods.

3.2.3. Subsurface geophysics

Subsurface geophysics study requires drilling of a pilot bore hole to conduct certain studies at the depths of the aquifer. This operation consists of running special probes upward/downward in the borehole in order to identify water bearing strata. A 100 mm reconnaissance borehole down to the bedrock is drilled to measure the entire thickness of the aquifer and its characteristics. Two types of subsurface geophysical surveys are widely in use:

- Electric logging – this is good for locating shallow aquifers.
- Gamma-ray logging – good for hard strata

From the borehole observation the geological strata, ground permeability and water salinity can be assessed.

3.2.4. Well site location

With the acquired data and information from the above, the suitable well location shall be identified following the sequences of activities as below:

3.2.4.1. Desk study

- Geological data and information,
- Topographic maps and aerial photography,
- Surface and Sub-surface geophysics,
- Existing data on wells and or boreholes (borehole logs),
- Existing data on down-hole geophysics and pumping tests,
- Hydro-geological information (groundwater levels and piezometric maps),
- Groundwater quality data (chemistry and salinity contours).

3.2.4.2. Site survey

- Accessibility of drilling rigs,
- Geotechnical and geological features,

- Co-ordination with authorities (right-of-way, ownership, other users, etc.)

In the Sultanate, most of these exercises are done by the Ministry of Regional Municipalities and Water Resources and the final recommendation on the well location and the depth to drill are provided along with the ''Well Drilling Permit'' which is mandatory before commencing the works.

3.3. Well Drilling

As a general rule, after finalizing the well location and the parameters, the production bore well shall be drilled around 50 m away from the pilot bore well to avoid interference. The choice of method of drilling depends primarily upon the geological strata and the depth to drill.

3.3.1. Drilling techniques

Following are the Drilling Techniques that are commonly used:

1. Rotary Percussion drilling used for aquifers with consolidated hard strata such as limestone, sandstone, plutonic or volcanic rocks. This technique would involve using foam to remove the drill cut materials.
2. Rotary drilling adapted to drill through unconsolidated loose strata. This technique would involve using ''Bentonite'' mud to stabilize the loose wall while drilling and to flush out the drill cut materials.
3. Percussion Drilling with bailer is suitable for both but not used generally by NWS.

3.3.2. Drilling procedure

Drilling Procedure in general includes the following steps:

1. Core drilling for surface casing
2. Installation of surface casing and concrete protection
3. Well core drilling to the designed depth
4. Installation of casing and screen
5. Installation of dip tube
6. Gravel pack surround and installing centralizers
7. Well development
8. Pumping test
9. Well disinfection and flushing
10. Chemical and bacterial water analysis

11. Well head platform construction

3.3.2.1. Core drilling for surface casing

For all the wells, Core drilling for surface casing shall be 16 to 18-inch diameter drilling to a minimum depth of 6 m to accommodate the surface casing.

3.3.2.2. Installation of surface casing and concrete protection

Material of surface casing shall be Carbon Steel, Stainless Steel, u-PVC or PE100 as per the design which shall consider corrosion, stiffness and other mechanical properties of the material. This casing shall be concreted to the full depth to give stability to the well and as well as to act as a seal against sub-surface pollution. The exposed head shall have a concrete base of a minimum of one square meter.

3.3.2.3. Well core drilling

Well core drilling shall be in accordance with the finished diameter of the bore well.

For the finished bore well diameter of 6 inch (150 mm) the drilling shall be 225 mm.

For the finished bore well diameter of 8 inch (200 mm) the drilling shall be 325 mm.

This shall be to the full depth of the bore well.

Suitable technique of drilling based on the strata shall be followed – either hard rock or loose strata and corresponding techniques shall be followed. Compressed air foam lift shall be used for removing the cut bits and in case of un-consolidated strata 'Bentonite Mud Clay' shall be used to stabilize the wall.

Following data shall be recorded as a must while carrying out drilling operation to the full depth:

- Time of start and stop of each session
- Depth of start and stop at each session
- Rate of penetration
- Soil samples at every meter – collected, numbered and bagged
- Depth at which water strikes
- Water sampling and analysis at every meter after water strike – pH and EC
- Rate of water flow measured through 'V' Notch at every 5 minutes or as required

At this stage, with the available data on quality and quantity of water, the suitability of the well to continue further shall be decided. If found suitable further steps in well construction shall be continued or else in the unfortunate event of poor quality and or poor yield, the well construction shall be discontinued and filled back as instructed by the Site Engineer.

3.4. Well Casing and Screen

Well casing and screen go together in a well construction. Well casing is basically a solid impermeable lining for the drilled hole to act as a wall to maintain the open hole from collapsing and withstand the back-flushing head pressure. The material of the casing shall be as per the design and the commonly used are Stainless Steel (type 304, 316, duplex), CS or uPVC. In certain hard rock formations well is left without casing but it is preferable to have well casing. This casing shall run through the entire depth of the well excepting for the places of the screen.

Well screen shall be made of similar material as the casing but shall have permeability in the form of slots to allow water from the aquifer to enter into the well chamber with a minimum of resistance and without letting the passage of sand during pumping. Casing and screen shall run through the entire well depth alternatively or in sections as per the well design based on water bearing strata.

3.4.1. Physical and chemical qualities of casing and screen materials

3.4.1.1. Physical

- Material should be in compliance with International Standard Specifications
- Shall have greater strength to weight ratio
- Uniform wall thickness
- Uniform nominal diameter (ND) and outer diameter (OD)
- Shall be straight without buckling and bending
- Shall be unruffled with the inner and outer diameter
- Shall have homogeneous texture and have smooth finish in and out to avoid hydraulic friction
- Shall stand against abrasion, erosion pitting and encrustation

3.4.1.2. Chemical

- Withstand pH range 7-10
- Chemically resistant and WRAS certified.

3.4.1.3. Joints

Casing and Screen shall have threaded joints preferably with ''Trapezoidal'' threads. Joint length shall be a minimum of 200 mm.

3.4.1.4. Screen

Selection of screens depends on the ground strata, soil type, grain size, available water quantity, rate of abstraction etc. Salient parameters of screen slots:

- Screen slots shall be commensurate with rate of abstraction of water.
- Combined slot area should not exceed 60% of the total surface area of the pipe surface.
- At any horizontal layer along the circumference, slots should not exceed 60%.
- Slot size shall be such that the flow through velocity in the range of 1 – 6 cm/sec.
- "Ribbed Screens" shall be preferred.
- In case of ''perforated-slot screens'' opening area shall be 10 - 12%
- In case of ''continuous slot wire wound screens'' opening area shall be 30-50%

The installed casing and screen shall be straight and truly vertical in status and any deviated casing shall result in disapproval of the well.

Table 2 Open Area of Slotted Screens.

Slot Type		Bridge Slot			Slotted Plastic			Continuous Slot		
Slot Size		Diameter								
Slot no.	mm	8''	10''	12''	8''	10''	12''	8''	10''	12''
20	0.5					4		18	18	14
30	0.8	3	2	3	8	6		25	25	16
40	1.0					8	8	30	30	21
50	1.3					10		35	35	24
60	1.5	6	5	6	14	11	11	41	33	28
90	2.3							48	43	37
100	2.5				21	16	16	52	46	39
125	3.2	13	12	13				51	51	45

3.4.1.5. Dip tube

A ''Dip Tube'' to the full depth of the well along the outer side of the main casing shall be installed to facilitate the water level measurements without interfering with the pump riser pipe and the power and sensor cables. This tube shall be of 25 mm ID PE100 solvent jointed and shall have perforations at equal intervals – preferably 2 Nos. 6 mm holes at 1 m c/c all along - to allow water raise through the tube. It is preferable to have this dip tube tied along with the casing while lowering the same.

3.4.1.6. Gravel pack

Gravel is packed around the casing and screen is necessary in all cased wells. It shall be of broken stone – preferably rounded - of nominal size 25 mm. To have optimum performance of the well, the gravel pack shall not be less than 75 mm thick around the casing and screen.

Gravel shall be homogeneously filled along and around the casing/screen. Packing shall be from the bottom to the top without any gaps. Verticality of the casing and screen should be maintained and ensured all along while gravel packing. While doing the gravel packing ''Well Centralizer Rings'' shall be installed at a minimum of 10 m center to center or as frequently as it may be required for the full depth of the well to maintain the verticality of the well. The top of the gravel pack shall be concrete sealed at the level of surface casing.

Commonly used diameters are:

1. Surface casing: 16”/400 mm

   20”/500 mm

2. Casing / screen: 8” / 200 mm

   10” / 250 mm

   12” / 300 mm

3. Observation wells: 4” / 100 mm

   6” / 150 mm

A finished cross section Well shall have the following features as shown in Figure 1.

3.5. Well Development

After installation of the casing, screen, gravel pack, centralizers and dip tube, the well shall be developed in order to:

- improve the productivity
- get rid of the drilling fluid and fine particles that may have remained in the well
- adjust the level of gravel if necessary
- provide clear water with no suspended solids or turbidity.

Preferred methods of development shall be:

- air-lifting
- pumping from low to high discharge rate (over pumping)
- pumping by steps
- jetting and swabbing
- chemical cleaning - adding acids, polyphosphates
- air-flushing -injecting pressurized air into the well

Two main parameters shall be monitored to assess the efficiency of the well development:

- solid content (sand particles),
- specific capacity (i.e. comparison of drawdown at constant pumping rate before/after development).

Upon completion of well development, pumping tests shall be carried out to assess the recommended yield of the well.

3.6. Pumping Test

Pumping tests are usually performed once the well is completed and properly developed so as to get a clear concept of how the well would behave and its performance and efficiency. It also helps in arriving at the recommended pumping rate and pumping pattern of the well.

Two types of pumping tests that are in vogue and shall be used:

- Step-drawdown test
- Constant discharge test

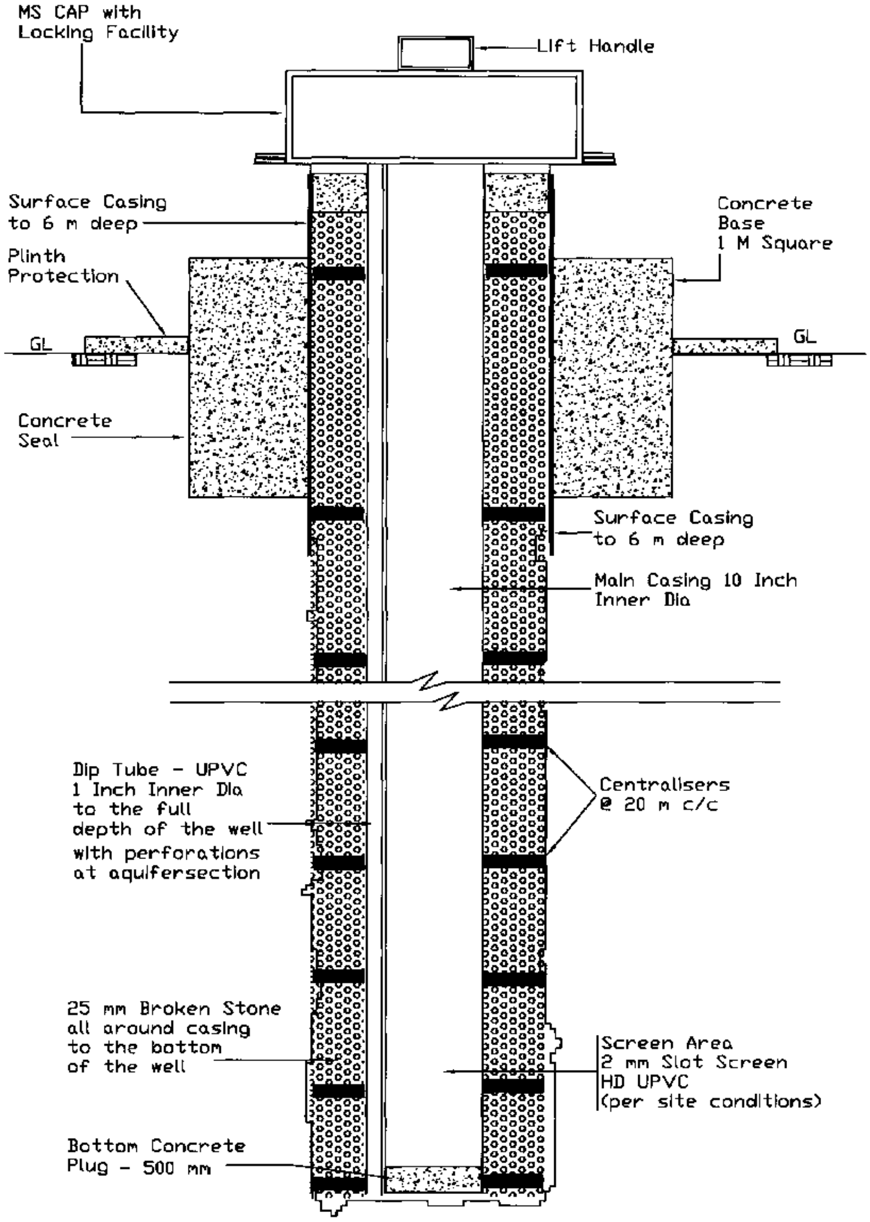

Figure 1 Typical cross section of a completed bore well.

3.6.1. Step-drawdown test

A step draw-down test shall involve pumping at different steps - a minimum of 3 steps - for a duration of 60 to 120 minutes each. It shall be performed at increasing discharge rates as detailed in Figure 2. Before commencing the test, it is mandatory to observe and record the Static Water Level with reference to a fixed datum and with date and time stamp. The same reference point shall be used in all the tests at any time.

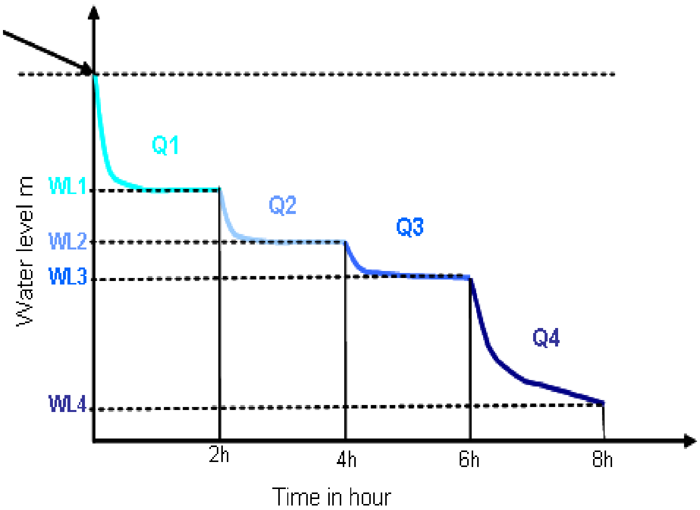

Figure 2 Concept of Step-Drawdown Test.

Temperature, electrical conductivity, salinity and pH of the water shall be measured and recorded during each step with time stamps.

The "Critical Flow Rate" is defined as that particular pumping rate beyond which the turbulent flow tends to increase the drawdown of the well in a geometric progression as shown in Figure 3.

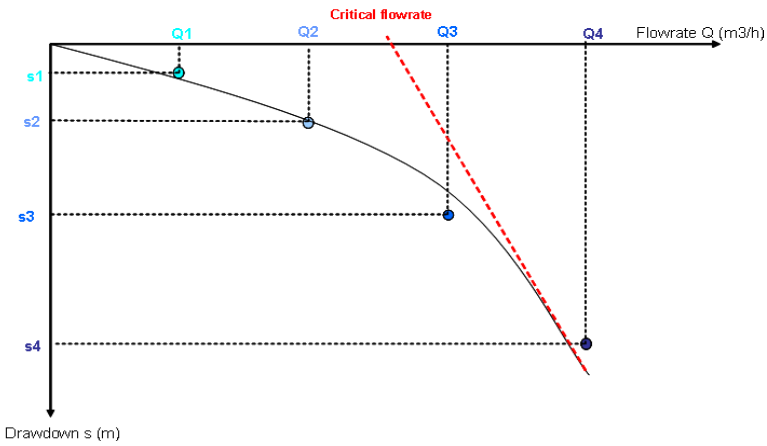

Figure 3 Critical flowrate.

All the findings and observations of the tests shall be clearly recorded figuratively and graphically for future reference and as a permanent history record of the well. These data shall be stored in hard copy and as well as in soft copies.

After completion of step draw-down test enough time shall be allowed for the well to regain its original status before starting subsequent test. and prior to starting the constant discharge test the water level must be allowed to return to its original level which is recorded prior to commencing of the step drawdown test.

3.6.2. Constant discharge test

This test will determine the ''hydrodynamic characteristics'' of the aquifer in general and well in specific.

During the test pumping rate shall be kept constant. The rate at which constant discharge test needs to be conducted shall be arrived at from the observed data from the Step-Drawdown test conducted earlier. The constant discharge test must be continued without any interruption for a minimum of 72 hours – draw down permitting.

Dynamic water levels/groundwater levels, temperature, electrical conductivity, salinity and pH of the water must be measured at pre-fixed intervals and recorded with date and time stamping during the pumping test.

The recovery must be measured at pre-fixed smaller frequencies for at least 12 hours as soon as the pump stops and at the same time intervals as when pumping.

3.6.3. General protocol to maintain during monitoring pump tests

- The water level must be taken with an accuracy of a centimeter from a fixed reference point whose reduced level is known.
- Water levels shall be measured before pumping starts - Static Water Level - SWL
- Water levels shall be measured during pumping - Dynamic Water Level - DWL, according to Table 3.

Table 3 Dynamic Well Level.

Elapsed time since the beginning of pumping or recovery	Frequency of measurement
From 0 to 15 min	1 min
From 15 to 30 min	5 min
From 30 to 60 min	10 min
From 1 to 2 h	15 min
From 2 to 4 h	30 min
From 4 to 8 h	1 h
> 8 h	2 h

3.7. Methods to Measure Discharge

Following methods are widely in use for measuring the discharge rate while performing a pumping test.

- Mechanical flowmeter
- Electromagnetic flowmeter
- Ninety-degree V-notch weir

Though V notch measurements have been widely used earlier, the most commonly used methods nowadays are mechanical flowmeters and electromagnetic flowmeters.

3.8. Downhole Video Logging

Downhole video logging shall be carried out to confirm the interior of the well. This operation allows to check if the inner structure of the well is in conformation to the design and expectations. Problems like bad position of the screen, deformation of the structure, or defective screwing, premature clogging or sanding can be detected at this stage. Very importantly verticality of the well can be confirmed.

3.9. Well Head Facilities

Concrete well head shall be constructed to prevent polluting intrusions from the surface in to the well. The different elements constituting a typical well head shall be as shown in Figure 4.

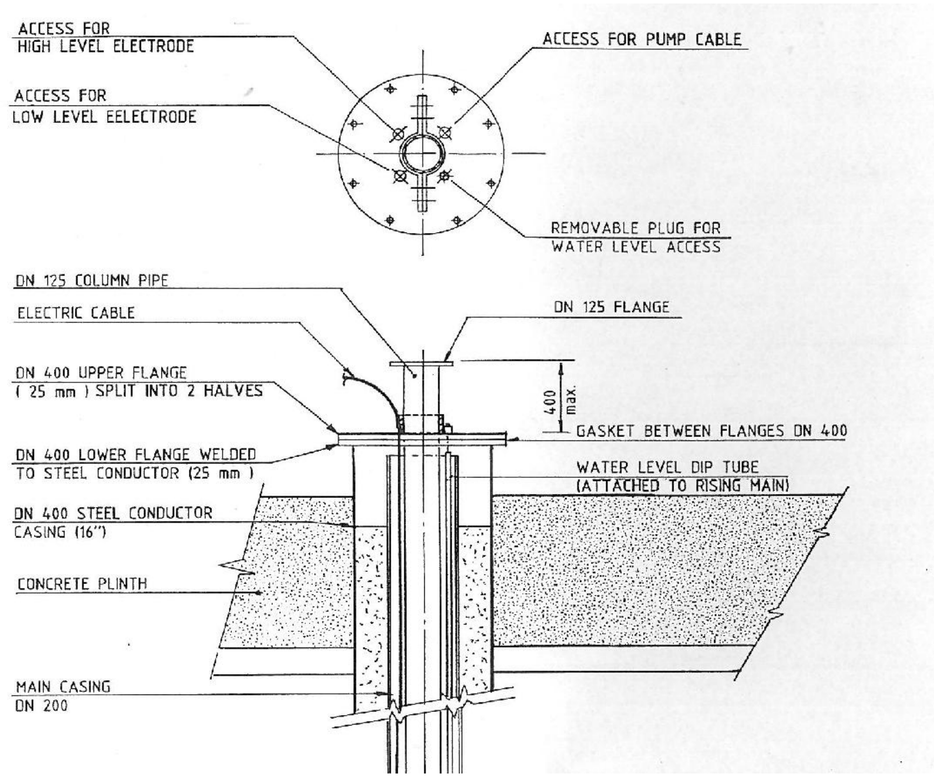

Figure 4 Typical Bore well, Details PAM-STD-401.

3.10. Well Disinfection

The most commonly used disinfectants are Sodium hypochlorite and Calcium hypochlorite.

The outline disinfection procedure shall be carried out as described briefly below:

- Calculate the volumes
- Add chlorine + contact time

- Check residual of chlorine
- Flush and take sample

3.11. Well Head Platform and Accessories

The well head platform and the fencing/compound protection shall be as below:

- The plot limits of the well site shall be at least 10 m x 10 m for 150 mm diameter wells and 15m x 15 m for 200 mm and above diameter wells
- Concrete slab 150 mm thick around the well casing over a minimum surface of 5 m^2 with a 45-degree slope with a platform height of 100 cm above ground level
- Top of casing extending 30 cm above finished well platform level with drilled flanged end.
- Cover designed to prevent all kind of intrusion into the well
- Lockable fencing/compound to prevent unauthorized access and equipped with a proper security device.
- A PVC coated chain linked fence of 2000 mm high with 3'' GI tubes shall be installed with fence holding concrete base of 600 mm deep all around (200 mm above and 400 mm below ground) with a width of 200 mm
- Fence top shall have a single armed barbed wire
- A block work compound wall with similar dimensions and safety features
- Around the wall water weep holes of 2-inch PVC pipe at 3 meters intervals shall be provided
- In case of compound wall, a grilled window shall be provided near the well head to facilitate crane operation during maintenance works.
- Well head platform shall be placed towards one side of the fence/compound to facilitate crane operation during maintenance works.

3.12. Pumping System

After calculating the specific yield of the well from pumping tests, pump and control system can be designed. In Oman, submersible pumps are mostly in use. Submersible pumps are quicker and easier to install and are more efficient for deep wells as mechanical losses are limited.

The pumping system in general shall consist of the following appurtenances.

- Upstream single orifice air valve
- Downstream double orifice air valve
- Upstream/downstream pressure gauge
- Wash-out system
- Strainer
- Flowmeter
- Non-return valve
- Gate valves in appropriate locations
- Water sampling point

The standard installation of well pumping installation is given in the standard drawings.

3.13. Instrumentation

The borewell shall be equipped with the following instrumentation:

- Water quality online analyzers for residual chlorine, pH, conductivity, TDS, turbidity, SAC254 (Spectral Absorption Coefficient) etc. depending on the groundwater quality hazards
- Water level transmitter
- Pressure gauge with cut-off switch
- Well water level protection switch for pump and motor
- Motor winding temperature
- Motors electrical protection
- overload relay
- low/high voltage relay

Refer to PAM-SPC-700 General Electrical Specification & PAM-SPC-800 General ICA & OT Specificaiton.

3.14. Well Completion Report

In general, the well completion report shall have the following documents in detail.

- Krookie (land sketch)
- Mulkiya (land ownership)
- Well drilling permit (given by Ministry of Regional Municipalities and Water Resources)
- Drill logs
- Soil samples
- Well development documents
- Lithology – report and graphics
- Pump test observations and recordings
- Pump test interpretations
- Recommended yield and pumping pattern
- Recommended pump settings and flow rates
- Downhole video logging – report and softcopy
- Water quality analysis report – chemical and bacteriological analysis
- Well location map with grid reference
- Recommended maintenance schedule
- Work photographs - softcopy.

3.15. Regulation for The Protection of The Ground Water

In the Sultanate of Oman, water resources are limited and groundwater resources are subject to over exploitation due to growing needs. The protection of the ground water became a major stake.

In 2000, Water Wealth Protection Law was issued by the Royal Decree no 29/2000 (replaced the Royal Decree No 82/88) which declares water resources to be part of the national wealth and contains a specific article dealing with protection of groundwater resources.

Furthermore, the law on Conservation of the Environment and Prevention of Pollution is the key law on environmental protection and the prevention of pollution in Oman issued by the Royal Decree No 114/2001 (replaced the original Royal Decree No 10/82).

The pollution protection of the wellfields is under the responsibility of the Ministry of Environment and Climate Affairs (MECA).

The law on protection of sources of potable water from pollution is issued by the Royal Decree No 115/2001 which addresses the protection of drinking water resources from pollution and empowers MECA to identify drinking water protection zones. Activities likely to pollute drinking water will be prohibited in these zones.

The overall objectives of wellfield protection are to safeguard and protect the existing groundwater reservoir against over extraction, pollution, and to prohibit and eliminate unauthorized land and water uses that might adversely affect the wellfield.

Wellfield protection zones are designated by ministerial decisions. Initially three ministerial decisions were published in 1988/89. These regulations were modified, updated and amended in 2003.

The well fields are drilled and operated by the NWS.

4. Desalination Plants

4.1. Introduction

This chapter, Desalination by Reverse Osmosis, is prepared considering the experience of the design team on detailed design of Sea Water Reverse Osmosis plants. The design team shall be capable of studying the following in detail, submission of accurate data and forecasts in the design report.

- Water quality study for sea water at the intended intake;
- Marine and bathymetric surveys;
- Relevant regulations and environmental acts;
- Seasonal biological behaviour of the intake area and possible hazards.

4.2. General Requirements

4.2.1. Principle of reverse osmosis

Osmosis is a natural process involving fluid flows across a semipermeable membrane barrier. It is selective in the sense that the solvent passes through the membrane at a faster rate than the dissolved solid. When a semi permeable membrane separates two solutions with different concentrations of dissolved salts, a flow of solvent from the less concentrated compartment to the other one occurs, tending to balance the concentration gap. This flow creates a differential pressure between the two compartments, called "osmotic pressure" – which depends mainly on the salinity gradient and on the temperature (for a given type of membrane).

If a pressure higher than that of the osmotic pressure is applied on the compartment filled with the saline (high Total Dissolved Solids - TDS) solution, the direction of the flow of solvent will change: solvent (pure water) will flow from the high TDS compartment to the low TDS side. This phenomenon is called "reverse osmosis" and may be used to produce drinking water from saline water. This is referred to as a pressure driven diffusion-controlled process.

From a pure hydraulic point of view: a flow of pressurized feed water is divided into a stream of permeate (recovered on the other side of the membrane, at low pressure) and a stream of concentrate (flowing out of the system from the same side of the membrane, as the feed water, and still at high pressure).

Key operating criteria to reverse osmosis is:

- The recovery rate: ratio between permeate and feed water flows;
- The actual working pressure – which must be higher than the osmotic pressure of the concentrate;
- The flux through the membrane: flow of water through the membrane, expressed in cubic meter per square meter of membrane per hour;
- The salt rejection: is a measure of salt or TDS that is rejected by the membrane (not allowed to go with the permeate).

For a given salinity:

- The higher the feed pressure, the higher the recovery rate.
- The higher the temperature, the lower will be the osmotic pressure and salt rejection rate.
- The higher the flux, the higher will be the salt rejection at a constant temperature.

There are several types and sizes of membrane elements from numerous brands available in the market differentiated by the material composition. These all have different properties.

4.2.2. Sea water / Brackish water reverse osmosis plants

RO plants are designed to produce potable quality water either from sea or brackish water.

It is a reliable process, suitable for a wide range of production capacities, well adapted to modular design; RO plants can be supplied in containerized skids and moved with trailers for a mobile plant version .

Main aspects of RO plant design depend upon:

- Raw water quality and variability and associated design envelope.
- Performances and reliability of the pre-treatment.
- Treated water quality standard

Distillation processes based on desalination can cope with significant variations in raw water quality without major operational issues. However, RO membranes are sensitive to water quality parameters, for which excess values may affect the performances of the treatment or even damage the membranes.

The design of the pre-treatment process of the feed water is, therefore, the most critical part of RO plant design. Membrane failures or bad performances are a result of a lack of accurate knowledge of the raw water quality and poor or inappropriate design of the pre-treatment.

The RO performance and energy consumption are directly linked to the salinity of the feed water. Lower salinity levels require less energy than high salinity levels i.e. brackish water is more energy efficient per cubic meter of potable water production than sea water.

The “rate of permeate recovery” is an important parameter in an economical system design and the efficient functioning of the system depends further on the following parameters:

- Energy consumption;
- Feed water consumption;
- Concentrate disposal;
- Permeate salinity and
- Consumption of chemicals.

Design flux is a critical parameter in the design of reverse osmosis (RO) systems, it represents the volumetric flow of the permeate water per unit membrane area, typically expressed in litres per hour per square meter (LMH). The design shall define the flux to ensure the membrane performance, operational stability and long-term reliability.

$$Flux\ (LHM) = \frac{Permeate\ Flow\ Rate}{Effective\ Membrane\ Area}$$

- The design shall note that the flux impacts the membrane fouling, scaling tendencies, energy consumption and cleaning intervals.
- The design flux shall be tailored to the feedwater type, pretreatment quality, membrane specifications and site-specific operating conditions.

Table 4 Typical Design Flux based on typical feedwater type.

Feedwater Type	Salinity Range (TDS) mg/L	Typical Design Flux (LHM)	Rationale
Brackish Water	1,000 - 10,000	20 - 30	Lower osmotic pressure permits higher throughput.
Sea water	10,000 to 45,000	12 - 18	High salinity increases pressure and fouling risk

The design shall also consider:

- The pretreatment efficiency regarding the effective removal of suspended solids and organics allows for higher design flux without compromising membrane life.
- Temperature correction as the design flux is temperature dependent; higher temperatures typically increase permeability but may stress membranes (normalise flux at 25°C).
- Membrane type as some advanced membrane technologies may support higher flux, but the designers must confirm compatibility with feedwater and cleaning protocols.
- Recovery rate and pressure limitations of seawater RO often operate at lower recovery and flux to reduce risk of concentration polarisation and scaling.

The staging ratio defines how membrane area is distributed between consecutive high-pressure stages to achieve the targeted recoveries while controlling fouling, scaling and energy consumption. It refers to how much membrane area is allocated between the first and second stage (any other additional stages) within a pressure vessel train.

Table 5 Typical Staging Ratio.

Parameter	**Brackish Water RO (2-Stage)**	**Seawater RO (2-Stage)**
Staging Ratio	0.6 - 0.8	0.5 - 0.7
Stage 1 Recovery	50 - 60%	35 - 40%
Stage 2 Recovery	30 - 40%	15 - 25%

Parameter	Brackish Water RO (2-Stage)	Seawater RO (2-Stage)
Overall Recovery	75 - 80%	50 - 55%
Design Flux	20 - 30 LHM	12 - 18 LHM
Operating Pressure	10 - 25 bar	55 - 70 bar
Notes	Higher tolerance for flux/recovery; lower osmotic pressure	Lower stage recoveries to control scaling and fouling

The design shall also consider the following when designing multistage RO facilities:

- Membrane selection - The use of high-flux membranes in Stage 1 with high-rejection membranes in Stage 2 for better permeate quality and stable operation.
- Concentration polarisation (CP) - Later stages face higher feed salinity; smaller second-stage area helps maintain cross-flow velocity to keep CP in check.
- Energy recovery - For SWRO, match staging to ERD feed/brine flow splits - a lower Stage 2 area helps optimise device efficiency.
- Anti-scalant and pretreatment - Adjust chemical dosing and pretreatment quality to manage scaling risks at higher recoveries.
- Politing - where possible, confirm staging with pilot plant data under site-specific feedwater conditions.

Problems that arise in RO manifest themselves in the following ways:

- Reduction in water flux (production);
- Reduction in salt rejection (permeate high salinity).
- Increase in Differential Pressure (ΔP).

The three main causes are:

1. Compaction;
2. Degradation;
3. Fouling/ Scaling.

For RO system sizing:

- Select system parameters resulting in the most cost effective design and economical operation.
- Select the design which will produce the required quantity and quality of permeate at the highest possible recovery rate.

Parameters affecting the RO system sizing are:

- Total cost of ownership;
- Permeate recovery rate;
- Flux rate;
- Arrays;
- Type of pretreatment;
- Feed water quality and temperature.

Other key parameters affecting an RO plant are Equipment and process unit redundancy and plant availability.

4.2.3. Different stages in RO treatment system

RO water treatment plants in Oman typically consist of the following main process stages:

1. Intake and outfall
2. Pre- treatment;
3. Pressurization;
4. Membrane separation;
5. Disinfection;
6. Post treatment process;
7. Pressurization & energy recovery
8. Disinfection, remineralization and pH correction.

4.3. Raw Water Source and Quality

There are basically three types of raw water used for drinking water production through RO in Oman:

- Ground water (brackish water);
- Sea water from beach wells;
- Sea water from direct intake.

Table 6 Comparison between typical raw water quality

Parameter	Typical Range		Units
	Brackish Water	Seawater	
Total Dissolved Solids (TDS)	1,000 - 10,000	10,000 - 45,000	mg/L

Parameter	Typical Range		Units
	Brackish Water	Seawater	
Salinity	1-10	~35	g/L
pH	6.5 - 8.5	7.8 - 8.4	
Temperature	15 - 40	10 - 30	°C
Conductivity	1,000 - 10,000	45,000 - 55,000	µS/cm
Calcium (Ca^{2+})	50 - 200	390 - 440	mg/L
Magnesium (Mg^{2+})	20 - 200	1,200 - 1,350	mg/L
Sodium (Na^{+})	200 - 3,000	10,500 - 11,200	mg/L
Chloride (Cl^{-})	200 - 6,000	19,000 - 20,800	mg/L
Sulphate (SO_4^{2-})	50 - 500	2,650 - 2,700	mg/L
Alkalinity (as $CaCO_3$)	100 - 300	120 - 140	mg/L
Silica (SiO_2)	1 - 15	1 - 15	mg/L
Turbidity	<10	<1	NTU

Note that brackish water TDS can range from 2,000 to 45,000 mg/l in the above waters influenced by either coastal salinity or local geology.

For the design of an RO plant, it is of high importance to have an accurate knowledge of raw water quality and its variability. Full water quality analysis should be done for the following:

- Microbiological parameters
- Physical parameters
- Organic and inorganic parameters
- Ionic balance

For design considerations, percentiles are typically used depending on the water quality data set. In Oman, particular attention must be given to new sources as geology has a significant impact in certain areas.

In addition to the water quality data, an environmental / catchment survey should be carried out to identify all potential point and non-point, pollution sources. Mitigation of these issues should be included in the proposed design. Earlier experience has shown that use of beach wells provides significant protection against marine algae blooms (red or green tide), jelly fish and seaweeds. The Water Safety Plan methodology should be followed.

4.4. Sea Water Intake and Sea Water Outfall

The detailed design report shall include sea water intake as one major hydraulic mechanical and civil structural unit. The following are some of the factors that shall be considered in detail design:

- Bathymetric study
- offshore geotechnical study
- Water quality - physical, chemical & biological properties;
- Sea water currents
- Seawater levels - high tide, low tide and mean sea level;
- Any possible hazards of security and seismic;
- Flora & Fauna in the out-fall area;
- Protection and facilities for desludging, screening & cleaning for unwanted matter entering the intake;
- Protection against biological growth in the intake chamber and intake pipes (using shock chlorination and the use of pigging systems);
- Allowance for future expansion of the intake capacity;
- Proper sizing of pumps and drives;
- Mechanical, electrical and instrumentation works for uninterrupted operations.

It is mandatory to conduct a brine dispersion and recirculation study in order to comply with the environmental regulation :

In accordance with Article 11 (5) of Ministerial Decision No: 159/2005, the treated effluent discharge is required to comply with the following criteria within a 300 m radius from the point of discharge:

- Increase of the temperature of surrounding water to be less than one degree C (weekly
- average);
- Reduction of averaged dissolved oxygen to be less than 10% (weekly average);
- Change in the pH to be less than 0.2 unit;
- Increase or decrease in salinity to be less than 2 salinity units (2 ppt) of the daily surrounding averages.
- impact of brine recirculation to the intake shall be very limited (< 0.2 g/l 95% and less than 0.3 100% of time (typical requirement from Authority in charge).

4.5. Pre-treatment

Seawater intake shall be equipped to alarm and protect from adverse chemicals, phenolic, petroleum or organic compounds, which can adversely affect membranes and the process.

The incoming feed water is pre-treated to be compatible with the water quality specified by the manufacturers of membranes by removing suspended solids, adjusting the pH, pre-chlorination, de-chlorination and adding a threshold inhibitor to control scaling resulting from precipitation of various salts (calcium sulphate, barium sulphate, calcium carbonate).

Design of special treatment processes including Dissolved Air Flotation (DAF) systems and specifications shall be considered for possible periodical threats of algae, jellyfish and seaweed, which are being experienced in existing SWRO plants operating in Oman.

Redundancy of pre-treatment for uninterrupted operations of the SWRO plant at the full plant capacity shall be considered.

Cartridge filters shall be provided as an ultimate barrier to protect the membranes and guarantee a Silt Density Index (SDI) < 5 at all times as required by the membrane manufacturers.

4.6. Pressurization and Energy Recovery

The application of high-pressure pumps, and ERD with a booster raises the pressure of the pre-treated feed water to an operating pressure appropriate for the membrane and the salinity of the feed water. In practice, this pressure is higher than the osmotic pressure. The pressurisation system may, also, include the energy recovery device - transferring the energy from the brine concentrate to the feed water.

Separation: two types of membranes are currently used for RO systems:

1. Spiral wound and;
2. Hollow fine fiber membranes.

They are generally made of cellulose acetate, aromatic polyamides, or thin film polymer composites. Both types are used for brackish water and seawater desalination. The specific membrane type and size, and the construction of the pressure vessel containing the membrane elements, vary according to the different operating pressures used for the feed water and the product water quality requirement.

Currently there are two different sizes of spiral wound RO membranes based on manufacturing methods, these are 4040 and 8040.

Table 7 Key Specifications for 4040 and 8040 membranes.

Parameter	4040 Membrane	8040 Membrane
Diameter	101.6 mm (4”)	203.2 mm (8”)

Parameter	4040 Membrane	8040 Membrane
Length	1016 mm (40”)	1016 mm (40”)
Surface Area	7.9 - 9.3 m^2 (85 - 100 sq ft)	33.9 - 37.2 m^2 (365 - 400 sq ft)
Typical Flow Rate	2,270 - 9,085 l/d	7,570 - 37,850 l/d
Typical Applications	Small to medium systems	Large industrial or Municipal

Any new technologies based on membrane separation can be offered for approval, after full compliance evaluation.

Disinfection: Although the RO membranes remove (>6 log) bacteria, viruses and other pathogens if the process is working correctly, water still needs to be disinfected as a precaution to membrane failure and other contamination sources and to provide a free residual chlorine during storage, transmission and distribution.

Remineralization and pH correction: The disinfected permeate requires remineralization and pH correction with the aim of producing a chemically stable water, and one which is non-corrosive to materials in the water supply system i.e. metal and concrete. Remineralization may be accomplished by blending with high mineral content water (typically less than 3% of feed water) followed by disinfection. pH correction is always the last stage together with disinfection and prior to final water storage or distribution.

4.7. Pre-treatment Design

4.7.1. Pre-treatment performance

Typical feed water quality after pre-treatment should be within the following indicative ranges and shall not affect negatively, by any means, the membranes manufacturers’ warranty limits:

- SDI 15 — < 4 (90% time and < 5 at all times) 3
- Organic matter — < 1 mg/L
- Turbidity — < 1 NTU on average, and always less than 0.5 NTU.
- TOC — < 3 mg/L
- Oil and Fat — < 0.1 mg/L
- Oxidants — absent
- COD — 10 mg/L
- AOC — 10 µg/l Ac-C
- BFR — 5 pg/cm2 ATP
- Ferrous iron — 4 mg/L (pH <6, Oxygen <0.5 ppm)
- Ferric iron — 0.05 mg/L
- Manganese — 0.05 mg/L
- Aluminum — 0.05 mg/L

- Temperature 40 °C

4.7.2. Pre-treatment processes

The design of the pre-treatment depends on the source of raw water characteristics and variability.

A risk assessment and mitigation strategy on raw water characteristics should be carried out when selecting the preferred option to ensure that any loss of production capacity is acceptable. Cost benefits of the various options may also be studied:

Option 1 - For good quality sea water, with open intake:

The basic treatment process stages may include:

- Screening;
- Continuous / shock pre-chlorination (to avoid bio fouling). Please note continuous chlorination is discouraged other than in MSF plants;
- Flash mixing and coagulation with ferric chloride;
- First stage dual media (sand and anthracite) filtration in gravity or pressure filters;
- Second stage (if required) dual media filtration.

The two stages of filtration may be replaced by micro-filtration or ultra-filtration; coagulation is always necessary. Sedimentation may be required before UF in case of high SS.

Typical standard design parameters are:

- Retention time in flash mixing coagulation tank: typical minimum is 60 seconds;
- Hydraulic loading on dual media gravity filters: less than 8 m/h – with one filter being backwashed.

Option 2 - For sea water, with open intake, subject to green or red tides:

If algae blooms present a risk to production as is the case of the coasts of Oman occasionally, dissolved air floatation (DAF) facility may be included before the filtration stage as it is easier to 'float' algae than to settle flocs and particulates.

Option 3 – Deep sea intake:

This option aims at mitigating the presence of algae, oils, jelly fish etc., as raw water would typically be drawn from a depth at which impact would be marginal. Care must be taken on the depth, a figure of 10 to 15 meters (draw off 2 to 5 meters above sea bed) at low tide may be a safe depth at which Algal blooms will not affect the intake. It must be noted that at the end of a bloom, the dead algae (or mass jelly fish death) may present an issue if they accumulate on the sea bed around an intake structure. Environmental considerations may prevent this option.

Option 4 - For sea water from submerged sea wells or beach wells and (brackish) ground water:

The raw water coming from submerged sea wells, beach wells and brackish ground water may mitigate many risks and provide a more secure raw water supply from both a quality and quantity perspective. Environmental constraints may present limitations. Submerged sea wells, although not currently in practice in Oman, may be a possible option.

Table 8 Key Environmental Benefits.

Benefit	Description	Constraint Mitigated
Reduced impingement and entrainment	Seawater passes through sand/gravel strata that trap plankton, eggs and larvae, so fewer organisms get drawn into the intake system.	Marine organism mortality; biodiversity loss
Lower pre-treatment chemical use	Natural sediment filtration removes turbidity and organics, cutting coagulant and biocide demands by up to 50–80%.	Chemical discharge; toxicity to non-target species
Improved feedwater quality	Seabed filtration typically yields turbidity < 1 NTU and lower colloidal load, reducing downstream membrane fouling and cleaning frequency.	Membrane fouling; frequent CIP operations
Reduced energy consumption	Cleaner feedwater allows for lower-pressure ultrafiltration or direct RO operation, trimming overall energy use by 5–10 %.	High pumping costs; carbon footprint
Minimized noise and visual impact	All intake infrastructure sits below the low-tide line, eliminating unsightly coastal works and reducing above-water noise.	Coastal zone aesthetics; disturbance to local communities
Enhanced brine dispersion	When paired with deep diffusers, saline reject is discharged below the thermocline, promoting rapid dilution and minimizing surface plumes.	Surface salinity spikes; stress on nearshore ecosystems
Simplified environmental permitting	Demonstrably lower ecological footprint and risk profiles can fast-track approvals in jurisdictions with strict marine-impact regulations.	Regulatory delays; costly mitigation measures

For these options pre-treatment requirements may be much simpler compared to other intake systems. A simple mechanical filtration with MMF may be enough to reach the required SDI (< 3) for the RO feed. Best practice is to include a single or multimedia filtration to protect the membranes.

Hydrogen Sulphide is sometimes present in well water (coastal or other locations), or even in sea water, with much decomposing weed in it, if it becomes oxidized it can produce colloidal sulphur, which can have disastrous effect on membranes.

H_2S may be dealt by one or two methods; the feed can be degassed, and then treated with chlorine, if the feed is hard it may require acid dosing before the degasser.

In many locations, odour control is used to get rid of the gas by air strippers, typical techniques included counter current scrubbing with sodium hypochlorite or caustic soda.

4.7.3. Chemical conditioning before RO

In addition to the physical processes, pre-treatment must include a series of chemical conditioning of raw water to get acceptable feed water. The two types of fouling i.e. chemical fouling (e.g. by carbonates) and organic / biofouling (e.g. bacteria, algae, jellyfish) must be addressed to protect the membranes and as well as from chemical damage (e.g. by chlorine).

Typical treatments are:

- Shock: pre-chlorination (pre-filtration desalination for organics, hydrogen sulphite & biological control; continuous chlorination is discouraged)
- De-chlorination, by sodium metabisulfite (SMBS) injection (for membrane protection from chlorine damage)
- pH adjustment: generally, by acid injection (protection for chemical fouling and anti-scalant optimisation);
- Anti-scalant injection (typically a phosphate based substance).

The combination of the targeted pH and the anti-scalant requirement is a key parameter of the design, as it impacts the operational costs. The lower the required feed water pH, the less anti-scalant will be required. Although it will increase the amount of acid injection required. The choice of the most suitable anti-scalant depends on the chemical composition of the water, on the brine concentration rate and on the water temperature. A cost comparison needs to be done between acid (including caustic soda for pH correction on the permeate) and anti scalant uses.

Pre-treatment for borehole waters, well water (brackish water) may include aeration, before coagulation and flocculation to remove dissolved gases.

Chemical conditioning is required for Fe^{2+}, Mn^{2+} and high TOC removal. The design shall consider aeration followed by sedimentation for iron removal while MnO_2 media filtration is widely used for iron and manganese.

The design shall assess the different types of anti-scalants to be implemented based on the raw water chemistry. This assessment shall include different types of ion that can lead to the production of sparingly soluble salts (SO_4^{2-}, PO_4^{3-}, SiO_2 , Ba^{2+}, Sr^{2+}, etc). The scaling potential for $CaCO_3$ needs proper assessment using the different Saturation indices (LSI and S&SDI) and adjust to prevent scaling with the system.

4.8. RO Stage Design

4.8.1. General layout

The main aspect of the RO System includes the following stages:

- Cartridge filters
- Pressure pumps
- RO membranes
- Energy recovery devices
- Booster pumps

The other main equipment required within the RO systems includes:

- Dosing systems (scaling and fouling),
- Water quality monitoring systems (conductivity, pH, etc),
- Pressure measurements systems (ΔP monitoring, etc), &
- Valving

RO membranes (elements) are housed in pressure vessels or tubes.

Pressure vessels are typically designed and manufactured for either 4040 or 8040 membranes. Housings are sized to accommodate a minimum of 1 membrane while having a maximum of 7 or 8 membranes per vessel / tube is considered as reasonable with the existing available types of membranes. Smaller installations may house 3 or 4 membranes per vessel / tube.

Table 9 Typical FRP pressure vessel Pressure Rating.

Vessel Type / Application	Common Diameter	Max Design Pressure (psi)	Max Design Pressure (bar)	Max Design Pressure (MPa)	Notes
Brackish Water RO (BWRO)	4″ or 8″	300 – 450	20.7 – 31.0	2.07 – 3.10	Lower osmotic pressure; 300 psi common, 450 psi for high-TDS brackish
Seawater RO (SWRO)	8″	1 000 – 1 200	68.9 – 82.7	6.89 – 8.27	High salinity feedwater; 1 200 psi typical for open-ocean SWRO

Vessel Type / Application	Common Diameter	Max Design Pressure (psi)	Max Design Pressure (bar)	Max Design Pressure (MPa)	Notes
Ultra-High Pressure / Brine Concentrators	8″	1 500 – 1 800	103.4 – 124.1	10.34 – 12.41	Used for ZLD or very high-salinity brines

A number of pressure vessels / tubes are installed on skids. One skid can be considered as an operational unit: all the pressure vessels / tubes of one skid are operated as a single unit i.e. run together, or washed together.

The number of parallel skids or units is determined as a trade-off between investment costs and operational flexibility. A plant may be designed such that a capacity upgrade in future can be achieved by adding an additional skid or unit to a parallel process train.

In general, each skid or unit is fitted with its own high pressure pump and its own energy recovery device. The modern "pressure center" design is deviating from this concept.

All the above shall be evaluated together with the recommendations of the membrane supplier.

Number of passes:

To filter out certain special elements, such as Boron or to guarantee TDS or chloride levels, it may be necessary to implement multi successive passes. In that case, the partial or full permeate flow of the first pass is pumped to a second pass in order to increase the specific or total salt rejection rate. However, due to a recent relaxation in Boron limits in both WHO and Omani water quality standards, the full flow second pass application for adequately removing Boron has become obsolete.

The second pass (if required) can be designed to treat part of the permeate flow of the first pass only, depending on the expected efficiency of both passes. In this case, the permeate of the second pass is blended with the by-pass first pass permeate.

In case a second pass is used, the concentrate of the second pass has generally a low salinity compared to raw water; therefore, it can be recycled in the feed of the first pass. This must be taken into account in the general salt balance, when designing the first pass.

Number of stages:

In brackish water, the TDS content of the concentrate may be relatively low. In such a condition, it can be economical to implement a second stage of reverse osmosis; it consists of sending the concentrate of the first stage to a second stage of RO membrane, to increase the total recovery rate. Unlike the two passes configuration, the two stages design does not improve the salt rejection rate, but only the recovery rate.

4.8.2. Main process parameters

The choice of the membrane type and design of the RO skids are based on the following parameters:

- Salt rejection rate;
- Recovery rate: ratio between the quantity of produced water and the quantity of feed water;
- Feed pressure;
- Flux through the membranes;
- Average age of the membrane;
- Yearly performance decaying rate of the membranes;
- Longitudinal pressure drop;
- Chemical and biological resistance;
- Membrane cleanability.

Membrane projections using manufacturer software and including their recommended safety factors shall be considered when designing the RO system.

Salt rejection rate:

The choice of the optimum salt rejection rate depends on the quality of the raw water and on the targeted standard for the permeate. Drinking water is never "salt free" water: hence, a very high salt rejection is not always necessary. RO membranes provide a constant average salt rejection, however, different ions have varying permeabilities through the membranes. This depends on valency, shape, polarity,etc. Usually, low pressure membranes have high salt passage.

The higher is the feed temperature; the lower is the salt rejection rate. For each project, the required minimum rejection rate must be specified at the highest possible temperature.

Typical figures are:

- For brackish water (TDS is feed water < 3,000 mg/l): rejection rate > 90%;
- For low TDS sea water (mix of brackish and sea water, with TDS around 25,000 mg/l): rejection rate > 98%;
- For sea water (TDS > 35,000 mg/l): rejection rate > 99%.

The membranes are designed for different feed (raw) water characteristics and permeate characteristics depending on the application.

Feed pressure:

The feed pressure is directly linked to the type of membranes (permeability), and to the targeted recovery rate, feed TDS and target TDS. Typical figures are:

- Brackish water (TDS < 10,000 mg/l): 12 to 18 bars;
- Sea water (TDS > 35,000 mg/l): 40 bars and above.

The feed pressure changes with the temperature and salinity. High pressure pumps are fitted with VFD to control within to set pressure limits to protect the membranes.

Impact of age of the membranes:

With the ageing of the membranes, their performances decrease and the required feed pressure for a steady recovery rate tends to increase. Salt rejection rates tend to decrease.

The design of HP pumps shall take into consideration the worst case scenario.

The actual feed pressure can be assessed against the design feed pressure to give an indication of deterioration. The expected life span and performance guarantees should be clearly documented so that on-going performance measurements and assessments can be made so that an early indication of quantity or quality deterioration can be identified.

Longitudinal pressure drop:

The pressure drop across the membranes is dependent on type and on the number of membranes per pressure vessel. A standard figure of 1 bar of pressure drop along the entire length of the membrane is common and acceptable in addition to the 1 bar pressure drop with new membrane. An increase in the longitudinal pressure drop may be the clue of abnormal membrane fouling; Cleaning in Place (CIP) is then required to recover the membrane's performances.

4.8.3. Energy recovery device

The pressure of the concentrate at the outlet of the membranes is equal to the feed pressure minus the longitudinal pressure drop. This represents a considerable amount of hydraulic energy, which must not be wasted.

Three technologies are in use in recovering this energy:

- Turbine - Based EDR - Turbo-pumps turbines including Francis and Pelton types - converts the hydraulic energy of the high-pressure brine into mechanical shaft power, which is then used to assist the high-pressure pumps.

Table 10 EDR Technologies

Type	How It Works	Typical Efficiency	Notes
Pelton Wheel Turbine	Brine jet drives a Pelton runner connected to a booster pump	80–85 %	Simple, robust; more common in older SWRO plants
Francis/Kaplan Turbine	Reaction turbine coupled to pump shaft	80–85 %	Better suited for steady flows

Type	How It Works	Typical Efficiency	Notes
Turbocharger	Brine turbine directly drives a pump impeller on the same shaft	75–80 %	Compact, but less efficient than isobaric devices

- Isobaric Chamber EDR - Pressure exchangers include ERI and DWEER types - transfer pressure directly from the brine to the incoming feedwater without converting to mechanical energy - the most efficient class for large SWRO plants.

Table 11 Types of Isobaric EDR.

Type	How It Works	Typical Efficiency	Notes
Piston-Type Isobaric	Alternating chambers with pistons transfer pressure between streams	90–92 %	Few moving parts; good for medium-large plants
Rotary Isobaric (Pressure Exchanger®)	Rotor with longitudinal ducts swaps brine and feedwater under equal pressure	95–98 %	Industry standard for large SWRO; very high efficiency and reliability

- Hybrid EDR - Combines isobaric and turbine principles to optimise performance under variable flows.

Table 12 Table 11 Types of Hybrid EDR.

Type	How It Works	Typical Efficiency	Notes
Dual Work Exchange Energy Recovery (DWEER)	Two isobaric chambers with synchronized valves; excess flow drives a turbine	90–94 %	Handles variable flow better than pure isobaric

Type	How It Works	Typical Efficiency	Notes
Hybrid Turbo-Isobaric Systems	Turbine for excess brine + isobaric for base load	85–95 %	Useful in multi-train plants with fluctuating demand

The best recovery efficiency is achieved with work exchangers. A design efficiency of more than 95% may be achieved. The benefits of energy recovery devices are less on low TDS brackish water RO processes.

4.8.4. Cleaning in Place (CIP)

Even with a correct pre-treatment and anti-scalant injection, the performances of the membranes tend to decrease due to the fouling and scaling mechanisms. Recovery back to design membrane performances by chemical cleaning (without dismantling the membranes) should be carried out in accordance with manufactures recommendations. It is not good practice to skip cleans which are typically required on a three-month basis. Although marginal deterioration in performance may be seen over a short time period, the rate of fouling becomes exponential. There is a point of no return, at which the membranes cannot be recovered, leading to premature replacement of membranes. Failure to recover membranes to an acceptable level can also indicate changes in raw water quality, which can, in some instances, mean that the anti-scalant being used is no longer effective and needs changing.

Membranes cleaning is usually performed with the following kinds of chemicals:

- Citric acid - dissolves $CaCO_3$, Metal Oxides (Fe and Mn) and retightening of membrane pores;
- Caustic soda - removal of organic fouling, biofouling, and some colloidal fouling;
- NaDDS (Sodium dodecyl sulphate) - removes similar to caustic soda but it removes colloidal fouling;
- Hydrochloric acid - dissolves $CaCO_3$ scale, removes metal oxides /hydroxide and other alkaline scales;
- Ammonia (free Ammonia) - removal of biofilm, proteinaceous deposits, polysaccharide layer and mixed organic-biological fouling;
- Hydrogen peroxide and Peracetic acid - removes biofilm, free floating microorganism and residual organic slime.
- EDTA or Na-EDTA - removes metal oxides/hydroxides deposits, metal based colloids, some inorganic scales and organic-metal complexes.

The waste solution from CIP must be neutralized before disposal. Disposal requirements in Oman are legislated under various environmental laws.

4.9. Post Treatment Design

4.9.1. Final disinfection

Final disinfection can be achieved by solutions from on-site-electrolytic-chlorine (OSEC) generation systems, sodium hypochlorite liquid or calcium hypochlorite (powdered made up into a solution).

Each type of chlorination system has pros and cons. The Designer shall conduct a feasibility assessment for selecting the type of disinfection process to be used in every case. A significant concern is for safety of the operator and local community particularly with the use of chlorine cylinders (which shall not be used in Oman) or electrochlorination. Prime importance is ensuring that the water is successfully disinfected, and free from disease causing organisms. It is equally important to maintain a free chlorine residual into the water supply system to maintain a healthy water to the customer tap.

- To ensure the appropriate Concentration x Contact Time (CT) values are achieved;
- To control within 0.1 mg/l of the required set point (after a period of stabilisation);
- To dose within the range of 0.2 mg/l and 1 mg/l;
- To have alarms to identify loss of disinfection and excess dosage;
- To ensure disinfection is carried out AFTER blending for re-mineralisation (if in place);
- To ensure disinfection is carried out BEFORE pH correction (or alternative arrangement by agreement only if CT values and log inactivation are achieved within the process constraints).

4.9.2. Remineralisation and correction of pH

After the retention of the HCO_3^- and CO_3 ions by the membrane, the pH of the permeate must typically be corrected.

The goal of remineralization and pH correction is to reach the calco-carbonic equilibrium expressed by the Langellier Saturation Index (LSI) having a value equal to 0 or slightly positive (0 to 0.2), while slightly increasing Hardness and Alkalinity levels. In accordance with APSR Water Quality Standards the LSI value shall be: $-0.3 \leq LSI \leq +0.3$, calculated at operating temperature.

Depending upon the characteristics of the raw water and the pre-treatment process applied, there may be an excess or deficit of CO_2 content in the permeate water.

In the case of brackish water with low salinity, it is recommended to carry out:

- Aeration in order to raise the pH;
- An additional pH correction, by injection of lime water or caustic soda. Also the use of calcite mineral is also used, in specific occasions to correct pH.

If there is a deficit of CO_2 in the permeate (which is generally the case with sea water), the chemical post treatment would comprise the following:

- CO2 injection;
- Final pH correction, by injection of lime water and caustic soda.

In case lime is injected, this should be in lime water form (maximum concentration: 1.5 to 2.0 g/l) and never in lime slurry form, as slurry can significantly raise the turbidity of the product water. To obtain the lime water, a settling and decanting stage in a "saturator tank" is required.

Hardness, Alkalinity and pH can also be increased while reaching the calco-carbonic equilibrium in limestone dissolving filters. Prior to entering these hardening filters, CO_2 shall be injected to the water to enhance the limestone dissolving process. The application of limestone filters is a good alternative to the injection of lime water. A proper selection of the limestone material quality is crucial.

4.9.3. Chemicals Handling, Storage, Preparation and Dosing

The types and quantities of chemicals used within the RO plant depends on the characteristics of the raw water and on the design of the plant.

The reliability of the operation, the consistency of the treated water quality and the life of the membranes are directly linked to the accuracy of the chemical's dosage, which depends on a proper monitoring of the injection pumps, a precise preparation of the solutions, and a good management of the chemicals stock.

Health, safety and environmental laws and international best practice MUST be followed. A HAZOP (Hazard identification and mitigation) review should be carried out as a part of the design process.

Examples include:

- The condition of chemicals storage: easiness of access for delivery, sufficient handling equipment (cranes, monorail, forklift, etc.) for heavy bags and drums, safety provisions (retention bunds) in case of leakage of spillage;
- Chemical solution preparation: transfer, make up, transfer, storage volumes and containment;
- Chemical dosing: duty / standby dosing pumps (rotational operation), secondary containment of dosing lines, catch pots and calibration pots.

4.9.4. Chemical Waste Disposal

Chemical waste from RO plants includes commissioning wastes, pre-treatment chemicals, cleaning chemicals and post treatment chemicals, as well as emergency disposal of chemical leaks, etc.

Typical methods of disposal are:

- Lined evaporation ponds;
- Deep well injection;
- Disposal after neutralization in surface bodies;
- Mixed with the brine stream in sea outfalls (dispersal characteristics and environmental impact assessment study would need to meet Oman & International best practice requirements / objectives).

- Concentration into solid salts

4.9.5. Instrumentation and Control

The level of instrumentation and automation will depend on the size of the plant. The minimum of on-line instrumentation must be implemented, for a correct and steady operation of the plant. This should include:

Raw Water:

- Temperature;
- pH;
- Turbidity;
- Flow;
- Flowmeter;
- Discharge pressure;
- Conductivity;
- Chlorine dosage;
- Post filtration turbidity;
- Post filtration chlorine;
- Differential pressure MMF/CF.

Feed Water (location dependent on chemical application):

- ORP;
- SDI 15 (if automatic measurement is selected)
- Residual Cl2;
- pH;
- Temperature;
- Pressure;
- Conductivity;
- Flow;
- Permeate pressure;
- Anti scalant dosage concentrate;
- PX feed water flow;
- PX feed pressure;
- PX outlet water flow;
- PX outlet water pressure.

Permeate:

- Concentrate pressure;
- Conductivity;
- pH;
- Flow.

Treated Water:

- pH;
- Residual free chlorine;
- Conductivity
- Boron
- Flow;
- Discharge flow & applicable quality instruments.

Parameters for high pressure pump and for booster pump after PX:

- Water flow;
- Suction pressure;
- Discharge pressure;
- VFD control signal.

If the RO system is a 2 staged system, the following instrumentation should be on each stage:

- RO feed pressure;
- RO concentrate pressure;
- RO permeate flow;
- RO permeate pressure;
- RO permeate conductivity.

All instrumentation must be fitted with data logging and out of range alarm management systems for local operational control & management.

The requirements for instrumentation, control and automation will be detailed in a separate document. Refer PAM-SPC-800 ICA & OT

All instruments should have a digital output, with sufficient de-bounce for manual logging purposes.

All instruments should have supplies in accordance with manufactures equipment recommendations e.g. dual (shared) feed water supplies may not be suitable in some cases.

All instruments should be clearly labelled.

Process flow diagrams and P&ID's must be provided with full detail of process stage conditions and criteria.

Other subsystems and components in the RO plant:

- Chemical dosing system.
- CIP system.
- Power supply HV, MV and LV primary substation.
- SCADA and DCS control system and data management.
- HVAC and service water network.
- Sewerage and effluent treatment system.

4.9.6. Materials

All materials in contact with brackish water, seawater and concentrate shall be of suitable design and shall have proven suitability in comparable installations. Corrosion due to inadequate combinations of different metals must be avoided. GRP, GRV or GRE (Glass Reinforced Polyester, Vinyl Ester or Epoxy may be selected for Low pressure application (< 16 bar) or HDPE.

1. Stainless steel material in contact with seawater or seawater concentrate shall have an actual Pitting Resistance Equivalent Number (PREN) of at least 40 and a Crevice factor (CF) of at least 35; to be calculated as follows:
   i. For super austenitic alloys: PREN = Cr (%) + 3.3 x Mo (%) + 16 x N (%)
   ii. and a Crevice Factor: CF = Cr (%) + 3 x Mo (%) + 15 x N (%).
   iii. For super and hyper duplex alloys: PREN = Cr (%) + 3.3 x (Mo+0.5W) (%) + 16 x N (%)
   iv. and a Crevice Factor: CF = Cr (%) + 3 x (Mo+0.5W) (%) + 15 x N (%).
2. 316L grade steel can be applied for submerged intake screen facilities such as bar screen and traveling band screen, applying a cathodic protection system by impressed current.
3. All stainless-steel piping shall be passivated in accordance with ASTM A380 after site fabrication has been completed. Socket welded fittings are not allowed when designing with stainless steel materials.
4. Where materials are subject to an elevated risk of corrosion (e.g. due to stagnant water or presence of crevices) additional protection shall be applied e.g. by coating or cathodic protection.
5. All stainless-steel welding shall as a minimum be tested in accordance with ASTM G48 (2015) 'Standard Test Methods for Pitting and Crevice Corrosion Resistance of Stainless Steels and Related Alloys by Use of Ferric Chloride Solution'.

5. Reservoirs

5.1. Objectives

The aim of reservoirs is to provide water balance for distribution networks or/and pumping stations during peak usage as well as to provide a reasonable undisturbed level of service.

This aim will be achieved through the following objectives:

- Locate reservoirs in prime locations in as much as reach distribution networks by gravity;
- Provide enough capacities to maintain the minimum volume, between any two programmable logic levels (HiHi to Hi), required to operate a pump within its minimum operational time;
- Maintain adequate level of pressure within a distribution network in as much as reduce the need for pressure controlling valves;
- Locate reservoirs downstream the main interconnected transmission system to secure storage capacities within network coverage area;
- Design holding capacities inasmuch as maintaining the level of service of the networks during major shutdown for about 48 hours.

The size of the reservoir in volume is determined by the number of hours of storage which refers to 24h of the **peak day of water demand.** *(Example : if the peak day water demand is 12,000 m3, 12hrs of storage corresponds to 6,000 m3).*

5.2. Classification

5.2.1. Strategic Storage Reservoirs

Strategic Storage Reservoirs are large-scale reservoirs (around and above 100,000m3) designed to ensure resilience to the water supply, particularly between Governorates or where emergency reserves are needed. Storage capacity shall be specifically studied by including the strategic transmission network and Desalination Plants internal storage capacity (which is recommended to correspond to one day of the plant capacity).

5.2.2. Transmission pumping Reservoirs

Transmission pumping Reservoir within a water transmission system. It stores water before it is conveyed over long distances and accommodates fluctuations in source availability or demand. Storage time shall be a minimum of 4 hours of the downstream peak day water demand.

Larger volumes can be considered by designers in order to enable energy optimization.

5.2.3. Distribution Storage Reservoirs

Distribution Storage Reservoirs are located within the distribution network and serve as a buffer to meet variations in consumer demand and the main resilience for the Level of Service. It helps maintain consistent pressure, provides emergency storage for firefighting or system failures, and allows for operational flexibility. Storage time shall be a minimum of 48 hours.

5.2.4. Elevated Pressure balancing Tank

During design the consultant shall evaluate the benefit of an elevated tank versus booster station in order to maintain the pressure during peak hours or in specific network situations.

It is designed to store water at a height that allows for gravity-fed distribution, ensuring consistent pressure throughout the network. Storage time shall be a minimum of 4 hours.

However, for small villages having less than 2,500 inhabitants, the elevated tank may be designed to satisfy one full day demand.

5.2.5. Break Tank

During design the consultant shall justify the need of a Break Pressure Tank instead of a PRV. Break Pressure Tanks are present in some parts of the network but progressively replaced. It serves as an intermediate storage unit where water is temporarily held before continuing through the network. Storage time shall be 2 hours.

5.3. Reservoirs Design Guidelines

5.3.1. Design and demand considerations

The capacity allocation for each type of reservoir shall be governed by the project's demand calculations based on the latest version of NWS master plan demands forecasting and trending according to National Centre for Statistics and Information (NCSI). In addition, the designer may refer to the General Design Guidelines.

In general, the designer is to consider two peak day demand, pumps off, storage capacity. Whilst designing water storage reservoirs, future developments should be taken into consideration by optimizing the footprint for the immediate phase and allowing space for future expansion. In addition to the two peak day demand and for Distribution Storage Reservoirs only, water required for firefighting shall be included as per the Public Authority for Civil Defense and Ambulance (CDAA) guidelines. This fire storage volume depends on the maximum flow rate and duration requirements needed in the supplying distribution system in accordance with CDAA guidelines.

For operations and maintenance purposes, Reservoirs (except elevated tanks) >200 m3 shall be composed of two or more similar compartments.

The Gross Volume (GV) of the reservoir consists of the volume between the overflow level and finished floor level (FFL) excluding the free board which is the distance between the overflow and the bottom soffit of the roof structural element (slab/beam).

Dead storage (DS) is the volume of stored water not available for distribution permanently. It is the total storage below pipe top level of the lowest discharge outlet from the reservoir. The dead storage usually contains accumulated deposits such as mud, silt and residues which should not enter into the potable water distribution system directly. The dead storage volume should not be more than 5% of the reservoir's effective volume.

The **Effective Storage (ES)** or the service storage for peak day is the volume of storage aiming to meet two peak day demands in a water distribution system and is excluding the water required for firefighting. The design engineer shall consider the water source, the hydraulic capabilities to determine the ES of each storage compound. If the project is to be phased and more storage reservoirs are required for the ultimate design period, the first phase design shall accommodate for the future expansions.

The effective storage volume is calculated as the difference in height between the High-High (HH) and Low-Low (LL) levels multiplied by the cross-sectional area.

For transmission pumping reservoirs, the difference between HH and LL levels shall confirm the minimum requirement for pump operation, e.g. suction volume, minimum on/off period and accommodate the maximum hour of a peak week.

The design must incorporate appropriate disposal procedures for chemicals used during disinfection (both during service life and periodic maintenance) to prevent harm to the surrounding environment.

Furthermore, the designer must ensure a flexible and maintainable design. This includes specifying periodic maintenance schedules for all components within the reservoir compound, covering structural, electromechanical, and instrumentation elements.

5.3.2. Alternatives

The designer has to consider performing study for the most cost and time effective choice for the types, volumes, location and elevation of the reservoirs based on serviceability, water quality, materials, constructability and OPEX. The study should be based on the present value analysis for the available alternatives.

5.3.3. Material Selection Matrix

Various types of reservoirs are existing in Oman and allowed for water storage:

- **Reinforced Concrete (RC): Circular or rectangular, ground or elevated Reservoir.**
- **Glass Reinforced Plastic (GRP) Modular Tanks: s**hall be designed according to NWS Standard Specification and obtain approval.
- **Carbon Steel (CS) Tanks: Circular** welded steel tanks shall be designed according to NWS Standard Specification (referring to AWWA D100-96) and obtain approval as an alternative to RC.
- **Glass Fused to Steel (GFS) / Bolted Steel Tanks:** shall be designed according to NWS Standard Specification (referring to EN ISO 28765 and/or AWWA D103) and obtain approval as an alternative to RC.

The selection of water tanks material is a complex process of optimizing an array of information to yield a particular design. Once the quantity and the type of water to be stored are established, the physical properties (such as temperature, potential for fires and explosion, operating pressure, etc..) determine the range of possible tank types (Table 12).

Table 13 Reservoir Material Selection matrix.

Storage Capacity (m^3)*[1]	Reservoir Material
≤ 1,000	GRP / GFS / Reinforced Concrete
≤ 10,000	GFS / Reinforced Concrete
10,000 to 50,000	Welded Steel / Reinforced Concrete
50,000 and above	Reinforced Concrete

5.3.4. Site allocation

The location of a reservoir is influenced by hydraulic considerations so that the Level of Service of the water network is satisfactory under the design peak demand conditions. The site allocation shall take in consideration the following factors:

- The safe access to the site in all weather conditions shall be considered.
- The proximity to the distribution area and proximity to the source.
- The land availability for reservoir, inlet/outlet/scour, overflow pipework, flow meter and pump station structure etc. The location shall not be near or next to heritage places and graveyards. Provision of sufficient buffer from existing houses, and future planned area development.
- If possible, reservoirs shall be kept on the mountain (Jabel) to supply water to the demand areas. Special considerations, calculation should be given to minimize the mountain's cuts and excavations ensuring the homogeneity of the soil foundation of the structure.
- The choice of ground reservoir at Jabel or elevated tank at ground shall be decided based on capital (CAPEX) and operational (OPEX) cost analysis.
- Pumps house location should be considered (Pumps Suction, noise).
- Electrical power supply facilities should be available or able to be provided economically to the site.
- Consideration should also be given to the potential likelihood of future development to enable a future storage of similar or required size within the site footprint. Minimum site backs shall be considered. Spacing between structures shall be the maximum of footing interaction or height of super-structure.
- Consider adequate storm water management and drainage for the site facilities so that flooding of the facility is avoided.
- Access roads shall be designed as per the Ministry of Transport (MOT) and ROP standards. The ascending gradient should not be more than 1V:10H.

To ensure that the proposed reservoir location and layout are acceptable, the proposed site shall be approved in advance by NWS.

[1] Effective Storage volume of the single tank.

5.3.5. Topographic survey

The designer has to consider having a detailed topographical survey for the selected reservoir sites, showing clearly the contour lines elevations and the contour intervals. Topo - survey drawings with topo-survey report shall be submitted along with the preliminary design report for the reservoirs. The drawing should include the volumes for cut/fill and calculations.

5.3.6. Seismic considerations

All types of the reservoirs recommended in this guideline shall consider having earthquake resistant structures according to Seismic Design Code of Oman – Version 1 and to Eurocode EN 1998(Eurocode 8 – Design of structures for earthquake resistance.

5.3.7. Soil investigation

5.3.7.1. Geotechnical investigation

The designer has to consider having comprehensive soil investigation works for the new projects sites as per the specified standards to verify the homogeneous, settlements and the strength of the soil layers underneath the reservoirs foundations before proceeding with the detailed design. The soil investigation works shall be conducted by an approved professional geotechnical offices/laboratory, registered in Oman. The design must count on the collected sites soil parameters. No design for the reservoirs and other site facilities foundations allowed based on assumed soil parameters.

5.3.7.2. Geophysical Investigation

The Designer has to consider having additional geophysical soil investigations tests to verify the homogeneous settlements and the strength of the soil layers underneath the reservoirs foundations before proceeding with the detailed design. Either type of the following GPS tests recommended, depending on the depth of the soil layer:

- Multi-channel speed wave analysis (active or passive)
- Seismic refraction test (compressive and shear wave velocity profile)

5.3.8. Reservoirs foundation requirements

All structures shall be supported by strong foundations where the maximum differential of uniform tilting shall not exceed 1/500. Reservoir on-grade-slab/foundation shall be designed based on the report of soil investigation (geotechnical) to withstand the minimum bearing capacity. Preferred to be constructed on hard strata of mountains but in case of lean soil, if alternative location is not available, the soil shall be improved according to guidelines included in PAM-GUD-201 section 6.1.2.3 and PAM-SPC-404 or Cast-in-situ pile foundation shall be designed according to Eurocode 7 - Part 7 (EN 1997-7).

5.3.9. RC Reservoirs structural requirements

Design of reservoir structure shall be performance-based design and checked for strength requirements. The structural requirement Alkali silica reaction (ASR), water absorption and equivalent sodium oxide shall be considered according to the international design codes. In order to control crack widths, reinforced concrete structures shall be made of hybrid conventional steel reinforcement and non-metallic fiber. A maximum crack width of 0.2 mm shall be considered (including presence of fiber) during the design according to Eurocode 2 (BS EN 1992-1) or fib Model Code 2010. Concrete shrinkage shall be limited to 0.05%. Providing that curing shall be carried out by the contractor immediately after de-shuttering.

The designer shall consider designing all structural elements using a suitable 3D structural analysis software (preferably finite element analysis) providing that the software is capable of integrating all elements together. Provision of expansion/contraction and construction joints shall be designed to enhance the durability of the structure. Congestion of reinforcement, at joints, shall be cleared during the design stage.

Expansion/ construction joints are weak joints within the structure and thus; reinforcement detailing within and adjacent to these locations is a must. Therefore, location of placement of waterstop bar or swelling waterstop bar shall be considered, i.e. congestion due to reinforcement, waterstop technology and aggregates shall be studied. Preferably, at these locations, details up to 90% (min. 60%) to be carried out using the aid of 3D modelling software.

The designer shall consider a minimum grade of 35 MPa for concrete with a maximum aggregate size of 20 mm and concrete cover of 50 mm (water side) and 75 mm (for concrete cast against soil). Supplementary cementitious materials can be considered to enhance the long-term performance of concrete and shall meet the requirements of AWWA or WRAS, etc. for use in potable water retaining structures. Reinforcement shall be made of a ductile/high tensile strength steel with a minimum grade of 500 MPa according to BS 4449. No epoxy coated reinforcement shall be used. The clear spacing between rebars shall not exceed 150 mm or be less than 110 mm.

The designer shall consider roof top finishing to consist of thermal insulation and screed with average thickness of 100 mm. Solar panels shall be considered to be fixed on top of the finished roof.

5.3.10. Reservoirs compartmenting and partitioning

The ground level reservoir ought to be designed with a partition to enable for the scheduled maintenance/cleaning of service reservoirs without causing any disruption of receiving and supplying of water. In general, division wall(s) shall divide(s) the total volume into equal compartments. For large storage capacity reservoirs (>50,000m3 per unit), the number of partitions and position of partition shall be decided based on qualitative and quantitative analysis. In case of expansion/contraction joints, water stop bars shall be located in the middle of the section in both retaining walls and on-grade-slabs. Partition shall not be required if the effective capacity of ground level reservoir is less than 200 m3.

5.3.11. Reservoirs Hydraulic testing

The reinforced concrete reservoirs have to be tested in accordance with BS EN 1992-3:2006. The tests are to be performed before the application of an internal liner.

Steel Tanks have to be tested in accordance with AWWA D100-2024.

5.4. Reservoirs lining and coating

5.4.1. New RC Structures lining

It is not mandatory to install a liner in a new RC reservoir. The selection of the internal liner is to be justified and can depend on several factors:

- Existing structure rehabilitation or new reservoir
- Water Safety: Prevents infiltration and contamination, compliance with drinking water standards.
- Structural Protection: Shields concrete from corrosion and cracking, extends service life.
- Operational Efficiency: Smooth surfaces reduce deposits and biofilm, simplify cleaning and maintenance.
- Resource Preservation: Prevents leakage, water volume.
- Regulatory Compliance: design and hygiene requirements.

The designer has to take into consideration that zero leakage is allowed in the design. The designer of a water retaining structure shall consider a water stopper or/and any other methodology, used in good engineering practice, to be provided at the joints during construction. In case of water tightness failure, Designer/Contractor may include internal liners based on the final use case (Potable Water / TSE) of the water retaining structures.

- Potable water internal liners require hygiene certification at 60 °C and comply with BS 6920, NSF 61, Kiwa or equivalent and approved by NWS.
    - Spray-applied polymeric liners (polyurethane, polyurea)
    - Integral leak proofing using admixtures to concrete - Crystalline growth admixtures and Pore-blocking hydrophobic admixtures,
    - Cementitious and polymer-modified linings
- TSE internal liners can also include composite laminate liners - Fiber reinforced polymers (GRP and GRE) and does not require WRAS certification

5.4.2. RC Reservoir rehabilitation by Internal lining

In case of structural failure and/or water leakage, root cause analysis (Reservoir structure diagnostics) shall be conducted before establishing a durable rehabilitation plan. This diagnosis shall include:

- As-built drawings analysis,

- Geotechnical survey or analysis,
- Inspection (visual, cracks, core-drilling, scanning of rebars, cover thickness, corrosion, 3D scan…)
- Hydraulic situation, and system role of the reservoir (out of order, need expansion, …)

The benefits and the principles of the rehabilitation solution shall be established by an experienced designer to enhance the reservoir's structure durability and increase the lifetime of the asset.

The following lining systems materials can be used:

a. Cementitious lining material

b. Epoxy lining material / hybrid coating

c. Polyurethane lining system

Lining materials must be WRAS/NSF approved for materials in contact with potable water at 60 °C ambient temperature with chlorine contents, tested according to BS 6920. Lining material shall feature:

- Outstanding physical properties
- Excellent chemical, abrasion and impact resistance
- Good low temperature flexibility
- Low gas permeability
- Excellent UV and weather resistance
- Does not contain solvents
- Easy to apply
- Providing a seamless coating with no vulnerable joints
- Can accommodate difficult surface profiles and shapes
- High bond strength to substrate
- Long and effective life
- On site quality assurance program
- Highest elongation offered
- Applied only by authorized by the supplier and trained contractors

5.4.3. Steel Tanks lining

For steel tanks, consider the materials used to prepare the surface of the tank, and the painting or coating water systems used to protect against corrosion. The design should provide cathodic protection as necessary.

Lining should be provided for steel tanks to protect from provision, in accordance with safety standards for drinking water, such as NSF Standard 61; and be acceptable to the reviewing authority.

Similarly, interior paint must be applied, cured, and used in a manner consistent with the ANSI/NSF requirements.

The main standards and guidelines for GFS tanks are:

- EN ISO 28765: Vitreous and porcelain enamels - Design of bolted steel tanks for the storage or treatment of water,
- EN 15282: Vitreous enamel - Design of bolted steel tanks,
- AWWA D103: Factory-Coated Bolted Steel Tanks for Water Storage,
- DIN 4034: Precast concrete manholes,
- BS 7777: Flat-bottomed, vertical, cylindrical storage tanks for low temperature service,
- API 650 for structural aspects,
- NSF/ANSI 61 for drinking water quality.

The preventive measures are:

- Strict factory quality control of vitreous coating,
- Specialized training for assembly teams,
- Assembly supervision by certified expert,
- Panel protection during transport and storage,
- Strict compliance with manufacturer's assembly procedures,
- Systematic leak testing after assembly,
- Proper installation of ventilation system.

The corrective measures are:

- Specific repair kit for small impacts,
- Stock of spare sealing joints,
- Systematic annual inspection,
- Preventive replacement of sealants after 10 years,
- Cathodic protection, if in aggressive environment,
- Regular cleaning and disinfection,
- Maintenance contract with specialized company,
- Detailed intervention documentation,
- Regular monitoring of coating integrity,
- Immediate repair of any detected defect

5.4.4. RC Structure external coating

In case of a high ground water table and roof slabs (RC), the Designer/Contractor shall provide the necessary protection (waterproofing) for the infrastructure to and shall consider the different methods and materials that can be used for waterproofing based on the site conditions. These may include:

1. Bituminous coatings and membranes

2. Synthetic sheet membranes (TPO, EPDM, HDPE, etc…)

3. Liquid-applied polymer coatings (polyurethane, polyurea, acrylic elastomers, etc…)

4. Cementitious flexible coating (polymer-modified cement slurries with flexibility for minor movements)

As the structures are designed to be watertight, crack width is less than 0.2 mm, and the expansion joints are filled/treated with sealants that are designed not to pass water under extreme conditions, no additional further surface treatments are required.

The protection of concrete roofs can be carried out using liquid-applied chemical waterproofing systems such as polyurethane or acrylic resins. These coatings are applied directly over the existing surface after cleaning and repair, forming a seamless and fully bonded waterproof membrane according to the Standard EN 1504-2.. The system bridges micro-cracks, tolerates thermal movements, and provides long-lasting resistance to UV and standing water. This method avoids heavy demolition, reduces downtime, and extends the service life of the roof structure.

It is recommended to use white to protect against heat because it reflects light instead of absorbing it, which greatly reduces surface warming.

5.5. Components of reservoirs

5.5.1. Reservoir leakage monitoring system

The design of any reservoir shall include a Leakage monitoring system. This system includes the following:

- Drainage system under the foundation of the reservoir split in areas according to the shape, compartments and joints location,
- Monitoring and accessible manholes with a water drop allowing measurement of gravity flow,
- Testing and the methodology of the measurement system is to be provided within the As-Built documentation.

The design of the leakage monitoring system of the reservoir is subject to approval from NWS.

5.5.2. Overflow site surface water drainage considerations

The design has to consider (based on site and vicinity topographical survey) having a draining of the accumulated rainwater on the reservoir’s roofs, the entire site, and the overflow water. Reservoirs, roofs and sites must be adequately sloped and drained offsite. The designer has to consider preventing any water pooling inside or offside the reservoir site. Designers should consider having approved offsite-draining locations from the local authorities. Flap valve to be designed at the end discharge points. Special consideration to the offside draining pipeline and drops off point, it should not affect, in any matter, nearby citizens and private or official properties.

5.5.3. Inlet pipes chambers

The inlet pipe diameter should be sized for the ultimate designed flows (considering one compartment is filling and other under maintenance). A 90° bend shall be provided on the discharge to allow proper mixing. The inlet pipes shall be located outside the water compartment. Top wall inlet shall be considered for inlet water pipe. Inlet pipe elbow rack (thrust block) shall be designed to be positioned within the inlet chamber. The designer has to consider having inlet embedded puddle flanges in the wall.

Minimum two inlets to be provided with additional one, redundant, for future connection. To be considered in liaison with NWS.

Inlet is advisable to be located above the Hi-Hi water level but in case that bottom inlet has to be done for certain reasons then, CFD modelling to be carried out based on the design.

5.5.4. Outlet pipes chambers

The outlet pipe diameter shall be sized for peak instantaneous (peak hour demand or fire flow scenario whichever is more). Outlet pipe shall be one size larger than the discharge water main to allow for a future increase in capacity. The designer can propose more than one outlet.

Minimum two outlets shall be provided for each compartment with additional (redundant) one for future connection. To be considered in liaison with NWS. The redundant outlet shall be located next to the main outlet within the same outlet chamber. It shall be closed with the blank flange and to be identified as "Redundant". The designer has to consider having outlet embedded puddle flanges in the wall.

The outlet pipe is to be fixed at a minimum 100 mm height above the outlet sump to minimize the risk of any accumulated sediment within the reservoir being detained during reservoir operation. This is typically achieved by positioning the outlet above the designed grade of the reservoir floor.

5.5.5. Overflow pipes

An overflow pipe has to be provided in each compartment with a suitable outlet arrangement. The overflow pipe diameter shall be sized for peak instantaneous flows (all pumping running at possible extreme flow regime). In general, the overflow pipe size is one size more than the inlet pipe. Bellmouth fitting is preferable for overflow arrangement, situated a minimum 100 mm above top water level with minimum of 300 mm freeboard between top of Bellmouth and underside of roof elements and it should be free of obstruction from roof members.

The overflow line shall be properly routed to nearby Wadi and sufficient diligence should be made in such that the overflow water should not cause any inconvenience by flooding nearby dwelling or public access roads or public or private properties.

5.5.6. Spilling response plan

A spill response plan should be developed for the utility in the event of a major reservoir spill or catastrophic failure. The response plan should be tailored to the reservoir design and site location. The plan should outline which design features are in place or need to be activated for minimizing the impact on life and property in the vicinity of the reservoir structure. Drawing should be developed to show the details of the plan. *Also refer to NWS' Emergency Response Manual.*

5.5.7. Water quality considerations

5.5.7.1. Water circulation modelling

The Reservoir must be designed to minimize stratification and have adequate circulation and baffling to prevent "dead spots" within the water column and short circuiting across the reservoir. The designer shall model the behaviour of the water in the reservoir using a Computational Fluid Dynamic (CFD) model and system behaviour data, including:

- Diurnal cycles and chlorine levels provided by NWS Operations
- Water flows, stratification and short circuiting minimization with and without measures
- Inlet location and outlet locations.

Recommendation whether to use baffle walls/curtains or agitators or combination of both.

5.5.7.2. Disinfection

On-site liquid hypochlorite generation system may be considered for disinfection system for the reservoirs compartments, before and after commissioning. The system shall be designed for minimum 20 mg/liter for disinfecting the reservoir before commissioning. The disinfection system shall be designed for maximum continuous residual chlorination of 0.5 mg/liter during operation.

The following systems may be considered for design:

- On site Liquid Sodium Hypochlorite production system.
- On site Liquid Calcium Hypochlorite production system.

Consideration should be given for designing cold chemicals storage for Sodium / Calcium Hypochlorite (liquid or dry) for safe operation sustainability. Dosing points shall be provided at inlet and outlet, noting that eyewash system should be considered with the design. Cost benefit analysis shall be performed and submitted with the design report for the best, cost effective adopted system.

5.5.7.3. Water quality taps

Water samples quality checking taps shall be provided for each compartment inlet and outlet pipes. The sample collections taps shall be easily accessible and marked according to NWS quality department requirements.

5.5.8. Ventilations considerations

Reservoirs must have adequate sanitary protection to prevent the water supply from being contaminated. Drinking water regulations require reservoirs to prevent entry by birds, animals, insects, excessive dust and other potential sources of contamination.

Finished water storage facilities must have dedicated screened vents designed to allow air in and out to balance internal pressures. Vent size must be adequate to relieve vacuum during peak-flow conditions. Vents must be sealed and secured to the roof to keep out contaminated surface water and to deter vandalism (typical for all vent designs). Overflows are not venting.

Vents should be constructed and maintained to avoid plugging or air restriction from dust build-up. Use reasonable security measures to protect the reservoir and stored water from possible damage and compromise by unauthorized persons.

Vents must have screens to keep insects and animals out of the reservoir. It is recommended using durable 24-mesh non-corrodible screens backed with 4-mesh screens. Further, vent openings must face downward or have shielding to minimize the entrance of insects, surface splatter, rainwater and excessive dust.

The designer should consider having one electrical ventilator in each compartment for providing the necessary fresh air during the maintenance according to international standards.

5.5.9. Reservoirs access

5.5.9.1. Hatches

Hatches are to be of the type raised in a single door and equipped with a second inner safety grid forming an internal compartment which provides insulation between the inside and the outside to protect the reservoir. This internal anti-contamination compartment includes a drainage system and mounts for the hatch.

The assembly of the wing(s) internal(s) includes ventilation to allow the free flow of air from the tank, preventing the accumulation of high levels of chlorine and protects the roof of the tank from the phenomenon of "sucking" during declining water levels. The outlet of the ventilation must be protected by a mosquito grid made of stainless steel 316.

On the cover of each hatch, an open sensor (linked to an alerting system at control room) is to be fitted. The output cables are made by passages through the wall bracket and terminating in the compartment between the two traps.

The hatches are sealed, vented and equipped with fixing points and concealed hinges. Opening of the doors is supported by stainless steel springs. The doors should automatically lock in the open position at 90 degrees. The outer cover is domed to promote runoff and lifting is done by an external handle located on one side.

Locking is done by turning the lock 90 degrees. Locks are hidden under a cover ensuring its external protection. The number of keys is limited and they are common to all sites. Access cover must be single man lift type assisted by a hydraulic piston. For additional clarification, refer to PAM-SPC-504

5.5.9.2. External Ladders

For security and maintenance of the exterior of the reservoir, a ladder equipped with a safety cage and hand rail must be installed. The ladder should be aluminum or galvanized steel or stainless steel. The ladder shall have a locking facility to restrict access to unauthorized personnel.

5.5.9.3. Internal ladders

The internal ladder (inside the reservoir compartment) should be constructed of completely stainless steel only. Four rungs of the ladder must protrude outside of the entry hatch. This can be achieved by part of the ladder being hinged beside the access hatch acting as a secondary barrier and ladder extension.

5.5.10. Site accessibility considerations

Access road is to be provided from nearest public road to the NWS site (reservoir/pump station), where the access from a public road is not available then a suitable access road shall be created. The design of the access road shall be subjected to approval and in accordance with the local ROP / MOT /Local Municipality requirements. The design of access is based on the size of operating and maintenance vehicles (min. 3T truck capacity).

The following shall be considered whilst designing the access road:

- External access with a minimum pavement width of 5 meters.
- Provide an access road around the full extent of the reservoir or extent up to which the maintenance is required until access to inlet/outlet/overflow chambers with a minimum width of 4 meters.
- Minimum turning radius shall be arrived based on the vehicle swept path analysis and simulations.
- Bench cutting and filling shall be provided.
- Loose/weathered rocks /stones may fall from the uphill, therefore it is recommended to protect the cut sloped sides with rich cement mortar grouting (with mesh reinforcement) as per the geotechnical investigation recommendations. In case of earth filling, the embankment portions shall be protected with geotextile membrane or riprap.
- Desirable maximum grade – 12%
- Preferred cross fall – 3%
- The access road shall be designed in one of the following materials:
    - Flexible pavement with asphalt seal
    - Concrete pavement

As part of the designing/planning phase, necessary approvals should be obtained for the access road from the following respective regional authorities (Table 5.1) or any other relevant authorities, which may be deemed required for the project, with a detailed drawing showing the followings:

- Project location
- Access road layouts including cutting/filling
- Cross sections at different chainage (shows bench cutting/filling)
- Longitudinal profile shows the percentage of ascending and descending slope
- Road markings and signs etc.
- Slope stability study for sites with cut and fill, to be prepared by the project designer.

Table 14 List of authorities for access road approval.

Authority	Requirements	Remarks
Ministry of Housing	Ensure availability of land for the access road	In the road/alignment layout extent of cut and fill boundary to be shown
Ministry of Transport	Ensure on ascending gradient, turning radius, junction details and signage, ascending/descending bay requirements	GI guard rail shall be provided where the difference in ground level exceeds 1.0 m
Electrical power supply provider	Ensure access road not affect any existing underground or above ground power utilities	Maintain the clear vertical distance of minimum 7 m between the permanent grade level and electrical OHL
Regional Municipality	Ensure cut/fill not affect the drainage pattern of nearby area and not affect nearby falaj system	
Telecommunication providers	Shall not affect the underground services	(All telecommunication providers)
Ministry of heritage and culture	Ensure access road boundary not affect any heritage site	
ROP	Ensure junction details and signage as per standards	

5.5.11. Site security

The protection of access to water facilities is important. Metal fences must be provided to prevent trespassing, theft and introduction of harmful products to the water from acts of domestic and non-domestic aggression.

The following elements should be installed on sites that are deemed high or medium risks:

- Secured perimeter with minimum of 2.5 meters height with barbed wire at the top,
- Security gates lighting and signing as required,
- Secured doors, secured reservoir hatches, secured chamber hatches with security locks,
- Secured ventilation grilles, secured access ladder(s),
- Secured electrical installation or other utilities,

- CCTV or other monitoring equipment deemed necessary for the site by risk analysis or advice from security advisors, ROP or NWS.

The following elements should be installed on sites with low risks:

- Secured entry point (ladder, door or hatch)
- Lighting and signage as required
- Secured perimeter could be of chain link type as it would normally be to stop animals and trespassers only.

5.5.12. Reservoirs architectural requirements

All reservoirs shall be designed for no additional loads due to external finishing unless otherwise stated for special requirements, like finishing with stone cladding or plastering. External walls shall be painted according to NWS approval, preferred thermal insulation coating.

5.5.13. Reservoirs site power supply

The designer should consider two sources for electrical power supply to the reservoir site compound. One from renewable (e.g. solar panels) power supply utilizing the available roofs and empty spaces at the reservoirs site and the second is from the domestic AC power supply. The designer should consider automatic switching between the two sources depending on the available electrical power intensity from either source. The design should include the tapping with the domestic grid for surplus generated power. Renewable power design capacity shall be according to the requirements of local distribution companies. Structural loads for the power generation ancillaries should be considered by the designer.

5.5.14. Reservoir site lighting

The designer should consider using the most effective energy saving bulbs, according to AER regulations, powered mainly by the solar energy while the domestic AC power shall be designed as a backup. This should be applied to the site external and internal lightening, reservoir and chambers internal lightening, SCADA and rotating equipment. Designer to select the right IP rating by following the IEC 60529. For site lighting general requirements refer to PAM-SPC-700.

5.5.15. Instrumentations

Water level categories are based on reservoir capacity, depth and daily demand. Furthermore, the criteria of the time required to intervene in the event of a high-high alarm must be a major consideration (travel time to site). For instrumentation general requirements, refer to PAM-SPC-800.

The standard control requirements for reservoir are as follows:

- Multiple levels of set point provision are available from Human Machine Interface (HMI) / SCADA for maintaining the level of water in the reservoirs by OPENING & CLOSING the inlet MOV.

- Standard settings are:

- LOW level set between 60% to 70%, of the effective volume, for OPENING the inlet MOV (settable, depending on Operations requirement).
- HIGH level set between 80% to 90%, of the effective volume, for CLOSING the Inlet MOV (settable, depending on Operations requirement).

Alternatively, flow control is settable for different desired levels so that MOV is regulated for controlling flow and maintaining set level.

PAM-STD-815 shows different water level categories and the required instrumentations, in each compartment, to communicate with SCADA.

5.5.16. SCADA control

SCADA and Programmable Logic Controller (PLC) shall supervise, monitor and control the reservoirs. All the data shall be collected from the different units and then transferred to SCADA-RCC according to NWS Standard Specifications. The following represents the basic minimum, which should be monitored, controlled functions in the OT (Operational Technologies) system.

- Continuous monitoring of level (data);
- Low Low level (alarm);
- Low level (alarm);
- High level (alarm);
- High High level (alarm);
- Overflow (alarm);
- Suction and discharge pressure low and high (alarms);
- Position and command of inlet and outlet valves;
- Water quality parameters (chlorination, turbidity, conductivity, pH and temperature (data + alarms) to be confirmed by Water Quality department;
- Flow inlet/outlet (data + alarm);
- Historic data;
- Fire alarm;
- Chlorine leakage (alarm).

The following signals have to be taken to PLC for monitoring & controls:

- Level Transmitter (analog signal)
- Level switch (digital signal for HIHI & LOLO)
- Electromagnetic flow meter (analog & digital for flow & totalizer)
- Pressure transmitter (analog signal)
- MOV signals for modulating, flow and totalizer
- Analyzer signals (analog and over Modbus)

6. Pumping & Booster Stations

6.1. Introduction

The purpose of this chapter is to provide a brief guideline for the design of pumping stations. It aims to provide a standardized guide containing requirements relating to the detailed design of potable water and Treated Sewage Effluent pumping stations and avoid variability in designs and keep improving overall the efficiency of new pumping stations.

The principal goals of the design are to ensure that the water pumping station is functional, reliable, fit for purpose, cost effective, able to maintain, future upgrades and complies with the requirements of NWS, shown in the Table below. In principle, pumping systems shall:

1. Fulfil the design criteria.
2. Comply with Omani regulatory guidelines & HSE requirements.
3. Minimize energy consumption by efficient operation. Operations scenarios shall be in conformity with the design and shall respect the design requirements in terms of number of duty pumps; operations diverting from the design considerations shall not be allowed.
4. Have reliable and long service life with minimal maintenance and least whole of life cost. Lifetime for non-structural mechanical installations shall be considered as 20 years.
5. Provide buildings & services, access roads, sewerage and wastewater drainage as per relevant international standards & Omani standards.
6. Provide infrastructure for operation and maintenance staff where necessary.

Table 15 Planning and design factors for water pumping stations.

Factors	Requirements
Functionality	– Efficiently deliver water from a defined extraction system to an appropriate receiving system. – Pump a range of flows from minimum to maximum of present and future demands with a number of duty, duty assisted and standby pumps. – Have minimum visual impact. – Incorporate remote monitoring, control and telemetered alarms. – Provide safe working conditions for operation and maintenance personnel. – Satisfy Civil Defense & Ambulance Authority (CDAA) requirements.
Pumping station layout	– Primary substation, HV/MV/LV power DBs and major power demand points (pumps) & ground reservoirs to be ideally located so that OPEX, CAPEX are optimized.
Design options & value Engineering	– Cost optimization by a CAPEX & OPEX comparison at NPV values.
Maintainability	– Be designed for minimal operator attendance and low maintenance. – Be easily maintained using standard maintenance practices.

Factors	Requirements
	– Incorporate features to allow for flexibility of operation. – Utilize standard components that are readily available and interchangeable.
Reliability	– Operate reliably, effectively and automatically with alarms, stop/start switches etc., i.e. normally unattended. – Have redundancy so that failure of any one item shall not cause total failure of the pump station. – Incorporate adequate security measures.
Cost optimization	– Mechanical equipment for direct and assisting for operations.
HAZOP	– HAZOP study, Risk level and mitigated risk level.

6.1.1. Basis of design report drawings

The NWS preliminary / concept design team or TOR normally develops the pre-design report for the detailed design works. The Designer shall develop the Basis of Design Reports in accordance with project specific requirements and the NWS guidelines.

6.1.2. Design specifications

The Designer shall produce Standard, General and Particular Specifications, necessary to meet the requirements of each project, and develop any required additional sections. Equipment specifications prepared by Designer avoid sole source equipment requirements and ensure competitive pricing for major equipment to be supplied by the Construction Contractor, such as pumps, motors, SCADA & instrumentation, hypochlorite generation plants, emergency power generation equipment, etc.

6.1.3. Design drawings

The Designer shall develop detailed design drawings to meet the needs of each project. The drawings must conform to the requirements of the NWS Standards. Drawings are to be compatible with GIS and asset requirements.

6.1.4. Future expansion/upgradation

The pumping station is to be designed expandable in future demands. The design shall ensure the proper operation for the current situation and flows as well as the future demands. The Designer shall ensure that adequate space is provided to accommodate the installation of future pump(s) and allied equipment (including surge suppression) for smooth expandability. The suction and discharge header pipework shall be sized and arranged to accommodate future flows without having to take the pumping station out of service when expansion is required. If it appears to be impractical or uneconomical to construct the pumping station building to house future pump(s), the Designer shall submit a comparative evaluation of cost, operability and constructability issues at appropriate design stage. This evaluation addresses alternative means of providing the desired capacity to meet future capacity requirements. Based on this analysis, NWS may direct the Designer before proceeding with final design of the pumping station relative to future expansion / upgradation.

6.1.5. Phasing of pump station design

A realistic design period shall be opted based on different indicators such as the increase in population, demand, strategic development projects and financial capabilities. Accordingly, the exercise of pump station phasing shall be carried out. For instance, the design period is 25 years; two phases can be planned. Phase -1 for the period of the first 15 years and second phase shall be started at the end of 15 years and cater to the requirements until the end of the design period.

The required design capacity (both the initial phase and future phase), including the maximum, average, and minimum flows to be pumped and shall be considered when selecting the type and size of equipment layout.

For transmission pumps, the following shall be followed:

1. The station layout should permit plans for periodic increases in demand.
2. If applicable, pumping higher flow with higher discharge pressure by increasing VFD speed by 15-20%.
3. If head and flow are expected to change significantly over the years, choose pumps with medium-sized impellers so that impellers only – not pumps- need to be changed to match the flow.

For booster pump station the following shall be followed:

- In the case of booster pumps in the distribution network, carefully consider the maximum daily and hourly flow rates or the requirements for firefighting flow at an interim period of 5 years at regular intervals.
- The normal demand shall be served with one duty pump while peak demand shall be served with two or three duty pumps.
- The expected sustained minimum flow is crucial as it shall dictate (1) the size and number of pumps, and (2) special considerations by the pump manufacturer in selecting pumps
- Frequent and extended operation at rates of flows below the minimum manufacturer's recommendation could result in serious damage.
- Limit the number of pump sizes. To reduce the inventory of spare parts, one size is best. Two sizes are acceptable in exceptional cases.

The above considerations shall be considered whilst arriving pump station size, besides a balanced approach to be made against initial cost and requirements. In some instances, instead of installing smaller impellers, the best solution may be to install small pumps initially and to replace them at some future date with larger equipment instead of adding more pumps. Note that the largest capacity cost item is the civil structure itself. In most instances, minimizing the number of pumps minimizes the capital cost of the station. If smaller pumps are used initially, the suction and discharge connections should be sized properly for the future units. Hence, whilst designing the structure, various options shall be worked out in such a way that in future, expansion can be made without affecting the pump station operation, as much as possible. Use reducers as necessary for the small original units. Full-size drivers and starting equipment may be preferable for the initial pumping equipment. In any event, the designer shall remember to allow space for future equipment and plan how these will be installed in future.

6.2. Design of Pumping Station

6.2.1. General recommendations

The architectural design should harmonize with the surrounding environment and respect the natural landscape as far as reasonably practicable.

The foundations of structures depend on the results of geotechnical investigation. Usually, the structure is of reinforced concrete, steel and masonry construction.

The building should be designed considering both the initial requirement and the potential need for space to accommodate future equipment for expansion.

The impact of noise on houses or other buildings near the station must be considered.

Windows should not be included in unattended locations or chemical storage including chlorine gas.

All equipment must be installed or built in such a way that flooding or other spillages cannot jeopardize the operation or running of such equipment. All electrical panels must also meet these requirements to prevent failure or electrocution.

Floors should slope to a sump fitted with a pump. The floor should be non-slippery to avoid accidents.

Welfare needs of operators, operations / maintenance, cleaning and security of employees should be considered in the design. In line with MD 286, the design team must carry out an assessment of all potential hazards during construction and operation phases. This must include a risk assessment detailing the future requirements for firefighting, manual handling, electrics, lone working, confined spaces, chemical control etc. NWS HSE is to be consulted at all stages of the design.

6.2.2. Pump station types

NWS operates a wide range of pumping stations of different configurations and capacities. A number of these are major pumping stations, most of which draw potable water from reservoir and pump to service reservoirs/elevated tanks. The pumping stations are located strategically within the transmission or distribution system. The pumps less than 100 kW in distribution systems may be designed for pump stations that are monitored by the operators in the control room at the nearest main pump station.

These pumps are designed to operate by motor drives of fixed speed or variable speed and shall be according to PAM-SPC- 601.

All pump stations shall contain duty, duty assisted and standby pumps. All pumps shall be sized to deliver up to the maximum design demand and shall be identical. The standby pump is programmed to operate in the event that the duty or duty assisted pump is out of service. Major pumping stations may have more than two pumps but will always incorporate a standby pump(s). Redundancy and overflows shall always be provided in all pumping stations.

This document is a guide only and sound engineering judgement must be applied at all times. It remains the designer's responsibility for all aspects of the design and the designer must justify any variation from these guidelines and duly recommended by Water Utility.

The Pumping Stations footprint shall be optimized and mainly for the small booster stations and shall be of the following types :

A- Booster Stations :

- BPS Type 1 for Vertical Pumps - skid mounted booster systems
- BPS Type 2 for Horizontal Pumps

B- Pumping Stations :

- PS Type 1 for Pumps capacity less than 100 kw
- PS Type 2 for Pumps capacity larger than 100 kw

Table 16 Pump Types and Design Parameters.

General Requirement	BPS Type 1 & 2	Pumping Station Type 1	Pumping Station Type 2
Minimum number of duty pumps	1	2	2
Minimum number of standby pumps	1	1	1
	Number of pumps depends on flow regime favoured		
Control philosophy	Pumps to duty rotate after each start/stop cycle and / or according to operating hours to ensure equal running hours		
Service rating	15 years design life		
Velocity through pipework at: Maximum flow Minimum flow	2.5 m/s 0.6 m/s standard		

General Requirement	BPS Type 1 & 2	Pumping Station Type 1	Pumping Station Type 2
Maximum velocity through valves	2.5 m/s (and according to manufacturer confirmation)		
Max speed	2,800 rpm		

Table 17 Pumps Arrangement.

General Requirement	BPS Type 1 & 2	Pumping Station Type 1	Pumping Station Type 2
Station pipework	Protective coatings internally and externally		
Suction line control	Isolation valves required.		
Delivery line control	Isolation valves and non-return valves (NRVs) required. Throttling valves are not recommended. Size and frequency of use of valve shall dictate whether valve should be motorised or manual.		
Station bypass	Provision to be considered for each installation.		
Sump pump provision	A separate sump pump is provided in the pump room for drainage.		
Access	Step irons/ladder	Step irons/ladder	Staircase
Internal concrete finishes	Protective liner or coating		
Lifting equipment	Monorail	Fixed permanently. Motorised lifting equipment may be required depending on weight, health and safety risks, and other relevant factors.	

Table 18 External Works.

General Requirement	BPS Type 1 & 2	Pumping Station Type 1	Pumping Station Type 2
Delineation of boundary	Preferably wall with pedestrian and vehicular access for O&M		
Markers	Underground services are to be appropriately signed.		

General Requirement	BPS Type 1 & 2	Pumping Station Type 1	Pumping Station Type 2
Signage	Required		
Exterior lighting	Required (switchable). Light pollution outside the boundary to be minimised.		
Internal lighting	Required for control room and other similar areas (switchable)		
Landscaping	Client's instructions to be obtained		
Watchman facilities	No facilities required	Toilet facilities required plus mess room fully equipped on larger stations	

Table 19 Ancillaries

General Requirement	BPS Type 1 & 2	Pumping Station Type 1	Pumping Station Type 2
Small power and lighting provisions	Full internal and external site lighting. All stairways and landings provided with emergency direct current (DC) lighting.		
Fire protection and detection (detectors, alarms, portable hose reel system, electrical, and protection)	Fire detection, alarm	Fire detection, alarm	Fire detection, alarm, and fire-fighting system
Earthing system	All pumping stations shall rely on earth roads. Recommendation is to use neutral as a protective multiple earthing (PME) system. All to subject approval of the power supply company.		
Standby generator	Always provide a socket for portable generators.	Always provide a socket for portable generators, but on larger stations, fixed generators shall be considered.	Permanent generator required.
Welfare facilities	To be provided, type to be agreed upon with the Client.		
Power supply company – Transformer requirements	Not applicable	Possible space requirement	
Air conditioning	Air conditioning of control panel rooms only		

General Requirement	BPS Type 1 & 2	Pumping Station Type 1	Pumping Station Type 2
Surge protection and auxiliary equipment	Always evaluated and provided if necessary		

6.2.3. Components of pump station

A pumping station should have at least the minimum but not limited to below:

- Pump hall room and pump piping arrangement;
- Pumps house dewatering and waste water pumps;
- Pipe supports, thrust blocks and any structural components required for safe operations and performance of the pumps;
- Location of surge vessel and associated equipment;
- Overhead cranes / hoist / monorails;
- Control room (optional);
- Walkways, safety railings, protections for rotating elements;
- Access platforms and ladders for O&M;
- Ventilation and dust protection for pump room;
- HV/MV/LV Electrical and switch gear room;
- Automation room;
- Battery room;
- Firefighting and industrial safety requirements;
- Office (Optional);
- Maintenance and workshop facilities (Optional);
- Storage for spare parts and equipment (Optional);
- Storage for chemical and substances (Optional);
- Welfare facilities (including PPE changing area)

Building design should be considered for minimizing reverberation sound fields, isolation of vibrations to steel works and sound levels to be less than 87 dB (A).

6.2.4. Site infrastructure

The location of a pumping station is strongly influenced by hydraulic considerations so that the pumps and pipeline operate satisfactorily under the design demand conditions. Adequate positive suction/flooded suction is preferred. A detailed study of the transmission & distribution scheme with options for favourable energy-economic efficiency shall be the procedure of determining the optimal location of storage and pump stations that can reduce the cost of electricity.

The use of intermediary pump stations integrated on distribution mains has the advantages of making the pressure uniform in large networks, avoiding the zones with exaggerated high pressure.

The choice of site is usually determined by the system requirements, land availability, availability of electrical infrastructures and aesthetic conditions, but the location should allow for a suitable layout for the incoming and outgoing water mains. Factors shown in Table 4.2 shall be considered during the site selection process. There should also be sufficient clearance from surface and subsurface obstructions to allow for construction. Sites located under electrical power lines should be avoided. Consideration should also be given to the potential likelihood of future development.

To ensure the proposed pumping station location and layout are acceptable, the site shall be approved in advance by NWS during the concept/preliminary design stage. Pump pedestal level or building floor, electrical transformers/ pad mounted substation or emergency generator are to be located above maximum flood level, with the floors being a minimum of 300 mm above the 1:50 year flood level. The Designer shall fully acquire the site info and metrological data during the preliminary/detailed design stage; properly design the surface/stormwater management considering the 1:50 ARI.

The size of a major pumping station is primarily dependent upon the ground reservoir, 33/11 kV Primary substation, MV/LV electrical room, number of pumps (present + future) and yard pipe works. In most cases, a pumping station is associated within the reservoir compound, which is located adjacent to the road or at mountain (Jabel). If the pumping station is to be located adjacent to the road, its entry and exit position should be determined in conjunction with the Ministry of Transport and ROP. As much as possible, it shall be located away or far apart from the residential area and sensitive areas like hospitals, Mosques, etc.

If the pump station is situated adjoined to Jabel, suitable protection measures are to be considered to protect the pump station from unstable boulders / loose soil that may fall from Jabel.

During the preliminary design stage, the Designer shall consult with the various regional stakeholders like MoH, Ministry of Heritage and Culture, power distribution companies, Municipality, MoT, all telecom providers, PDO, OGC, ROP, Falaj crossings, etc. This should be commended at the early stage of project as getting approvals/NOC can become prolonged that directly affects/delay the commencement of project execution unintentionally. Consideration must be given to key issues associated with the project and relevant stakeholders associated with these issues and bring to the notice of NWS as and when required.

Copies of relevant approvals from authorities/stake holders shall be included in the detailed design report, shall be annexed with the Tender Documents.

Table 20 Factors to guide in site selection.

Location Factors	Requirements
Site selection	– Must be located within the plot, ownership of which is dedicated to NWS including the easements. – Provide an all weather access road to the pumping station for routine, emergency operation and normal maintenance activities. – Power facilities should be available or able to be economically provided to the site.

Location Factors	Requirements
	– The site should have safe access and consideration should be given to construction requirements. – If it is located within the reservoir compound, the external pipe works should not affect/hurdle the future compound expansion and its pipe works.
Amenity and environment	– Provision of sufficient buffer from nearby existing built-up areas and future developments – Adequate entry and exit shall be provided with ascending and descending bay in consult with MoT and ROP
Design	– Proximity to pipeline and reservoirs – Accessibility – Site slope and soil conditions
Flooding	– Ensure that adequate storm water management drainage from the pumping station and site including access roads is designed so that flooding of the facility is avoided. – The pumping station site, electrical infrastructures, access road shall not be liable to flooding during a 1 in 50 year ARI
Supporting Systems	– In conjunction with determining the requirements for the site infra-structure, the designer shall consider the requirements for supporting systems to enable efficient and safe operation of the pumping station, as Requirements follows: o Electrical power and on site generator (if required) o HVAC systems o security o fire alarm with fire-fighting facilities o SCADA o Adequate Turning radius for maneuvering of maintenance vehicle o Access road and compound lighting

6.2.5. Access road

An access is to be provided from the closest public road to the pumping station. Where access from a public road is not available then a suitable access road shall be created. The design of the access road shall reflect the size and operating requirements for maintenance of vehicles. The road shall be an all weather-sealed road and have a suitable turning area for a 5 tons maintenance truck or as required for a larger vehicle. The minimum road width shall be as stated below with local widening as required at bends. Allowance for the parking of two 5t trucks shall be provided adjacent to the pumping station.

Minimum access road requirements:

- Minimum pavement width 5 meters
- Desirable maximum grade 12.5%
- Preferred cross fall 3%

The access road shall be designed in one of the following materials:

1. Concrete pavement
2. Compacted gravel pavement with two coat bitumen seal

The surfacing of the pumping station site and access road is site specific.

6.2.6. Access and protection

The protection of access to water facilities is important. Metal fences must be provided to prevent trespassing, theft and from acts of domestic and non-domestic aggression.

The following elements could be installed:

- Secured hatch, doors or covers to direct access points;
- Direct access door to the pumping station or technical room(s);
- Ventilation grilles;
- Access ladders.

Pumping stations are normally completely enclosed by high fences. These could be of the chain-link or iron railings type, depending on architectural requirements and risk level. Alternatively, a masonry wall could be constructed to meet architectural requirements.

The security fence is generally adjacent to the property line. Each site should have a boundary/ perimeter fence, which should comply with regulations and NWS security and HSE Policy.

6.2.7. Materials of construction

The design/supervision/project management consultant shall adopt careful materials selection based on application, design, worst environmental conditions and protection measures to minimize material degradation and frequent maintenance.

Preferably, approved products and suppliers registered with NWS shall be used.

6.2.8. Pump station building

NWS pumping station buildings shall primarily consist of RCC framed structure, RCC floor and cavity block wall construction with RCC flat roof. In some cases, the roof shall be metal roofing. The building sizing shall consider the followings:

1. Working heights for the selected pumps type;
2. Working area for pipe work and access to operation & maintenance of valves, instrumentations etc.;
3. Future expansion of pumping capacity;

4. Consider to optimize sound level less than 87 dB (A) of reverberating noise field at the entrance doors for operator-assisted pumps;

5. Appropriate layouts for equipment, flow velocities inside piping and appropriate structures for buildings must be devised in the detail design stage;

6. The length of the reservoir outlet pipe to the suction manifold of the pump station shall be with minimum possible number of bends and total pipe length.

6.2.9. Access and maintenance

All equipment should have horizontal and vertical clearances to allow for repairs and maintenance to be carried out efficiently and safely. Pumps, motors and equipment should be installed after finishing civil works.

Stepped platforms should be installed if necessary to access equipment.

If the pumping station has different floor levels, stairs and safety railings should be installed. To avoid the need to climb over pumps, motors or pipes, stairways shall be provided to provide access to all sides of pump and motor. Where required, lifting equipment specific to fitting new parts must meet all HSE requirements.

6.2.10. Overhead hoist

If temporary or portable hoisting is not practical, crane rails and electrical hoists should be installed for maintenance or repair works. These should cover all heavy equipment and allow for offloading of trucks, etc.

Hoists should be designed so that the equipment located inside can easily be taken outside for heavy maintenance or replacement. It is required to provide a parking bay for the overhead hoist for its maintenance works. Hoists must be marked and included in routine inspections by the relevant competent authority, which shall be conducted annually for the hoist and every 6 months for hoist equipment i.e. chains, straps, strop, hooks, shackles and D rings.

It is recommended that in case of constraints for space in small pump houses the design engineer shall opt for a monorail or tripod facilities for maintenance of pumps and fittings, which are of excessive weight for safety of workers and equipment.

6.2.11. Hydraulic equipment & instrumentation

A typical pumping station has at least the following equipment and that shown in Figure 4.1.Motor Operated Valves (gate valves, and butterfly valves) with electrical actuators;

- Non-return valves
- Pumps temperature and vibration sensors;
- Motors temperature and vibration sensors;

- Motors electrical protection;
  - overload relay
  - low/high voltage relay
- Pressure gauges;
- Surge vessels (bladder/air compressor type);
  - level transmitter
  - level switch
  - level gauge
  - pressure transmitter
  - pressure gauge
  - pressure switch
  - surge control system with HMI
- Anti-ram / surge tank;
- Air compressor (depends on type of surge vessel);
- Secondary or post chlorination (optional);
- Online water quality monitoring (optional);
- Sampling points (on inlet and outlet);
- Flowmeter(s);
- Pressure transmitters;
- By-pass for booster stations only.

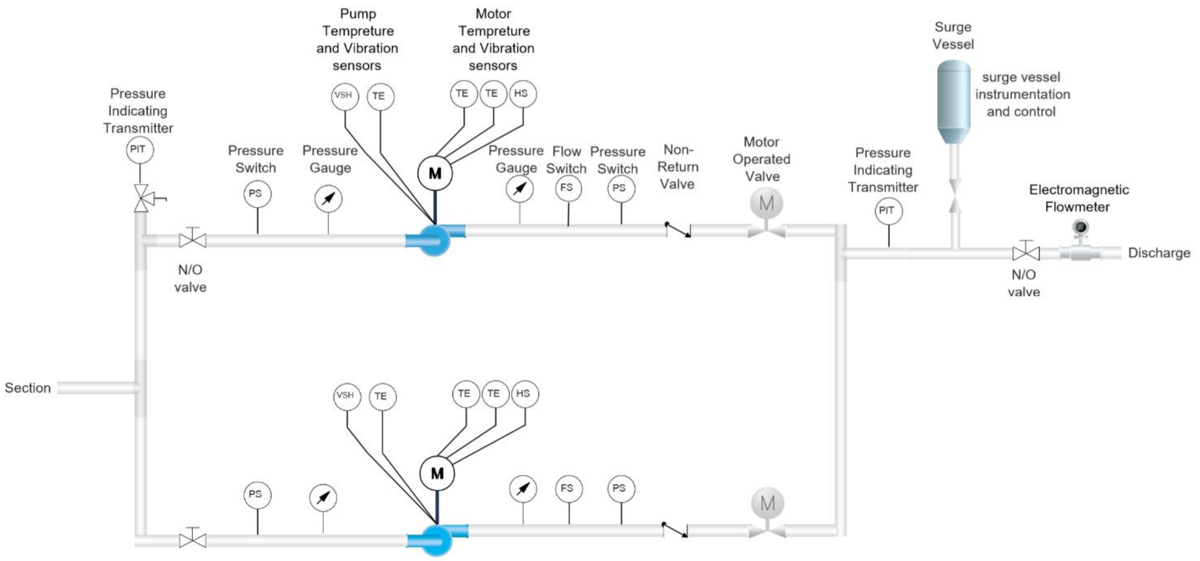

Figure 5 Schematic layout of a pumping station.

- If the pumping station is feeding water directly to the network, it is preferred to have a water network pressure transmitter be equipped with the pumping station to ease water flow and pressure control.
- If pumping station feeding reservoir(s), level transmitters & discharge flowmeter signals to be available in the pumping station.

6.2.12. Pipework arrangement and valving

All pipework within the pump station is to be tested to the structural integrity and leaks test. The test pressure is to be prescribed by the Designer and is to consider operating conditions (present/future), surge and shut off head. In addition, within the pump station building, the following pipe arrangement and valving are common:

a) Suction manifold & pipework

b) Delivery manifold & pipework

c) Miscellaneous fittings

a) Suction manifold & pipework

The suction manifold should be sized according to the maximum flow demand, including future flow demands, if any with a velocity limit of 0.9 m/s, the design shall prevent cavitations. The branches from this manifold to the individual pumps are to be similarly sized according to the range of flows for the pumps and shall be less than 1.5 m/s. Branch outlets should be at 30 to 45 degrees relative to main-line flow rather than 90 degrees.

The flow is to be directed into the suction of the pumps in a uniform manner without turbulence. Valves, tapers and changes of direction or pipe section are to be no closer than five times of pipe diameters upstream of the pump suction intake. Minimize head loss and turbulence, the use of long-radius bends in both suction and discharge piping is strongly recommended.

Inlet tapers (reducers) are to be eccentric, with the invert horizontal in order to prevent an air pocket developing. The sides of all tapers are to be straight and the taper is to be gradual with an included angle not greater than 15 degrees (i.e. butt weld tapers are not to be used on the suction).

The inlet pipe which connects directly to the pump is to be horizontal, straight and of the same internal diameter as the pump inlet and of three pipe diameters in length. Puddle flanges are to be incorporated to take thrust on suction pipework when passing through a wall.

b) **Discharge manifold & pipework**

The common delivery manifold is to be sized 1.0 m/s and individual discharge pipework of a pump is to be sized in order to allow for maximum flow velocity less than 2 m/s. The maximum flow shall consider future demand flows, if any, through the pipeline within a velocity limit of 2.0 m/s.

Tapers used on the pump delivery may be concentric or eccentric but are to be straight sided.

Delivery pipe work is to be restrained so that there is no loading back onto the pump flange. Puddle flanges are to be incorporated to take thrust on delivery pipework when passing through a wall. The outlet pipe which connects directly to the pump shall be of the same internal diameter as the pump outlet for at least three pipe diameters in length. It shall be considered below coupling where the stress travelling to pumps can be isolated. The design engineer shall perform a CFD analysis for the pipe work including supports and anchors in the design report.

c) Miscellaneous fittings

- Isolation valves are installed on suction and discharge pipes, and, by-pass pipelines;
- The discharge valve(s) should be modulating type and motorized and suction valve shall be manually operated;
- Washout, air release, air vacuum release or combination of air release and vacuum valves are to be provided at critical locations in the pumping station piping.

Tapping points for pressure monitoring should be installed at the suction and delivery sides of each pump as well as on the common header of suction and discharge with isolation ball cock valves.

a. Non-return valve (NRV)

In NWS, the most commonly used non-return valves (non-slam) are swing flap type, tilting disc, dual plate, and axial flow type etc. NRV shall be provided on the pump delivery side. Preferably, NRV shall be mounted in horizontal sections of pipe rather than vertical pipe. NRV is installed upstream of the delivery side isolation valve.

b. Dismantling joint

A dismantling joint is to be provided adjacent to at least one pump flange to allow removal and re-installation of the pump without significantly disturbing the pipework. As a minimum, the dismantling joint is to connect the straight section of inlet pipe to the inlet flange of the pump. Flange thicknesses are to be rated for the test pressure of the system.

c. Pressure gauges

Pressure gauges (digital and analog) are both required to measure the pipeline suction and delivery manifold pressures. Tapping points complete with ball type isolation valves are to be located on suitable locations.

Each pump shall be equipped with individual suction and discharge pressure gauges installed as per OEM manual.

d. Flowmeter

The flowmeter is to record all pumped flows; an electromagnetic flow meter is to be installed at the common delivery pipe only.

The flowmeter is to be located on the common transmission pipe within the pump house or inside a chamber. The flowmeter shall ideally be installed with reducers, dismantling joints and necessary couplings in such a manner no less than 5 clear pipe diameters upstream and a minimum of 3 clear pipe diameters downstream from any disturbance (for example, valve, tee, bend, change of direction etc.) in the pipework. The flowmeter is to be flanged and sized as per the manufacturer's recommendation (i.e. no tapers are to be used) so that the inside diameter of the flow meter matches that of the pipework. At least one of the pipe flanges joining the flowmeter must be a dismantling joint to the flow.

The signal converter and LCD unit are to be located above the flowmeter and readings displayed in flow rate (m^3/h) and cumulative total flow (m^3/h). Where the flowmeter is used for control purposes, it is important to specify the required flow range. The flow rate reading shall be sent to the local control room as well as to the regional SCADA control station.

Flowmeters are to be installed according to manufacturer's instructions and shall incorporate earth connections as required by the manufacturer.

e. Thrust restraint fittings

The Designer is responsible for the provision of adequate thrust restraint. Pipework, fittings and anchorages (including straps, puddle flanges and thrust blocks) must be able to sustain maximum pressures, including surge.

Thrust restraint for pipework can also be achieved by the use of thrust-restrained type dismantling joints. The use of adapter type flange is not permitted. Within the pumping station, particular care is to be taken to ensure that there is no loading onto the pump flanges caused by poor installation or pressure generated forces as this can cause distortion of the pump body and lead to premature failure of components such as mechanical shaft seals. The pump is not to be used to support the pipework. Flexible jointed pipes and fittings are to be adequately restrained. Flange thicknesses are to be rated for the test pressure of the system.

Where an existing pumping station is being upgraded, pipe pressure class, thrust restraint and possible fatigue of existing reticulation pipes due to cyclic loading must be checked.

6.2.13. Disinfection

Water quality studies are necessary to choose the type of disinfection, its installation, control storage, handling and transport requirements.

For the disinfection see **Section 12** of this document **PAM-GUD-202**.

6.2.14. HVAC system

Refer to **Section 16** of **PAM-GUD-201 General Design Guidelines.**

6.2.15. SCADA

The SCADA and Programmable Logic Controller (PLC) shall supervise, monitor and control the pumping station. For more detailed information of control and automation, refer to **PAM-SPC-800**.

All the data shall be collected from the different units and transferred to the Regional Control Room.

The following represents the basic minimum, which should be monitored in the SCADA system.

- Electrical actuated valves: open, intermediate, closed, % opening;
- Pump: start, stop, running, trip;
- Pressure alarm: low, high, normal;
- Pump temperature control;
- Pump discharge control;
- Pump operation mode (automatic/manual);
- Vibration sensors;
- Pressure control;
- Level: Low, high, normal;
- Water quality parameters;

- Energy consumption for each pump;
- Number of hours in operation (for maintenance);
- Flood alarm;
- Doors access control;
- Situation of electrical power;
- Historic.

The following indicators shall be present in fault inquiry status:

- MCC/Local;
- Auto Stop/ Auto Running;
- Auto Trip;
- Fault/ Out of service;
- Manual Stop/Manual Running.

Refer to PAM-SPC-800

6.2.16. Electrical equipment

For electrical detailed guidelines, refer to **PAM-SPC-700**.

6.2.17. Fire protection

After checking the CDAA requirements, the following fire detection/suppression systems should be installed: heat/ smoke detection in all rooms. Suitable fire extinguishing / suppression appropriate to the installed equipment and size of the site. For example, CO_2 for all electrical equipment, dry powder for generators and foam for fires on liquid fuels and oils. Alarm should be both, audible on site and relay back to the operations center.

6.2.18. Health, safety and environmental requirements

a) Emergency exit door

Emergency exit doors must be provided so that staff can evacuate the building during an emergency by the fastest safe route and all final exits must have push/paddle bar release. Emergency exit doors shall be designed to open from red zone (risk place) to green zone (safe place).

Where required, dependent on volume of staff these may need to be double doors.

b) Health and safety signage

All emergency exits, walkways, staircases, hallways must have emergency signage dictating the direction of travel to the emergency exits including pictograms on what actions to carry out on route or at the exit i.e. push bar release.

Other signage required on the site should follow best international practice and the following categories must be observed:

- Mandatory;
- Prohibition;
- Emergency and warning;
- Danger;
- Information.

All signage will be worked out on the hazard identified risk assessment and may change with further building design or alteration.

c) Firefighting equipment

Firefighting equipment can only be decided on the recommendations from a fire risk assessment in line with the completed design from the designer and further improved during the course of construction. Approval of CDAA shall be obtained. This may include but not be limited to fire extinguishers, suppression systems, fire blankets etc.

d) Confined spaces

Confined spaces are normally found where chemicals are stored or dosed (i.e. chlorine) or in an area with limited access and egress, or where there is a risk from flooding, free flowing solids or the design of the building or structure dictates.

Mandatory equipment requirements should be used in these areas and the following list should be included but is not exhaustive. This would be decided on a thorough and comprehensive confined space risk assessment.

- Gas monitors;
- Tripod and Winch;

- Harness;
- Breathing apparatus;
- PPE.

e) Welfare facilities

Welfare facilities must be included in all pumping stations for the benefit of staff and operational requirements. They should include the following as a minimum but this may change on the use of the site or risks that develop:

- Toilets and washing facilities;
- Changing areas;
- Pantry/Canteen facilities (depending on size of site);
- First aid facilities including a minimum of a 10-man first aid kit with 2 eye wash bottles;
- On all sites where chemicals or substances are to be used, a safety shower(s) must be fitted (see requirements in Other Facilities).

f) Safety shower and eyewash

Safety showers must be available in all areas where there is a high concentration of chemicals or substances with the potential to harm. This could be one or more on larger sites. The shower must be constructed of stainless steel and rise from the floor at least 2.5 meters with a shower arm at right angle to the main body "length of 0.5 meters" and a shower head 20 cm wide. It must have a foot pedal and pull handle activation that will release abundant amounts of water from a free flowing main or tank that is at normal body temperature between 16 and 26° C. The water must run for a minimum of 30 minutes. Around the base of the shower must be an interceptor that can take the residue chemical or substance away to safe storage (Remember that this storage must be emptied after emergency use). The shower must be green with white striping and must be signed above and from any substance source; it must be a maximum of 10 meters from any substance.

Attached to the safety shower fitted to the main stem at about 1.2 meters should be an eyewash, which is activated on opening allowing the injured person to place eyes in the flow of water, which must be controlled at a reasonable speed so as not to cause further damage. This should also empty to the interceptor.

6.2.19. Other facilities

- Access Doors: Doors giving access to the facilities and equipment must be equipped with an anti-intrusion system providing access only through use of a code or badge. The doors must open in an outward direction.
- Ladders: A chain must be installed across access ladders as a means of preventing further entry and as a visible sign unless authorized not to go beyond. In this instance, the chain must be replaced after the entry and exit.
- Sump covers should be installed and be of the gridded type, each grid not exceeding 30 x 30 mm.
- Trench covers: Steel or GRP trench covers for easy access for maintenance shall be considered. Whereas, chequered plates to be used elsewhere.
- Sump: The pump house drainage sump shall be equipped with float level controllers ''Pear shape type'' and a drainage pump that initiates automatically when the sump is full.
- For the security of the pumping station an independent flood alarm system shall also be installed. This system shall consist of a level controller ''Pear shape type'' positioned 20 cm above the floor which will inform the Control room.
- Communication: A telephone booth shall be provided inside the building adjacent to the pump room with a vision panel into the pump room. A flashing white light shall be installed to indicate the phone is ringing in the pump room. The operator must go into the telephone booth to operate the phone.

6.3. Hydraulic Calculations for Pump

6.3.1. Pump design parameters

The required capacity of the pumping station and its location is determined from hydraulic network analysis. The pump house building shall be as per the building standards guideline of NWS. For the selection of pump size, the flow rate or capacity, Q (m^3/h) and pump head H (mwc) are needed at the required duty point.

The capacity of a pump is the amount of water pumped per unit time. Capacity is also called discharge or flow rate (Q). In metric units, it is expressed as liters per minute (l/min) or cubic meters per second (m^3/sec).

Head is the amount of energy added to the water between the suction and discharge sides of the pump. Pump head is measured as pressure difference between the discharge and suction sides of the pump and the velocity head difference. Expressed in meters (m) of the water. Pressure and head are two different ways of expressing the same value. Usually, when the term "pressure" is used, it refers to units in kilo Pascals (kPa), whereas "head" refers to meters (m).

The selection of the appropriate pump curve depends on the specific duty point requirements and system head curve is a major task for design engineer.

6.3.2. Operating point

Pump characteristics are determined from the pump curve.

The point of intersection between the system curve and the pump curve is to be considered as the operating point during design.

There are several ways to change this operating point.

Operating point can be changed by changing the pump curve, which can be achieved by:

- Parallel operation of more pumps (more discharge);
- Serial operation of more pumps (higher head).

Another way is to change the system characteristics by:

- Altering pump speed;
- Changing pipe diameters;
- Altering impeller diameter;
- Changing pump model.

Whereas "Throttling valves" to meet pump duty point is not acceptable. It is a major concern and correct design is the responsibility of experienced design engineers.

6.3.3. NPSH and cavitation

The required net positive suction head (NPSHr) is the amount of pressure required to prevent the formation of vapour-filled cavities of water within the eye of the impeller. The formation and subsequent collapse of these vapour-filled cavities is called cavitation and is destructive to the impeller. The NPSHr to prevent cavitation is a function of pump design and is usually determined experimentally for each pump. The head within the eye of the impeller, the net positive suction head available (NPSHa), should exceed the NPSHr to avoid cavitation.

It is important to accurately differentiate the required NPSH indicated on the characteristic curve of the pump and the available NPSH that is determined from the characteristic of operations of pump suction against reservoir water levels.

A CFD analysis along with the hydraulic calculations showing safe operation against cavitation of selected pump curve at peak hour – lowest reservoir level shall be a part of the design report.

The CFD analysis for different options of pipefittings at the above scenario shall be produced to justify the best selection of fittings.

To avoid the phenomena of cavitation, it is necessary to comply with the rule:

- Available NPSH > required NPSH;
- For safety, the available NPSH must always be greater than the required NPSH of at least 1 m above required. In the worst-case operation.

6.3.4. Hydraulic transient

The detailed design report shall include a complete hydraulic transient study considering all possibilities of operations of start-up, sudden shut down, valves shut down in the piping. Transient models are mandatory and shall be done on a license version on an approved surge analysis software which shall be submitted to NWS in a soft copy. These models shall be updated after construction submitted(soft copy) to NWS. The hydraulic model shall be submitted with:

1. Calculations;
2. Pressure profiles maximum / minimum;
3. Transient properties for flow, pressure with time and along the pipe length;
4. Regions for column separation, negative pressure;
5. Selection of NRV;
6. Surge suppression system.

It shall include a separate chapter with operation philosophy for:

- Different scenarios;
- Necessary protections for safe pump operations;
- Recommended start up and shut down;
- Run on design scenarios.

6.4. Pump Design

Centrifugal pumps are most commonly used in water supply systems in NWS, which is a hydraulic machine that converts mechanical energy into potential energy. Centrifugal pumps are known as Rotodynamic pumps or dynamic pressure pumps. Detailed specifications are available in PAM-SPC-601.

6.4.1. Pump selection

Selection of the type and size of pump will be determined from the system duty requirement and specific speed for the pump. It will determine the final arrangement of the pumps, electro mechanical and other ancillary equipment within the pumping station.

A pump depends on the design requirements needed to provide a range of flows over a range of system pressures in order to satisfy system demand criteria. The main criteria when selecting a pump is operating efficiency.

The Designer is to ensure that there is a commercially available pump, which satisfies the required duty with high efficiency from at least three well-reputed suppliers.

Selection of a pump suited to the application, requiring an acceptable level of maintenance and efficiency in operation is critical to achieving minimum whole-of-life costs.

Each pump and drive unit are to be suitable for pumping water and for performing the duty throughout the specified range. The operating (flow and head) requirements are to be determined by the Designer and shall be approved by NWS.

Consideration is to be given to:

- NPSH (A), system flow and head for initial and ultimate requirements (where applicable);
- Pump efficiency within the normal duty range;
- Pump speed;
- Standby capacity, required;
- Best efficiency point to be as close as practical or within the normal operating range;
- Availability of spare parts, supported by adequate local service agent.

6.4.2. Field of use

The quality of potable water is an important factor for selection of pumps. The choice of the pump components for its materials of Construction is based on the chemical and physical properties of the water. The components of a pump are briefed in Section 6 of this guideline.

The selection or choice of the MOC for each of these components shall be:

- Corrosion and wear resistance against potable water;
- Resistance for decay and malfunctioning against physical properties of the potable water.

6.4.3. Shaft position

- Horizontal pump: The shaft of the pump is horizontal or parallel to the ground. Entire pump unit would be installed on the ground level;
- Vertical pump: The shaft of the pump is vertical or perpendicular to the ground. The motor is located above the pump head or at a higher level with an extended shaft.

Design of the shaft material shall be according to AWWA E103.

6.4.4. Number of impeller(s)

There are two types of pumps:

- Single stage pump: is composed of one volute with a single impeller;
- Multi stages pump: is composed of more than one impeller, each increasing pressure above its predecessor.

Impellers in multiple stages shall have the same diameter and trim.

All Impellers shall be hydraulically and dynamically balanced as per ISO 1940-1 latest version or as per design engineer recommendation.

5.1.1. Type of impeller

The type of impeller depends on the type of water to pump, total head and discharge. Impeller type can be open, semi open or closed.

Suction can be either single suction or double suction. A single suction impeller allows liquid to enter the center of the blades from only one direction. A double suction impeller allows liquid to enter the center of the impeller blades from both sides simultaneously.

Pumps can be classified according to the suction and delivery of the liquid with reference to the position of the access of the impeller.

1. Radial Flow–Normally a centrifugal pump is of radial type in which liquid enters at access and flows out radially. In radial flow pressure is developed wholly by centrifugal force;
2. Axial Flow–In axial flow liquid enters along the access and flows out along the access. In axial flow the pressure is developed by the propelling or lifting action of the vanes of the impeller on the liquid;
3. Mixed Flow–This is a combination of radial and axial flow in which the pressure is developed partly by centrifugal force and partly by the lift of the vanes of the impeller on the liquid.

6.4.5. Type of casing

In case of a volute pump a spiral casing is provided around the impeller. The water, which leaves the vanes, is directed to flow in the volute chamber circumferentially. The area of the volute chamber gradually increases in the direction flow. Thereby, velocity reduces and pressure increases. As the water reaches the delivery pipe, a considerable part of kinetic energy is converted into pressure energy. However, eddies are not completely avoided, therefore some loss of energy takes place due to continually increasing quantity of water through the volute chamber.

In case of a diffuser pump, a guide wheel containing a series of guide vanes or diffuser is the additional component. The diffuser blades which provide gradually enlarging passages surround the impeller periphery. They serve to augment the process of pressure built up that is normally achieved in volute casing.

6.4.6. Number of suction ports

Normally all centrifugal pumps will have a single suction entry for water inlet. In case of double suction, two impellers are set back to back. The two suction eyes together reduce the intake velocity, reducing the risk of cavitation. Mixed flow type double suction axial flow pumps are capable of developing higher heads.

6.4.7. Specific speed

Specific speed is a non-dimensional number characterizing the type of impeller in a unique and coherent manner and is determined independent of pump size, which indicates whether the suction condition of the pump is sufficient to prevent cavitation. This factor can be useful when comparing different pump designs.

The designer shall consider the required pump impeller considering the best efficiency, impeller type, etc. for the particular pump section.

6.5. Components of Pump

Pumps internal components shall be manufactured in a manner to ensure suitable and flexible assembly.

6.5.1. Motor

Pumps are to be designed to operate on AC induction motors and drives.

The power delivered to the motor shaft is known as brake horsepower (BHP).

There are three different types of motor drives as follows:

- Fixed speed: one speed, used in most cases;
- Multi-speed: generally, two choices of speed, rarely used in pumping;

- Variable speed: constantly adapting to the pumping need (direct delivery system with variable flow and constant pressure).

The speed of rotation depends on the number of pairs of poles where the common speed for a frequency of 50 Hz are:

- 3000 rpm with 2 poles
- 1500 rpm with 4 poles
- 1000 rpm with 6 poles

The following points shall be considered during the selection of the motor:

- Never oversize a pump but select with the required duty point considering limited margin;
- Selected motor shall be capable of running the pump in any point on the pump curve;
- Choose a motor with sufficient power reserve but not too oversized;
- Respect NEMA standard for allowable number of starts and minimum time between starts as per applied voltage and frequency in accordance with MG1, 12.44;
- Prefer a motor running 1500 rpm rather than 3000 rpm to maximize longevity and reliability;
- For all motors winding class should be class F with a service factor of 1.1 as a minimum;
- The ambient temperature inside pump house is generally 55° C;
- The selection of the motor is done according to the capacity of the pump. Motor selection shall be based on the impeller size, maximum capacity, specific gravity of the fluid, and service application. The selected motor shall be capable of operating the pump in any range on the selected pump curve. Motor capacity shall have run off power demand of the pump, sufficient margin min 10% over the maximum operating point. Motors shall be designed with proper overloading service factor minimum of 1.15%.

6.5.2. Impeller

Impeller is the rotating component of the pump. It is made up of a series of curved vanes. The impeller is mounted on the shaft connecting an electric motor. Impeller material can be phosphor bronze LG 1, LG 2, cast iron, stainless steel or as detailed by the design engineer.

6.5.3. Casing

Casing is an airtight chamber surrounding the impeller. The shape of the casing is designed in such a way that the kinetic energy of the impeller is gradually changed to potential energy. This is achieved by gradually increasing the area of the cross section in the direction of flow. The casing shall be pressure tested for the reference shut off head of the pump.

6.5.4. Shaft

The design Engineer may specify the pump shaft shall be Grade 431 stainless steel in accordance with AS 1444 with a ground finish over its entire length and shall have sufficient dimensions to transmit the maximum power of the motor. Alternatively, shall be confirmed to:

- Pump Shaft Stainless Steel to BS EN 10088 Designation 1.4021 /1.4401 / 1.4057
- Shaft Sleeves Stainless Steel to BS EN 10088 Designation 1.4021 / 1.4401

The design engineer shall specify high tensile steel shafts with stringent standards and finishes, protection from stress-induced corrosion, failures, etc.

6.5.5. Seals

Seals are leakage control devices. Seals are meant to control the escape of liquid from the volute along the shaft. The seals are designed to ensure that the point where the shaft passes from the inside to the outside of the pump does not leak.

They are of two types:

- Gland packing;
- Mechanical seals.

Packings are made of woven fabrics impregnated with various lubricating substances in order to lower friction. It is packed within a gland housing that needs manual adjustment to prevent leakage but still ensure lubrication by the liquid to be pumped.

Mechanical seals are another key component of pumps. They consist of two faces, one stationary on the outer side and the other rotating inner side and are located on the shaft between the impeller and the rear casing.

6.5.6. Coupling

It is a device used to connect pump and motor shafts together for the purpose of transmitting torque. The coupling is a unit, which connects the motor shaft and the pump shaft enabling the transfer of torque energy from water to pump.

The coupling must fulfil several functions such as:

- Absorbing and accommodating torque irregularities (shocks);
- Allow some misalignment between the shaft of the motor and the pump;
- Separate the critical systems (avoid the problem of resonance).

The choice of coupling requires knowledge of the following parameters:

- Nominal transmission torque;

- Stiffness, permissible misalignment, offsets;
- Dimensions;
- Atmosphere, temperature;
- Safety coefficient, nominal torque of the coupling.

6.5.7. Bearings

Bearings are intended to provide smooth friction free rotation of the shaft of motor and pump.

A wide variety of bearings is available. All bearings shall be designed for not less than 50,000 hours of operation.

Bearing must be able to:

- Support the loads between the rotor and the pump casing;
- Ensure a minimum friction with good lubrication;
- Ensure proper alignment.

Bearings consist of:

- Two rings: outer one linked to a fix element (body of the pump) and inner element related to a rotating part (shaft);
- A cage: to position and maintain the rotating balls in the bearing rings;
- Rolling body: balls or rollers ensure proper contacts between the rotating and the stationary rings.

The lubricant must fulfil the following functions:

- Reduction of friction torque;
- Reduction of the wear;
- Transfer of heat (oil lubrication);
- Reduction of operating noise;
- Protection of the bearing against corrosion;
- Quench of the bearing (grease lubrication) to prevent penetration of solid or liquid bodies;
- Compatibility with the seals;
- Tolerance or repellence to moisture.

There are two types of lubrication:

1. Oil lubrication: when speed and temperature are high;
2. Grease lubrication: When speed is restricted, (the maximum speed permitted for grease-lubricated bearings is 20% less than the maximum speed permitted in oil lubrication). When protection against corrosion is important and to provide life-lasting lubrication.

6.5.8. Suction pipe

It is the pipe connecting the pump to the reservoir outlet from where the water is flooded with positive suction. However, in unavoidable circumstances the design engineer can design detailed negative suction pumps with due care for NPSH available at the suction eye of pumps when running at peak hours with maximum demand and reservoir level is at its lowest.

6.5.9. Foot valve with strainer

The foot valve is a non-return valve, which permits the flow of the liquid from the sump towards the pump. In other words, the foot valve opens only in the upward direction. The strainer is a mesh surrounding the valve; it prevents the entry of debris and silt into the pump. The strainer shall be of not more than 20 mm maximum allowable passable passage.

6.5.10. Delivery pipe and delivery valve

The pipe connected to the pump to the outlet delivery connection and delivery valve is fitted on the discharge line to regulate the flow of liquid from the pump. The delivery valve shall be motorised butterfly (BF) valve and shall be detailed for its operations in the control philosophy by the design engineer.

6.6. Pump Performance Testing

A pump test is performed to measure various aspects of the pump's operation. The end result of a pump test is an estimate of the overall efficiency of a pump and to authenticate the manufacturer design guarantee are attained at rated parameters.

The pumps shall be designed, manufactured and tested in accordance with the last edition of one of the following appropriate standards:

- ISO
- BS EN
- DIN
- API
- Hydraulic Institute Standards (HIS)

The pumps and associated equipment shall be designed and suitable for the type of water being pumped and shall be adapted to the climatic conditions in the Sultanate of Oman.

The pumps shall be of a type as indicated in the particular requirements of the specifications.

The pumps shall be designed for continuous duty at rated parameters and should be capable of pumping the flow ranges specified in the specifications.

The pumps and associated equipment shall include all necessary provisions to prevent contamination of the water being pumped.

The Contractor/Manufacturer shall produce material test certificates for all components making up the pump in accordance with ISO 10474-3.1B certificate.

NWS reserves the right for shop inspection / stage inspection by their authorized representatives. The Manufacturer / Contractor shall provide full assistance and co-operation for such inspections. Transport conditions to safeguard all parts of the pump, motor including the bearings and seals should be clearly defined.

The Contractor / Manufacturer shall make suitable provisions for detailed shop testing of the pump and inform NWS sufficiently in advance to enable their representatives to witness the tests.

The pumps shall be tested in accordance with the ISO 9906 or relevant equivalent standard to be approved by NWS or their representative, at the Manufacturer premises to witness the pump performance as per the designed guarantee points as detailed in the specifications /contract.

6.6.1. Factory acceptance test (FAT)

a) Pump

The factory test for each variable speed pump shall include, but shall not be limited to the below:

- Factory performance tests should include at least five data points evenly spaced from minimum to maximum flow to define the shape of the pump curve. Pump manufacturers should guarantee pump performance at the flow, head, brake horsepower, and efficiency specified. Pump curves developed during the factory test should be certified to guarantee performance. Testing of large pumps to use the job motor for the testing. All submersible pumps should be tested with the job motors. NPSH as per the approved requirements;
- Test on power consumption and efficiency;
- Bearing temperature measurement;
- Noise measurement;
- Vibration measurement.

b) Motors

All motors shall be given a standard commercial test as defined by IEEE and the all tests shall confirm the followings:

- Temperature
- Efficiency
- Torque
- No load, full voltage vibration level
- Noise test
- Power consumption
- Winding resistance
- High voltage
- Locked rotor test
- Direction of rotation
- Insulation resistance

All listed tests shall be detailed and procedures shall be included for every test along with the acceptable tolerances according to international standards.

All instruments used during shop tests at Manufacturer's premises shall be duly calibrated by recognized laboratories and the calibration certificates shall be inspected and certified by NWS's representatives prior to starting the shop tests. The equipment for testing pump performance corresponds to the most common pump test standards ISO, HIS and DIN.

The calibration certificates should not be more than six months old. (required to be calibrated periodically).

6.6.2. Site acceptance test (SAT)

After installation and before commissioning, the Contractor shall carry out a site test and to be witnessed by NWS, their representative (Consultant) and manufacturer's representative.

The consultant shall detail required SAT for pre- commissioning and commissioning. It shall cover whatever required tests to perform for the best commissioning. It should be incorporated by the design engineer in his detailed design submissions. Some of the required tests are given below:

- Proper Installation and alignment, levelling of pump / motor and coupling;
- Head / flow characteristics of the pump with speed;

- Pump designed duty points with speeds;
- Vibration measurements;
- Bearing & motor winding temperature measurements;
- Test on relay settings, system earth and power redundancy;
- Test on fire detection system;
- Test on SCADA, PLC and RTUs;
- Noise measurements;
- Hipot, megger and earth testing for power cables, motor etc.

7. Transmission Pipelines

7.1. General

7.1.1. Definition for Transmission Pipeline

A water pipeline that interconnects source (desalination plant) and storage reservoir(s); storage reservoir(s) and break pressure tank(s) normally without direct consumer connections.

7.1.2. Sizing

The design of a pipeline is to be validated systematically by a hydraulic study. The design will be done taking into account:

- Design criteria;
- Pipeline material.

The following factors should be considered when sizing a pipeline:

- Minimum and maximum pressure;
- Allowable head losses based on the type of network, transmission or distribution as specified in Section 7.1.3.2and Section 9.1.1.3 of this document.
- The designed capacity (present and projected demand);
- Firefighting requirements;
- Optimum diameter to reduce pipe laying costs;
- Risk of stagnation due to low velocity or water hammer due to high velocity;
- Impact on the upstream and downstream network;
- CAPEX vs OPEX considerations.

Appropriate consideration must be given to the pressure ratings of pipe materials. However, all hydraulic calculations carried out during design shall be based upon nominal internal diameters.

7.1.3. Design Criteria

7.1.3.1. Velocity Criteria

The peak velocity as high as 2.0 m/s, at a horizon of 25 years for transmission, is possible but is seldom considered due to the high stresses this can cause and high pressure loss that can be generated over short distances. A peak velocity of 1.5 m/s is more common but will depend on pipe diameter (more acceptable for large pipes than small diameters). However, a safety factor shall be considered for future expansion of the network.

- $1.0 \leq v < 2.0$ m/s

7.1.3.2. Head Loss Equation

When calculating the head losses in for a water or TSE system the following equations are used:

- Darcy - Weisbach (larger pipes and high flow velocities) or
- Hazen - Williams (smaller pipe diameters)

The use of any other head loss equations must be approved by NWS' Hydraulic Team.

The Designer/Contractor shall make reasonable assumptions of head losses when developing hydraulic models. The table below presents typical values for both Darcy - Weisbach and Hazen - Williams.

Table 21 Typical head loss coefficient for Darcy - Weisbach (D-W) & Hazen - Williams (H-W)

Material	Pipe Age t=0 yr		Pipe Age t=5 yrs		Pipe Age t=10 yrs		Pipe Age t=15 yrs		Pipe Age t=20 yrs	
	H-W C-value	D-W ε (mm)	H-W C-value	D-W ε (mm)	H-W C-value	D-W ε (mm)	H-W C-value	D-W ε (mm)	H-W C-value	D-W ε (mm)
Ductile Iron	140	0.26	135	0.30	130	0.35	125	0.40	120	0.45
Unlined Cast Iron	130	0.26	120	0.35	110	0.45	100	0.55	90	0.65
Steel	140	0.15	135	0.18	130	0.21	125	0.24	120	0.28
Galvanised Steel	120	0.15	110	0.20	100	0.25	90	0.30	80	0.35
PVC / uPVC	150	0.0015	150	0.0015	150	0.0015	150	0.0015	150	0.0015
HDPE	150	0.007	150	0.007	150	0.007	150	0.007	150	0.007
Concrete (lined)	140	0.10	135	0.35	130	0.40	125	0.45	120	0.50
Asbestos Cement	140	0.10	135	0.12	130	0.14	125	0.16	120	0.18
GRP	150	0.005	150	0.005	150	0.005	150	0.005	150	0.005

Note: For treated effluent (TE) pipelines, the Darcy–Weisbach roughness coefficient (ε) values shall be increased by 30%, and Hazen–Williams C-values shall be reduced by 10% from the tabulated values.

7.1.3.3. Linear Head Loss (hf)

Limiting head losses with the potable transmission and distribution system is crucial for maintaining efficiency, reducing operational costs and ensuring reliable water supply. The maximum allowable head loss should not exceed 5 m/km.

Transmission networks: $hf < 5.0\ m/km$

7.1.3.4. Pressures

The pressure within a pipeline is determined by the pressure at the start of the pipeline and at the terminal point. Pressure at various points along the pipeline is shown on the hydraulic gradient that must be prepared for all transmission pipelines during preliminary design at the latest.

The hydraulic gradient can be used to determine the pressure rating of the pipe to be used.

It is important to ensure that:

1. Pipeline profile should not go above the gradient line at any point;
2. The pipeline does not empty into a receiving reservoir due to the hydraulic conditions at the end of the pipeline, e.g. a motorized valve or altitude valve should be placed at each end of the pipeline;
3. At no place must the pressure within the pipeline exceed the manufacturer's pressure rating for the pipe and the maximum strength of the material shall not be exceeded based on the total load (internally and externally due to surcharge).

To avoid the above happening, a break pressure tank may be required, or an elaborate control system installed at the terminal reservoir.

If branches are to be taken off the main under design, the adequacy of the pressure at the take-off must be determined and, if necessary, a booster pump(s) or a pressure reducing valve installed.

Normally, the aim of the design is to transfer a desired flow down the main whilst achieving a minimum pressure at the far end. This minimum pressure could be the top water level of a service reservoir or water tower. The pressure could be a service level required to feed properties or the desired suction pressure of a transfer pump at the end of the main being designed.

7.1.4. Material Selection for Pipelines

NWS generally recommends the following for pipe selection matrix:

	Pressurized Potable Water					
Material Type	HDPE PE 100		PVC-O		DI Ductile Iron	CS Carbon Steel
Nominal Pression	PN 16	PN25	PN 16/25		All PN up to 40 Bars	All PN
OD - 110 to 450mm	✓	✓	✓	ND - 100 to 1000 mm	✓	✗
OD - 450 to 710mm	✓	✗	✓			
OD - 710 to 1000 mm	✓	✗	✗			
OD Over 1000 mm	✗	✗	✗	ND Over 1000 mm		✓
Reference Specification	PAM-SPC-205		PAM-SPC-208		PAM-SPC-201	PAM-SPC-202

	Pressurized TSE - Treated Sewage Effluent				
Material Type	PVC-O	HDPE PE 100			DI Ductile Iron
Nominal Pression	PN 16/25	PN 16	PN25		All PN up to 40 Bars
OD - 110 to 450mm	✓	✓	✓	All ND over 100 mm	✓
OD - 450 to 710mm	✓	✓	✗		
OD - 710 to 1000 mm	✗	✓	✗		
OD Over 1000 mm	✗	✗	✗		
Reference Specification	PAM-SPC-208	PAM-SPC-205			PAM-SPC-201

Table 22 Pipe Material Selection Matrix

Due to the varying factors affecting the selection of the pipeline material, a thorough assessment of these factors is necessary. These parameters include the following:

- ➔ Properties of the fluid (Hardness, pH, etc...)
- ➔ Service conditions (pressure, ambient conditions, etc...)
- ➔ Soil (bearing capacity, soil characteristics, etc...)
- ➔ Properties (ductility, corrosion resistance, friction factor, etc...)
- ➔ Cost (life, maintenance cost, etc...).

7.1.4.1. Materials Selection

Multiple technical factors impact the selection of pipe material, with the following key considerations including: the characteristics of the transported fluid, the pipe size range, pressure, and soil conditions.

1. Ductile Iron (DI)
    a. Generally used for:

i. Pressurised systems (raw and finished water).

ii. Wadi crossings (with self-restraint joints)

iii. Water treatment interconnecting pipes.

b. Advantages:

i. High strength for supporting earth loads.

ii. Resistant to internal corrosion.

iii. Durability.

c. Disadvantages:

i. Require internal & external lining.

ii. May require wrapping.

d. Remarks

i. Recommended for diameter ≥ 300 mm - smaller diameters can be used based on NWS approval.

ii. For raw and finishing water pumping station (inside pumping station):

1. DN ≤ 300 mm; working pressure up to 40 bar
2. DN ≤ 600 mm; working pressure up to 30 bar
3. DN > 600 mm; working pressure up to 25 bar.

iii. Special lining for corrosive fluids (brackish water, brine).

2. Carbon Steel (CS)

a. Generally used for:

i. Pressurised systems (raw and finished water).

ii. Air piping.

b. Advantages:

i. High strength for support of earth load.

ii. Lightweight.

iii. Adapted to location where some movement may occur.

c. Disadvantages:

i. Poor corrosion resistance

ii. Requires special corrosion protection such as special coatings and cathodic protection.

d. Remarks:

i. Recommended for large diameters or high pressure over 25 bars.

3. High Density Polyethylene (HDPE)

a. Generally used for:

i. Water treatment plants internal networks.

ii. Distribution system networks.

iii. Corrosive fluids

b. Advantages:

i. Corrosive resistance.

ii. Flexible installation.

c. Disadvantages:

i. Limited structural properties.

ii. Greater wall thickness at larger diameter pipes.

d. Remarks:

i. Applicable for water treatment plants internal networks with diameters < 300 mm.

ii. Applicable for corrosive liquid and brine discharge.

iii. Derating must be considered for temperature above 20°C, as per BS EN 12201-1.

7.1.4.2. Protective External Coating Matrix

The following table displays a comparison of the different Environmental / Soil Conditions and the appropriate external coating for DI or CS pipe (more common) to be considered for each specific situation:

Table 23 Pipeline External Coating Matrix.

	Pipe : Ductile Iron / Carbon Steel					
Environmental / Soil Conditions	**ZnAl 85-15 200 g/m² + Sealer**	**ZnAl 85-15 400 g/m² + Sealer**	**PU 0.9 mm**	**FBE 350–500 µm**	**3LPE / PEX**	**3LPP**
Above water table – non aggressive soil	Authorized	Authorized	Authorized	Authorized	Authorized	Authorized
Moderately aggressive soil	Limited	Limited	Limited	Limited	Authorized	Authorized
Aggressive soil (pH < 5.5 or > 9)	Not Authorized	Not Authorized	Not Authorized	Limited	Authorized	Authorized
Very aggressive soil / permanent water table	Not Authorized	Not Authorized	Not Authorized	Not Authorized	Authorized	Authorized
Marine soil / water (ρ < 500 Ω·cm)	Not Authorized	Not Authorized	Not Authorized	Not Authorized	Authorized	Authorized
Presence of stray currents	Not Authorized	Not Authorized	Not Authorized	Limited	Authorized	Authorized
Presence of hydrocarbons (BTEX, fuels)	Not Authorized	Not Authorized	Limited	Limited	Authorized	Authorized
Rocky / abrasive soil	Not Authorized	Not Authorized	Limited	Limited	Authorized	Authorized

External Coating Systems Description

The external coating systems included in the *Protective External Coating Matrix* are intended for corrosion protection of buried steel pipelines and shall be selected based on soil aggressivity, environmental conditions and mechanical constraints.

ZnAl 85-15 coatings (200 or 400 g/m² with sealer) provide sacrificial electrochemical protection and shall be limited to non-aggressive, well-drained soils. Their use is not suitable in aggressive soils, permanent immersion conditions or in the presence of stray currents.

Polyurethane coatings (PU 0.9 mm) act as passive barrier systems and may be applied in low-aggressivity environments only. Their performance under immersion and chemical exposure is limited and depends on the specific formulation.

Fusion Bonded Epoxy (FBE, 350–500 µm) are factory-applied epoxy barrier coatings.

Three-layer coating systems (3LPE / PEX and 3LPP) consist of an epoxy primer, an adhesive layer and a polyethylene or polypropylene outer layer. These systems provide high mechanical and chemical resistance and shall be used for aggressive or water-saturated soils.

Given that the ductile iron pipes as well as steel ones can present a poor corrosion resistance according to the water quality transported, it is advisable to use specific internal coatings:

1. **Standard coatings**: suitable for the vast majority of raw and potable waters in compliance with the regulations of the World Health Organization, such as cement mortar lining or epoxy;
2. Reinforced protections: for waters aggressive to ordinary cement (soft and acidic waters, highly abrasive waters...): Polyurethane coatings (for very soft water combined with long dwell times (more than 3 days) when a standard cement mortar coating is unsuitable;
3. Special coatings adapted to very special conditions and water corrosivity: Cement Aluminium can be used for transportation of very demineralized water. This coating is however not suitable for the transport of drinking water and requires monitoring of the released aluminum in the water at the water treatment plant.

7.1.5. Pipe alignment and installation

The location of the pipeline and associated equipment must ensure that:

- They are adequately protected (against outside attacks such as traffic, ground contamination by chemical spillage, physical impact, weather conditions);
- The pipe shall not be exposed to ultra violet light during transport and storage where this would impair its integrity and projected operating life;
- Away from all existing and proposed utility infrastructures specially wastewater;
- accessibility for the safe operation and maintenance of equipment;
- accessibility for later maintenance and extension works without adverse impact on other networks belonging to NWS or others;
- minimum impact on the operation and integrity of the existing network during construction work;
- In case of future pipeline failure, the risk of flooding and third-party damage is minimized.
- In case the slope is greater than 15%, extra protection shall be provided against erosion and water flood.
- Inlet and outlet pipeline to reservoirs located uphill shall be considered with extra protection to withstand soil erosion during rains.

Based on the direction of flow, pipeline profiles with gradual ascents and rapid descents facilitate air collection at high spots, whilst preventing any air entrainment and the ready evacuation of the pipe for maintenance. And, as well, at high points after pumps.

Washouts provide for the discharge of water from a pipeline to facilitate emergency repair / maintenance works, periodical drain to clear deposits and stagnation.

These appurtenances must be sized to allow for the rapid evacuation of air / water. Air valves are meant for both letting air in and out at the time of refilling and emptying.

Consequently, the following precautions should be taken:

- Provide the pipeline with a gradient to facilitate upward movement of the air (install with a minimum gradient of 2 to 3 mm/m);
- Avoid excessive gradient changes caused by following ground contours wherever possible, particularly for large diameter pipes;
- If the profile is flat, create as many artificial high and low spots as possible, to give gradients of:
  - 2 to 3 mm/m in ascending sections;
  - 4 to 6 mm/m in descending sections.

Extreme care must be taken to ensure that air valve chambers are not flooded as this will permit contaminated groundwater to enter a pipeline, design shall follow PAM-STD-304.

Pipeline installations should incorporate:

- Appropriately sized automatic air vents at every high point, single/double/combined (AWWA Design manuals - M48 and M15);
- Adequate wash-out / drain facilities at every low point (if flooding of the washout / drain point is possible mitigating action must be taken to ensure their ready availability for use). The sizing of washout shall conform to AWWA Design manual M41. The maximum time for emptying any isolated section of pipeline shall be limited to 3 to 4 hours.
- In general, air valves must be provided at regular intervals of every 500 m - 700 m according to AWWA M51.

The position of pipe and accessories must be such that:

- Future operational activities can be performed safely;
- Emergency intervention is possible without special equipment;
- The assets (such as valves and hydrants) are accessible at all times.

Pipes should be installed along a pre-arranged route. The preferred location for the distribution pipeline in urban areas is under the footpath. Transmission pipeline must be carefully sited to avoid risk of damage to third party property in the event of pipe burst. If possible, pipes should not be installed under storm drains, sewers, major roads, vegetation or parking areas and minimum 5 m away from the compound premises.

Sometimes, pipelines will cross but in no case should they be installed in a way that makes future intervention impossible.

The depth of the pipeline should be chosen in such a way that the slope of the pipeline is as constant as possible. Unless it is unavoidable the cover over the crown on a pipe is to be as equal as or greater than 1.0 m. A deeper profile is justified if the environment requires it so as to avoid distortion of the pipe due to external stresses or the operating conditions are such that it may cause the pipe to be lifted upwards if there is insufficient weight on it. And to provide the required gradients referred to above.

Normally, DI pipes are able to withstand normal ground pressures. If a polyethylene pipe (PE100) is being laid, the design of the bedding and backfill is an essential element of the pipe design.

The Designer/Contractor shall clearly present longitudinal profiles and cross-sections for NWS review to verify the vertical alignment, utility conflicts, hydraulic performance and constructability of the system. These drawings explicitly document crossings, clearances and easements, alongside typical and special cross-sections at critical locations. These scope of for the longitudinal profiles includes:

- **Alignment baseline**: Show centerline geometry with stationing/chainage (e.g., 0+000) and PI/PC/PT data for curves; label tangents, radii, deflection angles, and lengths.
- **Stationing intervals**: Major ticks every 100 m, minors every 20–25 m; add callouts at grade breaks, fittings, high/low points, and crossings.
- **Survey control**: Reference control network, benchmarks, and survey date; note collection methods for longitudinal profile and cross-sections (e.g., total station, GNSS, LiDAR).
- **Terrain representation**: Plot existing ground along centerline from profile levelling and add offset spot elevations where needed to capture side slopes or constraints.
- Existing ground conditions: Continuous ground line along the alignment with annotations for roads, railways, waterways, culverts structures, and utility corridors; indicate flood levels where relevant.
- Proposed pipe: Pipe invert level, diameter (outside and nominal), pipe cover, material, pressure class, lining/coating, joint type, slopes between nodes and in some cases the allowable deflection.
- Appurtenances: High-point air valves/vacuum breakers, low-point scour/washouts, isolation valves, connections, chambers, and thrust/anchor blocks with chainage and elevations.
- Crossings and clearances: Roads, rail, utilities, and watercourses with casing pipes, encasement lengths, and minimum vertical separations per authority requirements; add notes for encroachment permits.

- **Geotechnical and corrosion**: Borehole logs projected to profile, groundwater levels, rockhead, weak/expansive layers, stray current risk zones, and specified embedment/protection systems.
- **Construction constraints**: Temporary works, by-pass pumping points, tie-in details, and sequencing notes where vertical alignment affects constructability.

The cross-sections shall present:

- Typical trench section: Pipe OD, embedment class and thickness, bedding material/spec, trench width/restraining system, backfill type and compaction, and minimum/typical cover.
- Clearances: Horizontal and vertical separations to adjacent utilities, structures, and property boundaries; show easement width and any working strip limits.
- Surface restoration: Pavement layer build-up, sidewalks, landscaping, and any special finishes required by road authorities.
- Special sections: Stream/ditch crossings, culverts, tunnels/casings, bridge attachments, elevated/above-grade segments, microtunnel/jacked casings, and crossings with surcharge loads; include protection (concrete encasement, riprap, sleeves) and monitoring provisions.
- Corrosion control details: Insulating joints, test stations, sacrificial or ICCP anodes, continuity bonds, and coating holidays repair zones.

7.1.6. Pipeline and broadband duct

The Oman Broadband Company (OBC) is determined to provide broadband services all across Oman. In order to expedite the implementation of duct and reduce the project managerial process, OBC joined hands with NWS. As per OBC requirements, pipeline shall be laid along with NWS pipeline in the same trench (including transmission and distribution). However, the requirements of OBC and its detailed layout/requirements should be discussed with NWS/OBC during the commencement of each project.

7.1.7. Identification of pipeline

7.1.7.1. Detectable warning tape

Warning tape indicates the presence of a pipe and the type of liquid being transported. It must:

- Be immediately recognizable;
- Be visible before excavation reach an area where there is a direct threat of exposing the pipeline;
- Be installed throughout the length of pipeline;
- Be detectable by pipe locating equipment;

- Have some identification / warning markings on it.

Detailed specification of warning tape is given in Standard Specifications PAM-SPC-506.

7.1.7.2. Marking

Marking and numbering of valves and hydrants should be done at an acceptable level. The marking should indicate all necessary information as mentioned in PAM-SPC-507 to ensure traceability.

7.1.7.3. Marking post

Marking posts for pipeline identification should be done at natural ground level. The location and marking should indicate all necessary information as mentioned in AM-SPC-507 to ensure traceability.

7.1.8. Pipe laying techniques

The choice of pipe laying technique is based on the following criteria:

- Pipe diameter;
- Soil types (mountain, gravel plain, tidal, etc.);
- Obstacles (highway, wadis, etc.);
- The length of underground structure requiring access;
- Traffic constraints and traffic loads;
- Proximity of other utility networks;
- Operating constraints of the network;
- Cost.

The different types of execution technique available includes:

- Standard open cut trenching: because of its simplicity and low cost;
- Narrow trenching: a variation of the open cut technique but using specially designed excavators to minimize the surface area disturbed;
- Trenchless technologies such as micro-tunneling, horizontal direct drilling, thrust boring, etc. are especially useful in urban areas; the trenchless pipeline works should conform with PAM-SPC-429.

7.1.9. Accessibility

Every pipe and its associated valves, hydrants etc. should be accessible for maintenance. In these latter circumstances, consideration must be given and documented to the ability to provide alternative supplies of water in the event of pipe failure. During the planning and design stages of a project operations staff must be consulted to obtain local knowledge with respect to this issue.

For large diameter pipes (DN ≥ 1200 mm), accessibility not only includes access to the outside of the pipeline but also the inside.

The purpose of this internal accessibility is to:

- Facilitate the diagnosis of external and internal pipe condition in the future;
- Allow the repair and maintenance of the pipe.

Pipes of diameter ≥ 1200 mm must be fitted with blanked off branch tees so that accessing into the pipe is possible for a man and/or an automated machine carrying a camera without the need to cut the pipe wall for remote inspections, maintenance, and rehabilitation. Generally, the inspection device can function even with the pipe full of water. Hence, it is highly desirable to locate the access manhole at local high points in the pipeline profile, preferably near an isolation/air valve, so that the amount of water to be evacuated would be less. Preferably, the nominal spacing is up to 2 km but should not exceed 3 km in any case. The typical arrangement of the access manhole is indicated in Figure 6.

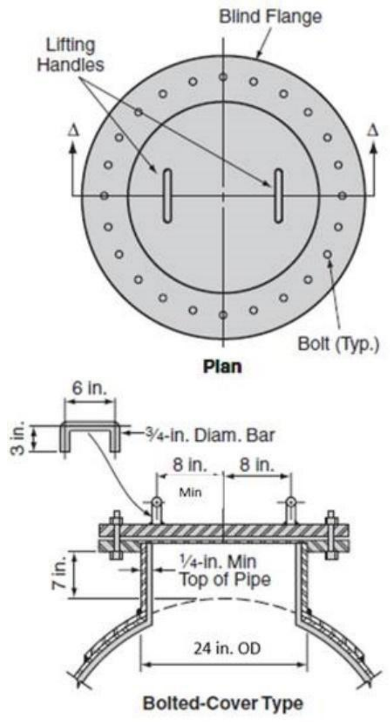

Figure 6 Transmission Accessibility Point

7.1.10. Materials

The choice of material must consider the following:

- Meteorological conditions which may lead to ground movement / exposure to direct sunlight;
- Soil condition that could cause external corrosion;
- Hydraulic capacity (internal diameter);
- Lifetime of the pipeline to ensure an adequate return on the investment;
- Maintenance and repair of the pipe material;
- Possibility of the future installation of additional branch connections, valves etc.;
- Suitability to use in potable water;
- Cost.

Pre-stressed concrete and glass fiber reinforced epoxy (GRE) pipes are acceptable alternatives. However, the value of the material shall be evaluated in a feasibility study showing life cycle cost analysis. In the past, other materials such as PVC, Asbestos Cement, Galvanized iron and MDPE were used but these are no longer in use.

In life cycle assessment, gate-to-gate concept shall be used, i.e. from procurement of the product to its end of life as an asset in NWS. This includes CAPEX, OPEX, environmental impact, mitigation of risk and salvage value (if rehabilitation or reuse worth).

Each type of pipeline has a pressure class also known as a pressure rating (normally measured in bar). This is a measure of the maximum operating pressure that a pipe can withstand without failing.

Pipelines can normally survive transient pressures in excess of this for short periods of time. These higher pressures are normally the result of pressure surge caused by sudden changes of rate of flow on the network associated with pumps switching on and off or valves suddenly opening and closing.

If operating pressures, pumping conditions, etc. are likely to cause surges of pressure in excess of the pipe rating or negative pressure; then surge protection facilities should be considered.

7.1.11. Pipeline connection

The connection between pipe elements must make a water-tight joint and, if a flexible pipe, allow for movement between the pipes:

- Ensure the durability of the installation;
- Minimize space requirement (space taken by the installation);
- Allow for future maintenance.

7.1.12. Self-anchored joints

The thrust is generated in a water supply system where pipe changes in direction, varying cross section, branch off-takes and dead ends. In general, the volume of thrust block (for acute bends and high pressure) is more and size may not fit in especially in urban areas where the space is a constraint. In this case, self-anchoring is an alternative to using concrete anchor blocks to absorb hydraulic thrust forces.

It is an attractive option when there are restrictions, notably in urban areas, when it is difficult to obtain sufficient quantities of concrete. It is also time saving when pipe laying and allows pipes to be tested immediately after being laid as there is no need to wait for concrete to set.

7.1.13. Anchor (Thrust) blocks

If self-anchoring is not used, then anchor blocks must be provided at each change of direction, horizontal and vertical, and pipe termination (bends, tees, end plugs, etc.). Also, at in-line valves, are to be made of concrete, designed to withstand the test pressure of the system and take into consideration the depth of pipe and type of soil against which the block is to thrust. Care must be taken to ensure that the soil behind a thrust has not and will not be disturbed by, for example, another service trench and any other nearby property.

The block is to be designed for submerged ground conditions, if appropriate. For large diameter pipes, thrust blocks are structural elements and should be designed as a structure. Thrust blocks can also be designed to be vertical so as to save the horizontal space for other utilities.

7.1.14. Joints

Installation of joints shall be according to AWWA M11. Joints should be flexible when rigid pipe elements are installed in unstable soil. However, when flexible pipes are used, the connection will be made by using electro fusion weld fittings (couplers, elbows and tees) or butt fusion welding.

7.1.15. Valves

Isolating valves (gate valves, butterfly valves) allow for the operation of the network, its protection and its hydraulic isolation when necessary. Gate valves with flanged joints shall be used on pipelines ≤ DN 400. For pipelines above DN 400, butterfly valves shall be used.

Specifications of all valves are given in the NWS Standard Specifications.

The frequency of in-line isolation valves should be determined for each pipeline and shall be located strategically in such a way that the evacuation of water from the pipeline between two isolation valves shall not exceed 3 hours. The maximum distance between isolation valves on the transmission pipelines is detailed in Table 24. Noting that the maximum distance is 1 km for distribution lines.

Table 24 Distance between isolation valves on transmission pipelines.

Pipe size (mm)	Max. distance "m"
150 to 500	5000

600 to 1200	3000
1300	2750
1400	2400
1500	2100
1600 to 1700	2000

Valves should be installed so that, where possible, all parts of the network can be controlled without adversely affecting another part. As a minimum requirement valves shall be installed:

- On all branch connections;
- On all branches from feeder pipeline;
- Not less than 2 valves at a tee;
- Not less than 3 valves at a cross.

The following types of valves are used in a water supply system:

- Isolating valves (Gate valves, butterfly valves);
- Non-return valves;
- Air release valves;
- Washout valves;
- Pressure regulating valves;
- Flow control valves.

Dismantling joints shall be used at each valve to facilitate the easy removal of a valve from the pipeline. This ensures consistency of equipment used on the network to facilitate maintenance.

Valves shall be operable from the surface using the means by which they were designed to be operated and equipped with a protection tube / chamber and surface box.

For DN > 400 the valve must have a smaller diameter by-pass valve. This allows ease of operation of the valve and the pipeline to be filled with minimum risk after repair or maintenance. Valves with integrated by-pass are also acceptable.

7.1.15.1. Isolating valve

Isolating valves shall:

- Be readily accessible for operations at all times;
- Be easily operable by the means they were designed to be operated (secure and safe when operating);

- Always have the same direction of closure;
- Be operated in a way that does not create a water hammer.

7.1.15.2. Manual valve

All manual valves should be clockwise operation for closing.

7.1.15.3. Motorized valve

The operation of valve using an electric actuator shall be considered in cases where:

- Operation of a valve could cause risk of injury to the operator;
- Remote control of a valve is desirable for control purposes.

If electric actuators are used, following conditions are to be considered:

- Identify if there is any flood risk to cables and switchgear;
- Provide electrical isolation switches in a position that permits their safe operation;
- Have the electrical system accessible for maintenance and verification;
- Make operation and testing simple and convenient.

7.1.15.4. Non-return valves

The non-return valves permit a fluid flow in one direction only. They are used in pumping stations in the pressure line to prevent a reverse flow in the pumps, and to ensure the correct flow of water into and out of storage reservoirs.

7.1.15.5. Air valves

Air valves are designed to perform the following:

- During filling, to remove the air present in the pipelines;
- Under working conditions to remove air that accumulates at high points;
- When a pipeline is emptied air must be admitted to break the vacuum and allow the water to be drained rapidly.

Air valves are located at high points of pipelines and in long ascending stretches of moderate slope (about 2 degrees) at intervals of 500 - 750 m. Also, where abrupt changes of slope exist, an air valve is to be installed. In distribution networks, the number and spacing of air valves can be optimized considering the density and locations of house connections, the design shall consider adequate simulations.

Air valves used in TSE systems shall comply with PAM-GUD-203 Section 8.4.1.

7.1.15.6. Washout valves

Washout valves should be installed at all low points and at the termination point of dead-ends to make water disposal easier and more convenient. If flooding of the washout / drain point is possible, mitigating action must be taken to ensure their ready availability for use. The sizing of washout valves shall follow AWWA Design Manual M41. The maximum time for emptying any isolated section of pipeline shall be limited to 3 to 4 hours.

7.1.15.7. Pressure regulating valves

Pressure reducing and pressure sustaining valves are installed following detailed modelling of anticipated pressures in the system. They are used to control the pipeline pressure downstream and upstream of the valve and facilitate the management of pressure to prevent over pressurization and/or to maintain the adequate supply pressures. For example, in distribution networks with large differences in ground level, the water pressure at low points can be higher than desired leading to higher rates of water loss and increased numbers of leakages.

7.1.15.8. Equipment for operating valves

All valves must be capable of being operated with a standard operating ''Tee'' key with standard square cap. Where this is not possible the square operating cap of a valve should be equipped with a telescopic / fixed length operating unit.

7.1.15.9. Metering

Metering is essential if the performance of transmission and distribution of the networks are to be calculated. The results of these calculations can then be used to prioritize where field investigations need to be carried out based upon the performance level of the associated networks.

Meters should be installed at the following locations in order that the performance of the networks can be calculated.

- Production sites;
- Reservoir inlet and outlet;
- Pumping station outlet;
- Tanker filling stations;
- Bulk off takes;
- District metered areas inlet.

7.1.16. Fire hydrants

Firefighting demands should be taken into consideration in the design of water distribution networks.

For each project, the location of a fire hydrant is determined in coordination with the Civil Defense and Ambulance Authority (CDAA).

After installation, each fire hydrant shall be checked and approved by the competent authority and certificate from CDAA is to be obtained by the Contractor.

Pillar type hydrants shall be installed adjacent to public roads. Underground fire hydrants shall be considered for confined spaces and specific areas subject to CDAA approval.

Fire hydrants shall be on a pipe not less than 100 mm diameter. The operating pressure of the nearby line should not go less than 1.5 bar when the fire hydrant is used at the rate of 60 m^3/h.

Specification of fire hydrants is given in the Standard Specifications AM-SPC-307.

7.1.17. Network protection

7.1.17.1. Protection against corrosion for pipes

This is a highly specialized area so the following points are only general guidelines. If corrosion protection in addition to that provided by the manufacture of the pipe / components is believed to be needed, expert assistance should be employed during the design stage.

Otherwise, all pipes and other components susceptible to corrosion shall be provided with adequate protection during manufacturing. For example, coating or resin with PPE layers. During installation great care must be taken to avoid damaging protective coatings.

7.1.17.2. Cathodic protection for steel pipes

Cathodic protection is a technique used to control corrosion of a metal surface by making it the cathode of an electrochemical cell.

A cathodic protection system shall be installed to protect and monitor the pipeline. All work related to cathodic protection systems shall be designed and supervised by a specialist. The work shall be based upon the guidelines for best practices as detailed in International Standards. A temporary cathodic protection system shall be provided during the construction period. Details are provided PAM-GUD-201 General Design Guidelines Section 17.2.

7.2. Accessories

7.2.1. Nuts, Bolts and Washers

Unless otherwise specified, nuts, bolts and washers shall conform to the requirements of International Standards. All nuts, bolts and washers shall resist corrosion and long-term deterioration. Where it is necessary due to ground conditions or the requirements of the pipeline specification, the nuts, bolts and washers shall be of stainless-steel grade SS316.

Bolts shall be of sufficient length that a minimum of three threads shall show through the nut when in the fully tightened condition.

All nuts and bolts that will be directly buried in the ground shall be adequately protected with an approved protection system applied in compliance with NWS standard specifications. Any such system must protect the nuts and bolts for a period equivalent to the design life of the pipe.

7.2.2. Gaskets and joint rings

All joints shall be made in accordance with manufacturer`s instructions.

Until required for incorporation in a joint, every joint ring, gasket, etc., shall be stored in accordance with the International Standards and manufacturer's recommendation. If there is any contradiction, the Project Engineer shall determine which is to be followed. Generally joint rings, gaskets, lubricants and jointing compounds shall be stored in a cool, dry place that is protected from adverse weather conditions.

Only lubricants and jointing compounds recommended by the manufacturer shall be used in connection with the joint ring and these lubricants shall be approved by the NWS laboratory manager as suitable for use on drinking water pipelines.

Flange gasket and joint rings should be manufactured from Ethylene Propylene Diene Monomer (EPDM) rubber conforming to International Standards and shall be a minimum of 3 mm thick.

7.3. Pipe Storage

Prior to use, pipes and fittings (valves, hydrants etc.) shall be stored in a safe, neat and tidy manner and in conformance with the manufacturer's recommendations. The open ends of all pipes and fittings shall be covered in such a way as to prevent the ingress of water, animals, insects and children.

7.4. Hydraulic Calculation

Hydraulic design of pipelines shall follow the principle described in this document, PAM-GUD-201 Appendix III that insists using a recognised hydraulic modelling software or using manufacturer`s tables or graphs. Detailed calculations are required to assess head loss through fittings, etc. within pumping stations and similar locations.

The head losses due to the flow through bends, tees, entrances, sudden enlargements, sudden contractions, valves, etc. shall be considered.

7.4.1. Hydraulic thrust

Hydraulic thrust forces occur at changes in direction, reductions in diameter (bends, tees, tapers) and at the end of pipelines carrying water under pressure. Thrust pressures can be high and, to prevent failure of the pipeline, shall be counterbalanced by appropriately anchored jointing systems, or by anchor blocks.

Various types of concrete anchor blocks can be designed, depending on the configuration of pipeline, strength and type of soil, presence, or absence, of significant amounts of ground water.

The anchor blocks shall be designed to withstand the thrust force, either by bearing on side of the trench, if the soil is sufficiently resistant, or by their own weight or can be vertically fixed in soil.

Standards for anchors and thrust blocks are shown in the Standard Drawings PAM-STD-410.

7.4.2. Water hammer (Surge)

Although some pipe materials are able to resist certain values of water hammer surges. It is usually more economic and reliable to provide the pipe system with a surge protection device.

Protective systems shall be installed to limit water hammer to an acceptable level. They shall act by slowing the change in fluid velocity or by limiting the pressure surge in relation to the pressure drop.

The designer shall determine the pressure surge and pressure drop envelope created by water hammer and judge, according to the pipe profile, the type of protection to be installed:

- Pump inertia impellor;
- Pressure relief valve;
- Air or "automatic air control" bladder (surge vessel);
- Auxiliary suction;
- A balancing column.

The anti-water hammer bladders are mostly frequently used. It has two functions:

- Limit the pressure surge (head loss controlled by a check valve);
- Prevent cavitation (balloon drainage).

In the event of a sudden pump shutdown, the pressure drop is offset by a flow rate provided by draining the balloon.

When the direction of water flow reverses, the energy in the water mass is transformed into a head loss by filling the bladder through a calibrated check valve.

Where surge vessels are used for surge protection, the pipeline pressures shall not fall below 5 m gauge at the pipe centerline level except at locations closer than 20% of the total pipeline length to a control such as a delivery reservoir or a surge feed tank where the pressures shall not fall below 2 meters gauge.

7.5. Pipe Installation

Any project should be subject to geotechnical study prior to its commencement. This study allows for difficulties to be anticipated and technical proposals to be submitted to counter them.

Pipeline route approval shall be obtained from the concerned authorities and land owners and those who rent.

A survey shall be undertaken to show the locations, dimensions and level of all services and obstructions in the pipeline route.

A resistivity survey (every 500 m) shall be undertaken along the metallic pipeline route and at all wadi crossings.

Representative soil surveys shall be carried out along the pipe route to identify contamination that may cause undue deterioration to the pipeline or contamination of the water it transports.

7.5.1. Excavation

All required authorizations from local authorities or other utilities should be obtained before starting excavation work. Based on information gathered from local authorities / utilities exploration and excavations should be made to determine the precise location and scale of obstructions and other utilities.

However, before any excavation can be carried out, the ground must be checked for the possibility of other utilities, third party intrusions or other risks, soft ground, shingle, proximity of structures, etc. This is normally undertaken at the planning stage where necessary utility plans and drawings must be obtained if this is impossible due to restrictions or lack of records the following should be carried out:

- Visual Inspection of the area to ensure that there are no drains, manholes, telegraph or overhead cable poles, ditches, trees or other obstacles that can cause problems during excavation. Move up to 100 meters away in a circular motion where practicable when checking;
- Cable location devices (cat and gen) should be used and any findings must be drawn on the ground prior to excavation;
- Trial holes should be dug on the line of excavation by hand at intervals of 5 meters or less if traversing major roads, junctions, paths etc.;
- Only when this is carried out should the ground be deemed safe to dig. Where an obstacle is found digging should be by hand only.

Bulk excavation is usually carried out by a machine. When this is not possible, then, hand excavation must be used. NWS Consultant shall decide whether and where hand excavation is required in line with international guidelines set out in New Road and Street Works Act 1991 (NRSWA). Where utilities or other buried services are known then hand digging should occur within the last 500 mm of the buried service ensuring that the service is dug around and supported by straps or strops every 2 meters in an open excavation.

All required authorizations from local authorities or other utilities should be obtained before starting the work. Based on information gathered from local authorities / utilities exploration and excavations should be made to determine the precise location and scale of obstructions and other utilities.

All the necessary precautions shall be taken not to disturb archaeological remains and historical sites in the vicinity of the works. If any important and historical things are discovered, the NWS Engineer and the department of Archeology and Antiquities of the Ministry of National Heritage and Culture shall be informed.

If the installation is under the road, and in order not to extensively damage the road surface, the trench edges shall be pre-cut using abrasive wheels. The width of this is to be sufficiently greater than the trench width to avoid disturbance to the remaining road surface during excavation.

Trenches shall be excavated to the required line and level as shown on the approved drawings.

The contractor shall deploy appropriate dewatering equipment to keep the excavations free from water from whatever source, so that the works shall be constructed in dry conditions. Only under exceptional circumstances will approval be given to proceed with water within an excavation.

In exceptional circumstances, the use of explosives will be permitted. The contractor shall obtain all necessary authorizations from the relevant authorities and make himself familiar with all current regulations of the Royal Oman Police. The explosives shall be stored in an appropriate and secure magazine. The contractor shall employ staff who are qualified and experienced in the use of explosives and who have all relevant approvals in writing from the applicable authorities, for handling explosives and blasting.

Where excavated material does not meet the requirements of NRSWA 1991, to be re-used in the excavation, the contractor shall be responsible for the disposal of surplus excavated material in an environmentally acceptable manner to an off-site location. The contractor shall present his proposals for such disposal to NWS for approval before commencing excavation works. No excavated material suitable for reuse in the works shall be removed from the site except on the direction, or with the permission, in writing of the engineer.

In all cases where there is a risk of subsidence from trench walls or other ground problems the use of trench support systems must be used.

In all cases trenches of over 1.30 meters deep and with a width equal to or less than two-thirds of the depth must, when their walls are vertical or substantially vertical be adequately supported.

7.5.2. Trench width

The trench width depends on the diameter of the pipe, the type of connection, the type of soil, the laying depth and the methods of shoring and compaction.

In all cases the trench shall be excavated with sufficient width to ensure that efficient laying and jointing of the pipes and associated pipeline components is possible. The clear width of the trench throughout its depth shall be as per PAM-STD-202. To reduce the cost of excavation, and backfill, narrow trench techniques can be used.

During the execution of the work:

- Trench walls should be stabilized either by battering or shoring (supporting);
- Edges of the excavation should be cleared of lumps of rock or clods of earth to prevent them from falling;
- Excavated material should be deposited at least the same distance away as that of the trench for example if the trench is 1.2 meters deep the excavation material should be 1.2 meters from the trench`s edge, if a trench is more than 4 meters deep and wide then the Excavated material should be moved at least 10 meters from the excavation or removed completely from site to a storage area;
- Where the excavation is known to impinge on neighbouring properties or businesses then paths shall be designated around the works and be signed and guarded in line with NRSWA (only in exceptional circumstances should the works be broken by bridging walkways) in this instance temporary bridges shall be installed to allow workers and pedestrians to cross the trench safely;
- Ladders shall be provided to allow access and egress from trenches but must be secured and protrude at least 4 rungs above the height of the trench they must not be worked off.

7.5.3. Trench depth

Pipes shall be installed at minimum 1.0 m cover from the top of the pipe to the finished ground level and shall not exceed 5 m, except with a special permission from –Water Utility. In certain cases, with the written permission of the engineer, pipes can be raised or lowered to cross obstructions. In the case of reduced cover depth, the pipe shall be encased in concrete as specified by NWS engineer. In case of deeper depths, the pipeline shall be designed to withstand the surcharge loads even by providing external protections or by modifying the characteristics of the pipeline.

Where trenches are required to be deeper than the general depth, the trench shall be excavated to the required depth with a gradual slope to permit the proper laying of the pipeline.

Additional excavations will have to be carried out to provide extra space around joints when necessary and to suit local circumstances.

The minimum horizontal distance between a water pipeline and other utilities shall be 0.5 m except sewer and Oman electrical standards. For more details see standard drawing Typical Pipe Laying PAM-STD-202.

7.5.4. Safety in excavations

At all times, the safety of those working in trenches is paramount. Appropriate protection against trench collapse must be used in cases prescribed in existing regulations, or, in general, when demanded by the nature of the soil.

7.5.5. Trench safety

a) Battering

This method is most often used in soils prone to slippage and consists of giving the trench walls an outward slope which must be close to the internal friction angle (angle of repose) of the soil.

This angle varies with the type of soil and will need to be determined on site through trials before allowing workers to enter the trench. This solution is rarely used in urban areas because of the space needed.

Battering of trench sides is an accepted method of protecting workers against the risk of trench collapse. No excavation with battered sides shall be made in roads, footpaths, private gardens, or within safe distance nearby buildings or structures.

b) Shoring

There are numerous shoring techniques. It is important to study and adopt the best system for the location before starting the work.

The most common shoring techniques are:

- prefabricated wooden panels (joined or single);
- wooden or metal sheeting;
- pile driven sheets;
- hydraulically operated trench boxes.

Whichever technique is used, the earth pressure has to be taken into consideration. The shoring must be capable of resisting the full thrust of the ground over its full height.

For the installation of shoring permanent overload, vibrations from vehicles, excavation machines and work equipment movements should be considered. The shoring must be modified to suit the nature of soil. Sufficient drainage must be provided in case of infiltration and runoff.

The trench support must be inspected prior, during and after use to ensure integrity.

7.5.6. Pipe bed

If the material in the trench bottom is considered unsuitable as a bedding material, it shall be removed and replaced to a depth of at least 0.15 m beneath the bottom of the pipe. A material such as pea gravel shall be used for pipe bedding. Exceptionally sand may be used subject to this being evenly spread across the bottom of the trench to provide full support along the full length of the pipe. Depressions shall be formed at each joint to prevent these creating stresses in the pipeline. Bedding materials must not be sourced from any location where they may have been subject to salt water infiltration or contamination with chemicals or any other substance.

In all cases care must be taken to ensure that the external wall of the pipe is not in contact with hard items such as rock, buried masonry, etc.

7.5.7. Pipe laying

All pipes should be laid according to International Standards and manufacturer's recommendations.

The trench bottom must be levelled to comply with the approved drawings. Ensuring that the pipe rests on the uniformly distributed pipe bedding material as described above. The pipes shall be laid accurately to the lines and levels shown on the drawings within a tolerance of ± 5 mm.

All pipes shall be laid and maintained to the approved alignments and grades ensuring that the pipe is properly bedded along its whole length. Pipe alignments shall be straight between bends or curves. Deviation from this principle shall only be allowed, following approval of thrust restraining proposals, where shown on drawings, to comply with detailed specifications to the contrary, or with the written agreement of the project engineer.

Pipes must be handled, stored and clad in conditions that will not allow deterioration of the pipe material to the extent that it may adversely impact on its future operational life. Pipes must be lifted and hauled using devices that will not cause any damage to the pipe itself or to its protective coating / cladding. Particular attention is to be paid to maintaining the original geometry of the pipe sockets and spigots.

When the requirements of the installation make this necessary it is allowable to make cuts of pipes. Every precaution must be taken that this is done using approved tools / equipment for the pipe material in question and is only done when necessary.

Cuts are to be made using equipment approved for use on water networks that do not contaminate the internal surfaces of the pipe and do not reduce the physical strength of the pipe. Cuts of appropriate geometry shall be made that are formed to fit with the adjacent assembly and are of the same quality as the original spigot of the pipe.

Before installation, the pipes are to be checked both internally and externally for damage to pipe coatings and linings. The interior of the pipe shall be cleared of all debris that could have been introduced and the ends of the pipe cleaned.

During installation pipes and other pipeline components shall be aligned as per the recommendations of the manufacturers and in all cases within the limits detailed in best practice manuals.

During excavation, pipe laying, backfilling and reinstatement temporary support shall be provided to all other utility infrastructure in consultation with the relevant utility managers.

Where installation of the pipe to the defined grade or alignment is obstructed by existing utilities, (such as conduits, ducts, pipes, etc.) the owners / managers of these utilities shall be consulted to determine the most effective way of resolving such obstructions. If the obstructing utilities are to be supported, removed or realigned this will be done to the specification of the owner / manager of the utility in question.

At the end of each working day, the ends of the pipes being laid are to be sealed using water tight caps to prevent ingress of water, animals, etc.

Wherever pipe coatings, internal and external are damaged these shall be repaired to the specification of the manufacturer. If this is not possible, the pipe / pipeline component is to be discarded and disposed of in a manner that prevents it inadvertently being used elsewhere.

Where manufacturers specify that deviation at pipe joints is permissible the deflection at each joint shall not exceed the manufacturer's recommendation.

7.5.8. Backfilling

When backfilling the trench, the Highway Authorities and Utilities Committee (HAUC) report (NRSWA 1991) should be used for guidance of the fill material and required strengths of ground support. All materials must meet the requirements of Class A Graded Granular Materials that should not contain any particles greater than 75 mm nominal size.

As soon as practicable, trench sections of installed pipes shall be partially backfilled to avoid pipeline flotation whilst, where possible, leaving joints open. Completed sections of the pipe shall be closed by blank flanges or cap ends that are restrained to resist test pressures.

Backfilling shall be done in two stages:

1. Backfilling with pipe surround, leaving joints open for testing procedures where possible. The pipe surround material shall be the same as that used for the pipe bed and shall provide a minimum of 150 mm protection to the at the sides and 300 mm (after compaction) above the pipe, the pipe surround shall be compacted in 150 mm deep layers to ensure uniform pipe support and protection;

2. Final backfilling of all sections still exposed after successful testing.

In cases such as where narrow trenching or trenchless installation techniques are used that preclude the provision of the pipe surround as stipulated above the contractor shall formulate proposals for the equivalent protection of the pipeline. These proposals are subject to approval by NWS before being implemented.

Appropriate backfill material that complies with the requirements of the relevant street managers shall be imported and compacted in layers not exceeding 150mm.

Road surfaces shall then be replaced using materials and methods that comply with the requirements of the street managers / land owners.

All material shall be of a consistency that permits its adequate compaction around and under the pipes and fittings and to support the trench surface without subsequent subsidence. (For further guidance see Road and Wadi crossing section below).

If the pipe is to be laid in open ground the compaction of backfilling material may not be required.

The balance of the backfill to the final ground level shall only be carried out using approved fill. Backfill material shall not be unsuitable / unwanted excavated material. This should be disposed of away from the site as described earlier.

The minimum and maximum depth of backfill depends on the pipe characteristics, laying conditions, bed zone and surround zone conditions, pipe stability and/ or protection. Its execution must meet variable demands depending on the:

- Pipe characteristics (rigid, semi-rigid, flexible);
- External loadings (depth of cover, wheel loads);
- More or less rocky or heterogeneous nature of the ground.

The main backfill zone varies according to the area involved (rural, semi-urban or urban), and should take road stability into account.

7.6. Utility & Infrastructure Crossings

Refer to **Section 9 of PAM-GUD-201** General Design Guidelines.

7.7. Valve Chambers and Surface Boxes

Valves shall be placed at the locations specified on the plans unless agreed in writing by the project engineer in consultation with NWS operations staff.

Valves shall be installed in one the following ways:

- Buried with a surface box for isolating / shut off valves;
- In a chamber of appropriate design and construction for its location.

Buried valves shall be installed with an access tube and surface box. Only those valves mentioned below shall be installed in a chamber:

- Motorized operated valves;
- Isolating valves either side of major obstructions (wadis, major roads, etc.) where facility is made to temporarily maintain supplies in the event of pipe failure under the obstruction;

- Air release valves;
- Washouts;
- Valves on deep lines (> 2.50 m) where it is difficult to maintain access for operation through surface box;
- Valves located on unstable ground, embankments etc.;
- PRVs.

For all valves, a metal surface box shall be installed that closely fits the tube / chamber and restricts the ingress of dust, debris, insects and animals. Surface boxes shall be designed to be height adjustable.

All surface boxes and chamber cover are installed to provide access to valves and shall be capable of withstanding the anticipated traffic loading for the location.

Valves in chambers shall be installed using dismantling joints unless the design of the valve allows replacement of internal parts without removing the valve from the pipeline. Dismantling joints, if fitted, must be in the chamber.

GPS coordinates (X, Y, Z) of each valve shall be shown on drawings and recorded in the GIS.

Valve chambers shall be constructed with reinforced cement concrete (RCC) or pre-casted RCC components, and are based generally on the ground.

If man entry is required into valve chamber, the chamber shall be of sufficient dimensions depending upon the size of the valve/s, waterproof, equipped with heavy duty Ductile Iron (DI) cover and frame and a sump of 300 x 300 x 300 mm dimensions.

Manhole covers shall be with Class D 400 in DI with a round frame at vehicles access areas and Class B125 (at pedestrians and non-vehicle accessible areas). The minimum opening dimensions is 600 mm. The cover shall have lifting eyes, a reliable locking system, sound proofing ring for stability and silence and optional anti-theft system.

7.8. Hydraulic Testing

Hydraulic testing is used to confirm the water tightness of pipes and fittings. A minimum of 48 hours' notice must be provided to NWS of the proposed date of a hydrostatic test. These tests shall be carried out during normal working hours unless agreed in writing by NWS.

After a new pipeline has been filled hydrostatic tests must be performed to ensure that the pipeline is properly constructed before final acceptance. The testing shall be carried out according to ISO 10802.

Before any testing is commenced, pipelines to be tested shall be filled with water and left to stand for at least 24 hours under a static pressure equivalent to the intended working pressure in that section of the pipeline. After this, the line shall be subject to pressure and leakage tests.

Testing shall be carried out in two stages:

1. Test of sections as construction proceeds (500 m maximum), leaving joints open and thrust anchoring of cap ends if required,
2. Final test of the whole system on completion.

The pressure test is applied to the pipeline by means of a continuously operating pump, equipped with a by-pass valve for regulating pressure. Attention must be given that the pipeline is filled at a rate which will not cause any surges, and which will ensure a proper air release.

The section being tested is gradually filled with water from the lowest point to aid the release of air.

All air must be released through hydrants and air valves installed for that purpose. Physical inspection of washouts and air valves must be made during pipe filling to confirm that all entrained air is released.

After checking for visible leaks, the full pressure test must be maintained for four continuous hours. The recommended test pressure is working pressure x 1.5 subject to a minimum of 10 bar at the highest point of the pipeline.

The test shall be recorded using electronic pressure data loggers capable of operating within the pressure test range.

The contractor shall design and document the testing procedure for approval by the Engineer at least 7 working days before the proposed date for pressure testing. All the material for testing shall be supplied by the contractor and submitted to the Engineer for approval.

Before takeover of a distribution network, hydraulic testing should be carried out either:

- If the working pressure is less than 7 bars, the testing pressure should be 10 bar,
- If the working pressure is more than 7 bars, the testing pressure should be working pressure x 1.5 or 1.25 x surge pressure, whichever is greater.
- The pipe shall be tested between valve chamber or into sections not exceeding 400 metres in length or otherwise approved by the Engineer.

Additionally, the system will undergo a 60-day trial before it is fully taken over by NWS.

7.9. Disinfection and Flushing

After the hydraulic test and swabbing is completed and approved the pipes shall be disinfected, flushed and sampled.

The pipe will be filled with potable water and supplemented with liquid chlorine (using a suitable pump), at a chlorine concentration of 25 ml/m³. Once full, the pipe must remain in contact for 24 hours.

After 24 hours, the pipe must be thoroughly flushed (potable water velocity of 1 to 1.5 m/s, following AWWA recommendation).

When all chlorine (disinfection) residue has been removed, the systems shall be admitted into service only once sampling results confirm correct disinfection.

Disinfecting the new pipe is unnecessary if it is not filled with water immediately after this operation.

It is important that this disinfection phase does not exceed 24 hours, as the high chlorine concentration could damage the pipe joints.

Any component used in the water system shall be fully disinfected and have the hygiene clearance from NWS.

Procedure shall be repeated if sampling results show contamination.

7.10. Design of Water Transmission Pipelines

The primary function of transmission pipelines is to transport large volumes of water between sources and distribution systems possibly over significant distances. They normally have few connections made from them.

Water transmission pipelines transport water from the source (water treatment plant, storage reservoir) to the service / storage reservoir. Their operational functionality dictates that each transmission pipeline is individually designed to meet the requirements of the storage systems associated with it in balancing it.

The flow of water in transmission systems is normally more constant than that in distribution systems in that they are used to fill reservoirs and when the reservoir is full the pumps that drive the water through the transmission system are shut down. The effect is that the water is either flowing through the transmission pipe at a consistent rate and pressure or there is no flow at all.

The design should be done by a software that implements hydraulic modelling according to CHAPTER 1 taking into account pipe material, laying, maintenance costs and energy costs (pumping).

For optimal size of pipe, the parameters of length, static pump head and quantity of water being transferred should be considered. Changes in elevation along the length of the transmission pipeline are critical in its design.

7.10.1. Valves on transmission system

a) Isolating valves

See also section 7.1.15.1

Isolating valves should be placed according to the following guidelines:

- Preferably at a uniform distance from, and adjacent to, pipe intersections;
- At intervals with a maximum interval of 3 - 5 kilometers depending on the terrain and ease of access.
- For DN ≥ 400 the valve must have a smaller diameter by-pass valve. This allows ease of operation of the valve and the pipeline to be filled with minimum risk after repair or maintenance. Valves with integrated by-pass are also acceptable. Bypass valves have the following sizes:

Valve Size	Bypass size
DN 400	DN 50
DN 500 – DN 600	DN 80
DN 800 – DN 1000	DN 100
DN 1200 – DN 1600	DN 150

Isolating valves shall be butterfly valves manufactured with double flanges.

b) Air valves

See section 7.1.15.5

c) Washout valves

See section 7.1.15.6.

d) Pressure regulating valves

See Section 7.1.15.7.

8. SCADA

The following represents the basic minimum which should be monitored/collected in the SCADA system and shall be transferred to the Regional Control Room:

1. Pressure at valves (data + alarms);
2. Position and operating of electrical actuated valves;
3. Flowmeters DMA (data + alarms);

4. Water quality analyzers for strategic points at long distances (data + alarms)
5. Historic data.

Refer to PAM-SPC-700 & PAM-SPC-800.

9. Distribution Networks

9.1. Design of Water Distribution Pipelines

The design of a distribution system is governed by following factors:

- Pipe size.
- Reservoir location.
- Minimum and maximum pressure required.
- Pressure variations.

A proper design involves minimizing investment and operating costs by optimising these factors. Due consideration should also be given to possible staged development of the network to manage expenditure profiles.

9.1.1. Distribution Pipeline Design Criteria

9.1.1.1. Velocity Design Criteria

The minimum design velocity is set at 0.4 m/s. This value represents the minimum that is required for maintaining water quality and must be validated using hydraulic and water quality models. The maximum allowable velocity within the distribution network is set at 1.5 m/s.

- $0.4 \leq v < 1.5$ m/s

In special cases, where the velocity is below 0.4 m/s water quality assurances (water age) must be presented and justified with the use of a water quality model which clearly shows the residual chlorine concentration and the water residence time. The minimum residual chlorine concentration, according to WHO Guidelines, is 0.2 mg/l.

9.1.1.2. Head Loss Equations

See section 7.1.3.2.

9.1.1.3. Linear Head Loss (hf)

Limiting head losses with the potable transmission and distribution system is crucial for maintaining efficiency, reducing operational costs and ensuring reliable water supply. The maximum allowable head loss should not exceed 3 m/km.

Distribution networks:

- hf ***< 3.0 m/km***

9.1.1.4. Pressure

In a distribution system water should be supplied with adequate pressure and flow. However, pressure is lost due to friction at the pipe wall and in pipe line components such as valves and bends. The amount of pressure loss is also dependent on the water demand from customers, pipe material, length, gradient and diameter.

To deliver sufficient quantities of water the pressure head in the network should, wherever possible, be at least 1.5 bar (15 mwc, worst point peak day, peak hour) in all parts of the network, including the remotest and highest points. The maximum pressure should not exceed 4 bar (40 mwc).

In case of firefighting flows, pressure in the network shall be maintained as minimum positive value i.e. negative pressure should not be developed in the network assuming zero parallel domestic use during fire.

9.1.2. Distribution Reservoir Locations

Ideally the reservoir should be located on a strategic point to equalise distribution flows and pressures.

The flow of water in distribution systems can vary greatly from the peak hours, when a lot of water is used for washing or drinking, to the minimum consumption hours, which are normally at night.

Service reservoirs play an important role in the relationship between transmission and distribution systems to balance the fluctuations of the water demand from customers that a treatment plant may not be able to meet.

Reservoirs should be placed at a higher elevation than the distribution area, but as close as possible to it to avoid unnecessary pumping and transmission line cost. If the area is flat, elevated tanks shall be used. Elevated tanks are generally smaller than ground reservoirs and serve to balance demand variations in localized areas.

9.1.3. Supply Methodology

The pressure needed for the distribution system can be obtained using the following systems:

- Gravity system: water service reservoirs can be elevated or situated on a hillside on such a level that sufficient pressure is generated in the distribution area. In these reservoirs, storage and pressure head functions are combined.
- Pumped system: these have a low-level ground reservoir and the required pressure is generated by pumps. Water only enters the storage vessel when the pressure in the network increases above the height of the tank. When demand for water increases water flows from the tank to supplement the volume supplied by the pump.

The steps for designing a water distribution system generally are:

- Choosing the preliminary source scheme and type of network;
- Division of the distribution area into districts;
- Estimation of the population of each district;
- Calculation of the water demand for each district at the nodal point;
- Calculation of the required pipe diameter.

Two main types of distribution networks are:

1. Branched distribution; &
2. Looped Distribution.

9.1.3.1. Branched distribution

Branched networks are predominantly used for small capacity water supplies delivering water mostly through standpipes. Their drawbacks are lower reliability (a single failure can cause an interruption of supply to all customers), increased danger of contamination, possibility of sediment accumulation in dead-ends, larger pipes are often required than in looped systems to overcome friction losses. Furthermore, a fluctuation of water demand can cause large pressure variations in the systems. Their advantages are that they are cheaper and easier to design and install and flow direction and rates can be easily determined for all pipes. Fire hydrants shall be installed at dead-ends of the pipe for the purpose of flushing.

9.1.3.2. Looped distribution

Looped distribution networks allow for the supply of customers from more than one direction making them less prone to overall failure of the supply. They have the advantage of better hydraulic performance which leads to smaller pipe sizes and the reduced possibility of water supply interruption during maintenance activities.

9.1.4. Pipe Material

NWS recommends distribution pipes:

- PE 100 up to 1000 mm diameter (OD 110, 180, 225, 355, 500, 630, 710, 900 and 1000 shall be used);
- Ductile Iron (DI) above 300 mm diameter;
- DI for special locations like road/wadi crossings.

9.1.5. Pipe size

To determine the pipe size, the design criteria for section 9.1.1.1. shall be used in conjunction with an approved licensed hydraulic modelling software, which is a requirement of NWS. The copy of this model shall be submitted to NWS.

When designing a pipeline network, it should be ensured that there is sufficient pressure at the point of supply to provide an adequate flow to all customers. If available, network models of the system should be used to check that all pipes will be operating to the required standard.

To maintain water quality, it is important to minimize transit times and avoid stagnation and low flow rates. Therefore:

- The system should not have excessive capacity unless this is required to meet a known increase in future demand;
- Ideally low-flow dead-ends and loops should be avoided. Situations that may give rise to negative pressures should always be avoided as in the event of a pipe failure contaminated water can be sucked into the pipe.
- Minimum chloride concentration, WHO Guidelines, shall be 0.2 mg/l and greater for maintaining good water quality in the networks.

The recommended diameters for the distribution system are from DN100 mm to DN 400 mm, unless otherwise specified.

9.2. Methods of supplying potable water

9.2.1. Metered house connections

A house connection distributes the water to individual consumers through a pipe connected to the distribution network as per PAM-SPC-431.

The connection to the water distribution pipeline is through a tapping saddle (electro-fusion for PE100 pipes or mechanical for others) to a valve adjacent to a water meter, generally installed on the wall of each property. The material for house connections is PE 100 SDR 11 which has a pressure rating of 16 bars.

A water meter (see water meter section below) is installed together with a strainer on the wall of each property. The meter should be installed at a sufficient height (approximately 1.5 m) to allow ease of reading and fitted with a ¼'' turn lockable ball valve installed before the strainer.
The approved meter is provided by NWS.

9.2.2. Water flowmeters

A water flowmeter is a device that records the volume of water passing through a particular point in the global water system, example: pumping station, reservoir, consumer connection.

The function of the flowmeter is:

- Accurately record volume of water passing appoint in the network.
- Accurately record with the customer's consumption.

- Provide an index as a basis for calculating the consumption billed to the customer.
- Allow the recording of flow rates being distributed.
- Assessment of network efficiency.

The general specifications required for water meter are:

- Recording accuracy even under low flow conditions.
- Resistance to available water pressures and pressures variations;
- Cause minimum pressure loss.
- Retain its operating and accuracy characteristics over time (sustainability);
- Withstands the conditions of the environment (resistance, inviolability);
- Be easily readable.
- Does not alter the quality of the water.
- When necessary to allow the calculation of rates of flow passing through the meter.

There are two main types of meters, which are:

- Electromagnetic meters;
- Ultrasonic meters.

Currently NWS follows the type of flowmeters as shown in Table 25.

Table 25 Type of flowmeters.

Flowmeter size (mm)	Technology used
Less than 40	Ultrasonic flowmeter
40 and above	Electromagnetic flowmeter

Note: In special conditions, ultrasonic flowmeter can be used at sizes of above than 40 mm.

There are also meters with automatic and remote meter reading devices. Water meters shall conform to ISO 4064 standard and OIML R 49.

Domestic meters are to be installed vertically, on the outside of the property wall without any protection installed at 1.5 m above ground level.

9.2.3. Leakage and wastage control

The main factors of difference between the total water input to the system and that is consumed are:

- Leakage in reservoir and water pipelines.

- Leakage in network pipes.
- Unduly high pressures in distribution systems intensifying leakage and bursts.

The following methods of leakage control are in use:

- **District metering**

The Designer/Contractor shall present DMA designs at the earliest stage of concept designs as this will prevent the creation of cascading DMAs and aid in optimising pipeline lengths. The main objective is to eliminate cascading DMAs on all new water systems.

Flow meters are installed at strategic points within the system comprising areas of 500 to 2500 connections.

There are different reasons for dividing networks into zones. One is to have a greater control over the distribution of water through monitoring the volume of water entering into an area. A network can be divided into District Metered Areas (DMA) for network assessment and control purposes. Other benefits from Water Quality point of view are:

- In case of contamination, it is easier to identify the source of the contamination quickly,
- Zoning reduces the extent and complexity of mixing water.

For the follow up of a network, a flow meter should be installed to measure the inlet and outlet from a strategic point of the distribution network. This enables continuous network performance assessment, which enables the identification of changes in network efficiency and thus triggering field investigation exercises.

The following factors should be considered for the installation of a district meter:

- Numbers of connections (domestic + commercial) should be between 500 and 2500.
- Minimum inlet and outlet points - One main inlet point with a secondary inlet which shall be used in emergencies,
- Handover network pressure test if taking over a private network.
- Maximum inlet diameter = 300 mm,
- Electromagnetic flowmeter + pressure linked via telemetry to access flow and pressure data remotely,
- The inlet of the DMA shall also have a pressure management system (pressure regulating valves - pressure reducing valve). This allows for lowering NRW at minimum consumption hours. This system shall be automated and linked to NWS SCADA.

- Minimum cascading network - not accepted for new system and may be allowed for only one cascade pending NWS approval.
- Number of boundary valves < 5.
- Water quality analyser to be located based on the hydraulic model, water quality analysis, to give the overall water quality within the DMA. The water quality parameter to be monitored includes free chlorine/total chlorine, turbidity, pH, conductivity, temperature, dissolved oxygen, oxidation-reduction potential (ORP) and shall be based on NWS requirements. These analysers shall be connected to NWS' SCADA system providing real-time information.
- The inclusion of several pressures, including the critical pressure point, for the monitoring of pressure across each DMA. This shall be linked to NWS' SCADA system.
- The change in elevation across a DMA shall not be more than 30 metres, to allow for stable pressures within the DMA.

- **Field Investigations**

These are normally triggered by the network efficiency decreasing below the target level. This decrease could be caused by opening of a boundary valve, new connections, leaks or burst.

- Passive System Detection

This uses the public and meter readers to report leaks, illegal connections, leaking fire hydrants etc.

- Pressure control

Reducing the system pressures to the minimum level reduces the force behind leaks and extends the working life of the network due to the reduced stress placed on the network. Ways to accomplish pressure reduction are reducing pumping heads and the use of pressure reducing valves.

In order to monitor the pressure in the network, pressure monitoring points should be located at the following points within a DMA:

- Inlet point of the DMA, shall include valves, flow meter, strainer, pressure regulating valves, air valve and bypass system as shown in PAM-STD-804 and PAM-STD-311;
- Internal pressure points at the average point, located according to the area's geographical and network's pipeline hydraulic analysis.

Specification of pressure transducer is given in the Standard Specifications.

- Water quality monitoring point

In order to assess and monitor the water quality, a sampling point and online water quality analysers shall be installed at the inlet of DMA. The sampling point and water quality analysers shall be installed as per the standards.

9.3. House Connection Header

Table 26 presents the corresponding heading size based on the number of house connections for the distribution system.

Table 26 Header size per No. of connections

No. of House Connection	Header Diameter (mm)
≤ 3	32
4 – 11	63
12 - 16	110

10. Surge Analysis & Protection

Refer to Appendix III of PAM-GUD-201 General Design Guidelines

11. Service Connections

11.1. Purpose

This document sets out procedures for the design and installation of service connections to all types of premises. Its objective is to aid NWS staff to maintain a good standard of service connection installations and to assist builders, developers, engineers, and architects in adopting approved service layouts. Aspects of service connection installations including tapping from service main are included. The guideline presented in this document forms part of the requirements for water services connection and installation which is in compliance with NWS' *Specification for Service Pipes (2024)*. The purpose is to prevent misuse, waste, undue consumption or erroneous measurement of water and most importantly to prevent leakage of drinking water.

The following should be followed, assisting with the standardisation of water service connection procedures.

Materials and fittings used should comply with the relevant NWS' *Specification for Service Pipes* and/or British standard where no NWS standard exists.

11.2. Scope and Application

This document shall apply to:

I. A new water connection.
II. Rearranging an existing connection, including a meter installation, or for disconnection from NWS' system.
III. The person, company or organization, including water fittings manufacturers or suppliers/agents, engaged in any activities relating to the supply, erection, maintenance and construction of water service connections.

This document covers that part of the water supply installation between a NWS system and a customer's installation, which generally consists of a service connection, water fittings, meter, storage tanks, and single/multi metered Buildings. The scope does not include the water distribution system belonging to NWS, except for the fittings that are required at the interface with Customers.

11.3. Definitions

All words and expressions shall have the meanings assigned to them at the beginning of this integrated document.

11.4. Existing Arrangements

Any connection arrangement installed prior to the date of issue of this document shall not be subjected to this Guide unless that arrangement contravenes material of construction, water leakage or contamination requirements as given in the Regulations. A prior consent shall be granted by a NWS before any additions or modifications to an existing connection arrangement are carried out.

11.5. Quality of Installations

NWS has an obligation to ensure that hygiene procedures related to the installation of water fittings are followed during installation work, therefore request that individuals involved in such installation work undergo a competency test prepared in accordance with its practices and standards in order to ensure the quality of this work.

No person shall install a water fitting to convey or receive water supplied by the NWS, or alter, disconnect. Only NWS approved fitting shall be used from ferrule connections until water storage tank(s).

No person shall be allowed to install or permit to be connected a water fitting in a manner that causes or permit to be connected or used, a water fitting which is damaged, worn or otherwise faulty so that it; causes or is likely to cause, leakage, undue consumption, misuse or contamination of water supplied by the NWS.

The applicant must ensure water fittings and storage tanks are inspected and maintained according to the procedures set out in this Guide and the Code of Practice for the Inspection and Cleaning of Customer Water Storage Tanks.

If during an inspection by NWS, it is found that a storage tank, including its water fittings, does not comply with the Regulations, it shall be considered non-compliant, and an 'Improvement Notice' shall be issued to the applicant. Failure to comply with such an Improvement Notice, or with any follow-up inspection report or instructions from NWS, shall result in the applicant being issued with a final notice, upon which the water supplied to the Premises shall be disconnected. Following compliance by the applicant, and upon further inspection and approval by the NWS, the drinking-water supply shall be reconnected to the Premises.

11.6. Service Pipe Location and Requirements

The followings to be adopted whilst laying service pipe connection:

- Where possible, lay underground service pipes at right angles to the main and in approximately straight lines to facilitate locations for repairs.
- Locate service within legal access to premises – avoid crossing land owned by others private/public.
- Underground piping shall have a minimum cover as follows
- 750 mm under roads;
- 625 mm inside premises under paver block/roads

- 300 mm inside the building
- Service pipes must not run beneath the wall or foundations.
- In designing and planning the layout of pipework, attention should be given to the maximum rate or discharge required, the economy of materials, accessibility, protection against damage and corrosion, and avoidance of airlock.
- Changes in diameter and in direction should preferably be gradual rather than abrupt to avoid undue loss of head.
- Where piping has to be laid in any ground/wall liable to subsidence then special consideration should be given to the type of pipes and joints to be used in order to minimize the risk of damage due to settlement. Where piping has to be laid across recently disturbed ground, continuous longitudinal support should be provided and not merely supporting piers at intervals.
- Ground tank inlet level at a height of one meter above the road level and not more than 30 m distance (pipe length) from the NWS meter.
- Service pipe from water meter or ground/underground tank/sump shall be taken to roof tank(s) through ducts and it shall be easily accessible for repair and replacement of appurtenances.
- For single house connection, refer to the latest standard detail drawing "PAM-STD-205" titled "Typical house connection". For multiple connections refer to drawing "PAM-STD-205", noted that the multiple connection drawing is applicable for grouping of water meters installed either at ground level or terrace level.
- All pipes and fittings installed for the potable water use shall be approved by the international hygienic institutes like WRAS, NSF or Kiwa.

11.7. Ground Tank Storage – General Requirements

11.7.1. Tank placement and location

- The tank shall be located away from any source of pollution, particularly if it is buried in the ground, and properly protected from ingress of foreign objects.
- The distance between the tank wall and any part of the wastewater drainage system shall not be less than 3000 mm, in case unavailability of the distance then the wastewater drainage system/pipe shall be protected in such a way that the pollution dispersion would not affect water quality.
- Tanks in basements or contained in other below-ground infrastructure must be located in such a way that they are accessible for inspection, maintenance or replacement without the need to disturb the associated infrastructure.

- The storage tank location must be positioned to avoid potential flooding.
- Storage tank access and inspection openings shall be at least 300mm above ground level or the highest known flood level.
- Storage tanks used for industrial, livestock, agricultural and other purposes that may come into contact with any other fluid or foreign material should be dedicated for that use only and provided with suitable backflow prevention devices, overflow arrangements and an air gap of not less than 50mm.

11.7.2. Fittings and accessories of the storage tank

- The maximum height of the inlet to the Premises' Ground Storage Tanks shall not exceed 3000 mm from road level and or tank base shall not be more than 1000 mm from road level.
- Float-controlled valves or equivalent flow control inlet devices should be securely and rigidly attached to the cistern or the tank and installed so that the valve closes when the level of the water is not less than 25 mm and preferably not more than 50 mm below the overflow level of the tank or roof tank.
- All inlets to storage tanks and roof tanks should be provided with a Servicing Valve to facilitate maintenance and a float-operated valve, or some other no less effective device, which is capable of controlling the flow of water into the tanks. The Servicing Valve should be fitted as close as is reasonably practicable to the float-controlled valve or another device.
- An alarm system, which is either audible or visual, shall be fixed in all underground/aboveground and overhead tanks of capacity more than 10 m^3.
- An overflow water pipe should not be connected to the drainage, it must be in a visible location, where the discharge of water can be seen or noticed vividly.
- All water pipelines (back and forth from the ground tank) should be open or installed in a sleeve (pipe inside a pipe) to facilitate easy repairing or replacement in case of any water leaks or damage of pipes.

11.8. One Connection Per Premise

The followings to be adopted:

- There shall be one service pipe connection and meter for one consumer. If an existing service pipe becomes inadequate to provide the required flow, it shall be closed and abandoned and replaced by a new service.
- For large consumers such as Malls, Mass housings/hospitals, etc., where several connections already exist, NWS shall negotiate with them to have one connection only. Distribution within the property shall be the responsibility of the consumer. Except for one connection, all existing connections shall be removed.

- Each consumer shall have a separate service pipe, where building subdivided into more than one units, the property owner/developer shall install all necessary piping to allow separate metering of each unit according to NWS requirements.

11.9. Water Meter Provision

- All potable water outlets supplying water to Customer Premises and connected directly or indirectly to the water main, or sub-main of the NWS network, shall be metered in a manner approved by the NWS using an approved measuring device.
- Any building, part of building or Premises divided into isolated flats (separate occupation) shall be separately metered.
- Open commercial spaces with one or more self-contained units located separately shall also be metered separately.
- Service or utility water requirements: air conditioning, housekeeping, garbage room, public toilets, filtration system, and general services must all be metered by one or more meters as approved by the NWS.
- Swimming pools (both above and below ground level) larger than 25 cubic meters with a fixed water supply connection shall be metered separately. Similarly, separate meters shall be installed on fixed water supply connections for any non-domestic use e.g. for cooling systems. The size and type of meter shall be appropriate for the individual Customer's overall consumption pattern and maximum flow rates.
- Access to metering installations shall be made available to authorized officers of the NWS for the purpose of meter reading, installation of a remote reading device, maintenance, etc.

11.10. Cross Connection Control

Adopt the following guidelines:

- There shall be no interconnection or cross-connection between any pipe or fitting containing NWS supplied water and a pipe or fitting containing water from any other source. The provision of backflow preventer or closed and sealed stop-valves is not a permissible substitute for the complete absence of connection.
- The design of pipework shall allow no possibility of backflow towards the source of supply from any tank, whether by back-siphonage or otherwise. Valves cannot be relied on to prevent such backflow.
- Where a supply of NWS water is required as an alternative or standby to a supply of water from another source or is required to be mixed with the latter, it shall be delivered into a spate tank by a pipe of fitting discharging into the air at a height above the top edge of the tank equal to twice its nominal bore, and in no case less than 150 mm.

- All pipework shall be so designed, laid, fixed and maintained to be completely watertight, thereby avoiding waste of water, damage to property and the risk of contamination of the water conveyed.
- No piping shall be laid in or through any sewer or drain or any manhole connected therewith, nor in a ground contaminated by sewage. Farmyards, animal pens, and cesspools should be avoided.

11.11. Illegal Tapings

Illegal tapings such as direct tapping before water meter, installing direct pump before and after the water meter, using water before the tank for irrigation and car wash.

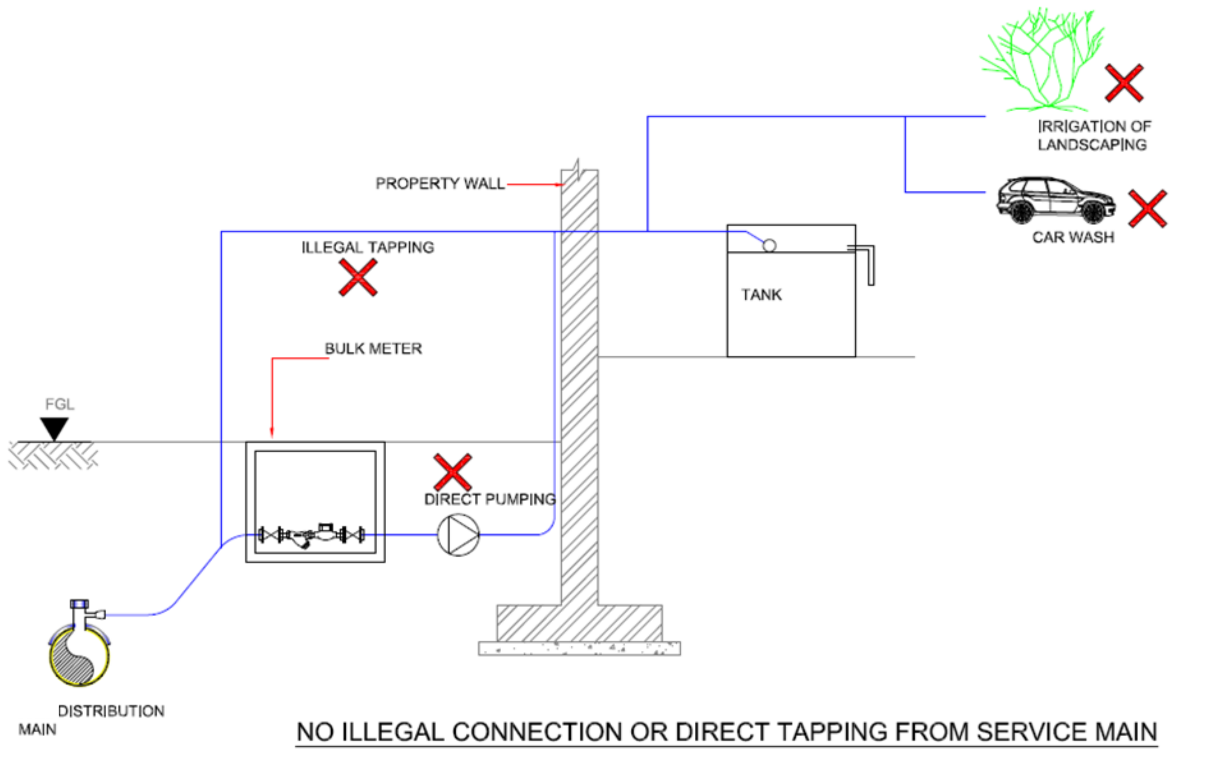

Figure 7 Illegal taping.

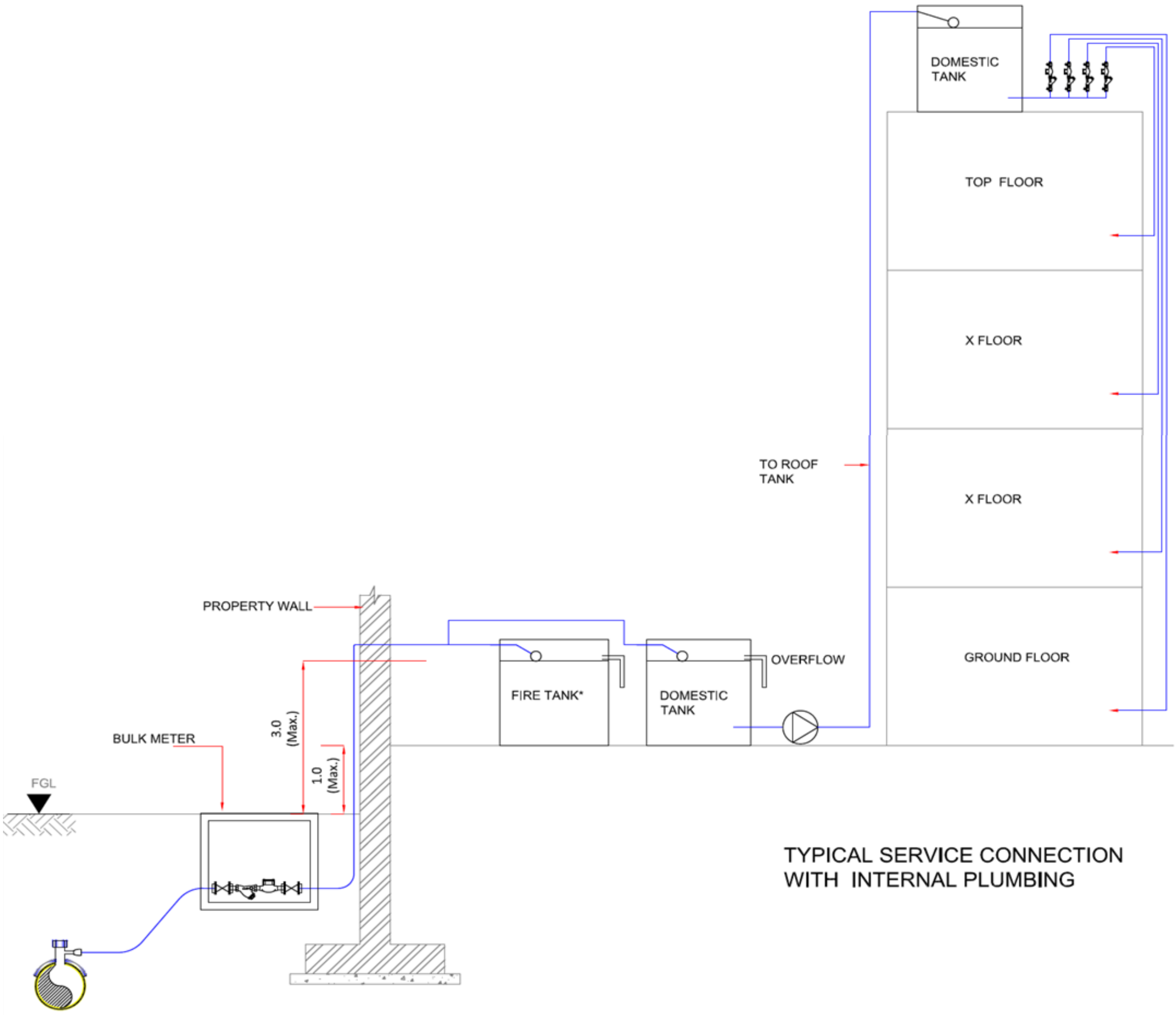

Figure 8 Typical arrangement of piping from bulk meter to individual domestic meter.

12. Tanker Filling Stations

12.1. Definition and Scope

A Potable Water Tanker Filling Station (TFS) shall be defined as a designated facility for the controlled filling of authorised blue tankers with potable water for distribution to consumers or dispatch to tanks in some remote area contexts.

Tanker Filling Station (TFS) is provided to supply potable water to remote dwelling areas where water cannot be supplied through distribution network as laying of distribution network is not feasible. Besides, TFS can be provided in addition to the reticulated distribution network in order to distribute water during an emergency situation like pipe burst or planned shutdown.

Similarly, Treated Sewerage Effluent Filling Stations shall be defined as a designated facility for the controlled filling of authorised green tankers with TSE for distribution for irrigation and industrial uses.

This section provides the design requirements for new Tanker Filling Stations, both potable water and TSE, and the upgrade of existing facilities to ensure efficient, safe, and controlled potable water and TSE distribution via tankers.

12.2. General Requirements

All Tanker Filling Stations shall be:

- Designed in accordance with the water quality standards
- Located to optimize the efficiency of water distribution by tankers
- Constructed with materials suitable for potable water applications
- Equipped with appropriate security and control systems
- Integrated with the monitoring and management systems

Tanker Filling Stations shall comply with all applicable regulations including Regulatory Authority Decision No. 31/2025 regarding the regulation of independent tanker operation in the water and wastewater sector.

12.3. Site Selection Criteria

12.3.1. Location Requirements

Tanker Filling Stations shall be located to:

- Minimize the distance from the water source and upstream reservoir,
- Minimize travel distances to service areas

- Provide easy access to major transportation routes
- Avoid residential areas where tanker traffic would cause disturbance
- Min. 200 m apart from the residential plot and 500 m away from public amenities viz., school/Park/Mosque/Eid prayer area
- Optimize the number of filling stations within the service area
- Vicinity to existing electrical power supply
- Provision to gravitate the spilled water to existing wadi/low level that will not flood to the adjacent area.

NOTE : No water TFS shall be installed or connected to a Transmission line.

The minimum distance between Tanker Filling Stations shall be determined based on:

- Population density and distribution
- Expected number of tankers to be served
- Availability of water supply infrastructure
- Future development plans and emergency situations.

For areas with existing or planned water distribution network coverage, any proposed Tanker Filling Station must be supported by a comprehensive needs assessment that provides strong evidence of necessity, clearly identifies which specific areas require tanker service despite network presence, quantifies the population that cannot be served by the network, and demonstrates why extending the network is not a viable alternative in the medium term.

12.3.2. Site Characteristics

The selected site shall:

- Have sufficient area to accommodate the design capacity
- Be relatively flat with good drainage characteristics
- Have secure perimeter boundaries
- Allow for appropriate buffer zones from sensitive land uses
- Have stable soil conditions for structural foundations

12.4. Design Capacity and Hydraulic Requirements

12.4.1. Capacity Determination

Design capacity shall be based on:

- Number of consumers to be served by tankers in the area
- Average daily water demand in the service area
- Peak hour factors (typically 1.5-2.0 times average flow)
- Future growth projections for a minimum of 10 years
- Seasonal demand variations

The minimum design capacity shall accommodate:

- Peak hour flow rate during the maximum demand day
- Concurrent filling of multiple tankers (minimum of two)
- Operational reserve capacity of 20%

12.4.2. Hydraulic Design

Supply systems shall provide:

- Minimum flow rate of 1 m3 per minute per filling point
- Working pressure of 2-4 bar at the discharge point
- Pressure regulation to prevent water hammer effects
- Flow control valves for operator safety

Storage facilities, if required, shall:

- Provide minimum 2-hour peak demand storage
- Include appropriate level monitoring
- Maintain water quality through proper circulation
- Be sized to accommodate diurnal demand variations

12.5. Site Layout and Circulation

12.5.1. Hydraulic supply

For Potable water TFS, the hydraulic supply shall be any of the 2 option below depending on the land available and the environment (urban or not) :

- Ground reservoir + VFD pumping station + diesel generator for 48hrs of peak demand,
- Elevated tank for 48hrs of storage capacity.

For the TSE TFS, the hydraulic supply can be adapted :

- STP or Ground reservoir + VFD pumping station for 12hrs of average demand.

12.5.2. Traffic Management

The circulation layout shall include:

- One-way traffic flow where possible
- Separate entry and exit points
- Queue lanes sized for peak hour demand
- Adequate turning radii for largest authorized tankers
- Clear traffic control signage
- Speed control measures within the facility

The minimum queue capacity shall accommodate:

- Peak hour tanker arrivals without backing onto public roads
- Minimum of 3 tankers per filling point during peak periods

12.5.3. Filling Bay Design

Filling bays shall:

- Be arranged to maximize throughput efficiency
- Include overhead canopy for weather protection
- Have impervious concrete surfaces with appropriate drainage
- Include curbing and barriers to protect filling equipment
- Be sized to accommodate the largest authorized tankers

- Have clear markings for tanker positioning

The number of filling bays shall be determined based on:

- Peak hour demand requirements
- Average filling time
- Minimum of 2 filling points for redundancy purposes

12.6. Water Supply and Quality Systems

12.6.1. Supply Connection

Connection to the water supply network shall:

- Be sized for peak flow requirements
- Include isolation valves for maintenance
- Have bypass capabilities for emergency operations
- Include appropriate backflow prevention
- Be metered for water accountability

If multiple supply connections are available, they shall:

- Be configured for automatic switchover
- Include non-return valves
- Maintain equal pressure at filling points

Direct connection to water or TSE transmission mains shall be avoided whenever possible and shall only be considered as a last resort; any proposal for such connection must be accompanied by comprehensive technical documentation demonstrating why connection to the distribution network (downstream of service reservoirs) is technically unfeasible.

12.6.2. Control and Metering Systems

Each filling station shall be equipped with:

- One control panel fully automated for each TFS
- The panel shall be equipped with an identification system (magnetic key or card system) with a digital screen. The identification system shall be waterproof, secured and impossible to forge. All the information shall be readable in both English and Arabic languages;

- The panel shall be in a metallic panel closed by a lock near the delivery point with sun protection
- The panel shall work on a power supply of 230 V, 50 Hz, 1 phase;
- The panel will monitor the tankers through identification system to control the quantity of water delivered by tanker;
- Automated tanker identification system
- The unit shall be suitable for the climatic conditions in Oman
- Electronic metering in line with SCADA requirement and connected to it, with 1% accuracy or better, before and after tank if any
- Filling operation for each filling point shall be started by push button operation (from the control panel). The push button shall open an electrically operated valve which shall allow the water to flow towards the filling point;
- The system shall have a button (above mentioned) to choose the bay where the water will be delivered. Each bay shall have a number (1, 2, 3, etc.) which will be clearly visible;
- The control panel must be installed in a safe and secured location, to avoid any damage, but still has to be accessible for the users.
- Automated data recording of volumes dispensed
- Digital display visible to the operator
- Capability for electronic receipt generation
- Integration with the Authority's billing systems

The control system shall include:

- The unit will deliver only the allotted quantity of water authorized by the system. After each filling the information (volumes) will be updated in the control unit and in the regional SCADA.
- Display access of the control unit can be secured by password for each tanker, each contract.
- The control panel shall be self-powered during 4 hours.
- The control unit shall be able to send alarms by SMS to NWS technicians’ mobile phones.
- The control units shall be the same for each TFS control panel, and shall be in modular conception and upgradable.
- Control units shall be able to communicate with other devices: PLCs, controllers, smart sensors and other units for the needs of process controls.

- The language program shall be standard. The units shall be parameterized with no specific language.
- A connection on the local unit with a laptop shall be anticipated and allow an intervention in the program.
- User authentication for authorized operators
- Pre-payment or credit approval mechanisms
- Automated filling control with preset volumes
- Emergency shut-off capability
- Transaction logging and reporting

12.6.3. Tracking Systems

In accordance with Regulatory Authority Decision No. 31/2025, the filling station shall include:

- Registration and verification of authorized tankers
- Systems to ensure only permitted tankers can obtain water
- Integration with the tanker tracking system administered by the licensed service provider
- Data exchange capabilities with central monitoring systems
- Real-time recording of tanker filling operations
- In case of data transmission or SCADA breakout, the Control Unit shall be able to store all data during 15 days.
- The data shall be sent to the Regional Control Room through GPRS or GSM data, in accordance with available signals on site. As an option, optical fiber medium could be proposed if available on site.

12.6.4. Water delivery point (bay)

The delivery point shall have the following functions:

- The start button shall open an electrically operated valve which shall allow water to flow towards the filling point.
- An emergency button is to be installed to stop the delivery point.
- An electrically operated valve with low power (24 V or 48 V) shall be installed at each bay. The valve shall be in a normally closed position in case of power failure.

- The required volume of water shall be recorded automatically by opening the valve and shall be measured by a water meter which shall gradually start the valve closure when the selected volume is close to be reached and shut it off completely after the required volume has been supplied.

12.6.5. Data report to control room

- The following shall must be sent on daily basis (once in a day at 00:00) to the main regional control room:
    - Total volume delivered by tankers per day
    - Volume delivered by each tanker (name, tanker number registration plate, contracted quantity, date)
- The following alarms shall be sent to the regional control room:
    - Low level of water reservoirs
    - Low low level of water reservoir
    - High level of water reservoir
    - High high level of water reservoir
    - Intrusion water reservoir
    - Pressure alarm
    - Water delivery failure to tanker
    - Power failure
    - Electro-valve failure
    - Water meter failure
    - Report failure
    - Communication failure
    - Electrical Valve failure
    - Identification failure

- Each filling station shall have its SCADA backup (historic, curves), with a local connection with a laptop.
- The monthly reports must be automatically generated by the system last day of the month at 00:00.

- The SCADA report shall contain the following information:
  - Daily water delivery in each TFS
  - Monthly water delivery in each TFS
  - The total water delivered from the beginning of the year (from 1st January)
  - Monthly quantity delivered by each contract with its number (Contractor and contract number)
  - Annual report for water delivered in each TFS
  - Annual report for water delivered by each contractor
  - Annual report for water delivered by tankers per Governorate
- 3 years data must be available all the time.
- A consumption graph must be provided for each TFS
- The SCADA system must be able to print the report of all activated alarms
- The remote management system which will be installed in each regional main control room shall reflect the following requirements:
  - Remote Management master station running on a Microsoft Windows PC-compatible microcomputer Multimedia graphical displays, Curves.
  - Fully integrated management of Excel (text or chart reports),
  - The communication interface shall also be by Web server and SMS
  - Data presentation can be customized by Diam,
  - Fully secured database and confidential access to the data with different access levels and passwords.

12.6.6. Water Quality Assurance

All Tanker Filling Stations shall include:

- A dedicated water sampling box/cabinet with controlled access for authorized personnel to collect water samples

Storage tanks, where incorporated into the design, shall be equipped with:

- Automatic chlorination dosing systems proportional to flow rate

Water quality shall comply with all applicable standards and regulations.

12.6.7. Driver's resting area

Each tanker filling station shall include a designated driver's resting area to ensure the well-being, safety, and efficiency of transport operations. Providing a rest area allows drivers to take necessary breaks, reducing fatigue and promoting alertness, which is crucial for road safety. Additionally, a comfortable and well-equipped resting space aligns with regulatory and occupational health standards, fostering a safer working environment. This facility should be adequately maintained, offering basic amenities such as seating, sanitation, and refreshments to support drivers during their mandatory rest periods, ultimately enhancing overall operational effectiveness and compliance with industry best practices.

13. Potable Water Quality & Disinfection

13.1. Chlorine Disinfection

Designs shall use the appropriate disinfection (chlorination) methods to eliminate harmful microorganisms in adherence to Oman's drinking water standards. The main processes used for disinfection in the Sultanate of Oman are the use of sodium hypochlorite (either bulk sodium hypochlorite or on-site chlorine generation). The use of calcium hypochlorite may also be possible for small and isolated sites.

All chemicals, reagents and internal coatings for piping shall be certified for drinking water applications by an internationally recognized institute for regulating potable water quality; e.g. NFS, etc.

All chlorine added to drinking water must meet American National Standards Institute (ANSI)/ formerly the National Sanitation Foundation (NSF) standards 60: Drinking Water Chemicals—Health Effects covers water treatment chemicals. The design shall also prevent the formation of disinfection by products (DBPs) such as THMs (trihalomethanes) and HAAs (Haloacetic acids).

The World Health Organization (WHO) has set a provisional guideline value for bromate of 0.01 mg/l. Bromate is introduced to drinking water as byproduct of:

- Presence of bromide in commercially produced sodium hypochlorite;
- Bromide in the salt used for electrolysis to produce hypochlorite.

Hence, the detail design engineer shall specify required stringent standards for chemicals used in drinking water production.

Since Chlorine is a powerful oxidant, it will react with some elements and it will also have a biocide action. The two main forms of chlorine are the free chlorine and the combined chlorine.

- Free chlorine will be divided, after dissociation in the water, into 3 different types of chlorine (dichlore, hypochlorous acid, hypochlorite ion), with a different level of biocide action efficiency for each type, depending on pH values.

- Combined chlorine will be combined with some oxidable element and will not have any biocide action.

It can be noted from the Figure below that at pH = 7.5, HClO and ClO- concentration are close to 50%. The most efficient disinfectant form is the hypochlorous acid (HClO). The best range of pH for a good disinfection will be between 6.5 and 7.5.

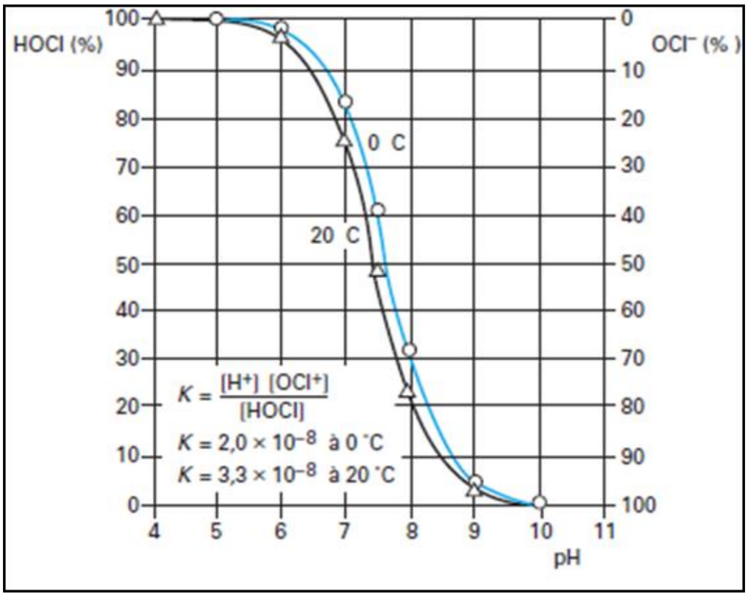

Figure 9 Dissociation of hypochlorous acid in the water.

High temperatures will favour the bactericide effect, although low temperatures favour HClO against ClO^-, high temperatures will speed up the disinfection reaction. However, one major drawback is that high temperatures decrease the reagent stability in time, and therefore its efficiency.

The main factors affecting drinking water disinfection with chlorine include:

1. Chlorine concentration, which depends on the type of disinfectant. Typically, granular calcium hypochlorite has a chlorine content of 65-70%, bulk sodium hypochlorite has a chlorine content of 10-12% and sodium hypochlorite produced in On-site Electrochlorination 0.4 to 1%.
2. Contact time, mixing and distribution: Longer contact time allows more complete inactivation of microorganisms, whereas proper mixing ensures uniform chlorine distribution
3. pH level: Lower pH (6-7) increases effectiveness as hypochlorous acid (HOCl) is more germicidal than hypochlorite ion (OCl-); effectiveness decreases as pH rises above 7.5
4. Temperature: Higher temperatures generally increase disinfection rates and chlorine reactivity
5. Chlorine demand: Suspended particles (turbidity) can shield microorganisms from chlorine and consume disinfectant. Organic matter, ammonia and other compounds consume chlorine through reactions, reducing available free chlorine for disinfection
6. Type of microorganism: Different pathogens have varying resistance (bacteria < viruses < protozoan cysts like Giardia and Cryptosporidium). Minimum treatment efficiency for disinfection shall be 99.99% removal.
7. The chlorine dose must be calculated to eliminate harmful pathogens based on different log removals and CT values (disinfectant Concentration (mg/L) × Time (minutes) at a given pH and temperature) with free chlorine. Some reference values are shown below.

Table 27 Typical Log Removal Values for Typical Micro-organism

Micro-organism	Size	Typical Log Rem	Approx. % Removal	CT @ 20°C (mg.min/L)
Total coliform bacteria	0.5–3 µm	2 - 4	99 - 99.99	0.02 - 0.05
Escherichia coli	0.5–2 µm	4 - 6	99.99 - 99.9999	0.03 - 0.05
Enterococci	0.6–1 µm	3 - 5	99.9 - 99.999	0.3 - 0.6
Giardia lamblia (cysts)	8–12 µm	2 - 3	99 - 99.9	25
Cryptosporidium parvum (oocysts)	4–6 µm	Not effective		
Enteric viruses (polio, rota, norovirus surrogates)	~70 nm	~4	~ 99.99	3

13.2. Secondary Chlorination Requirements

13.2.1. Water quality standard and risks

Water quality standards:

The current Omani un-bottled drinking water standards specify a free chlorine concentration range of 0.2 - 0.5 mg/l as an acceptable range for drinking water through public distribution systems. This standard is currently used to monitor compliance at all points from the treatment plant outlet to customer's tap within NWS perimeter of operation.

Risks:

It is frequent and common to have long lengths of pipes for transmission & distribution between storage sites and customers. The residual chlorine (RCl) may decay and read to zero well before the distribution ends without any post chlorination facilities, increasing the risk of water quality degradation. Thus, depending on the network configuration, it may be necessary to proceed to boost the chlorination by designing satellite chlorination units in strategic points as required. Therefore, the detail design engineer shall follow the following guidelines:

- RCl level boosting in distribution networks where necessary to maintain above 0.2 ppm in the lowest case.
- The design Engineer shall be responsible to produce network model in the distribution zones for RCl lowest 0.2 ppm worst-case scenario, and not to exceed 0.5 ppm in upper-case.

- Upper-case scenario is peak hour flow in peak day of the year and worst case is lowest elevation, sluggish flow condition in lowest demand hours and the days of the year.

The other factors that could be potential hazards in water quality are repairs in network and undetected leaks in transmission & distribution networks as well as adding new assets to the water supply system (e.g. new reservoirs, elevated towers, distribution networks, etc.). Hence, strict disinfection procedures shall be detailed in designing improvements or relaying of sections shall consider this effect.

13.2.1.1. Water supply systems, water age & transfer time

Three main configurations are mention in this guideline, depending on the type of asset:

- Post Chlorination on transmission lines (pumping stations or transmission reservoirs) and/or service reservoirs;
- Distribution networks: satellite Chlorine booster stations in distribution network as per the detail designers free chlorine decay models.
- Well water disinfection, straight after water abstraction from water tables.

13.2.1.2. Transmission systems

In transmission systems, location of the post chlorination units will depend on two main aspects:

- Water age and transfer time;
- Chlorine decay throughout the transmission (depending on both the pipe material and the chlorine decay in the water due to initial concentration, temperature and organic matter content).

Design of extension, expansion of transmissions in networks to existing distribution systems requires field trials with laboratory tests and calibration of the distribution / transmission model. It shall be performed and validated by the detail design engineer for chlorine decay time. A model shall be provided in a separate chapter in the detail report with a more accurate updated chlorine decay model and related data for the entire transmission improvement.

13.2.1.3. Distribution networks

The distribution network residual chlorine decay model corresponding to a DMA are prone to run out of residual chlorine standards causing poor water quality in certain branches due to:

- Poor branch velocities in off peak hours;
- Network Chorine decay time exceeds design;
- Under-utilized network;
- Operations of periodical flushing of distributing network lines through washouts.

The network for a DMA related chlorine decay co-coefficient, wall reaction order and diffusivity are required to submit in the detailed design and model of the network for RCl levels contours, min & max Chlorine decay kinetic factors shall be determined through laboratory tests and on-site sampling campaigns.

The upgraded chlorine decay model for entire distribution networks results showing residual chlorine (predicted for) below the required 0.2 ppm shall be highlighted. The detailed design shall cover for chlorine boosting in strategic positions with satellite chlorine boosters so that the chlorine booster will operate to maintain required chlorine levels based on the feedback from upstream RCl measurements. Therefore, during network detail design, the engineer shall follow the following guidelines and further applicable standards:

a. Expansion of distribution networks / adding new reservoirs: the existing distribution network updated with GIS shall be used in determining RCl (0.2 – 0.5 ppm) zones and off quality zones.
b. Identified zones, pipe networks and nodes shall use methods of optimizing (i.e. use of dynamic programming) to identify satellite injection points.
c. Develop injection philosophy to optimize the desired RCl (min 0.2 ppm) in identified zones.
d. In case of direct injection to the transmission pipeline or distribution nodes, it shall be considered to design & use on line static mixers.

13.2.1.4. Wells

NWS wells' water shall be correctly disinfected. Some groundwater sources may present a high risk of microbial contamination and may also contain more organics than water produced by desalination plants. Thus, special attention shall be paid to the disinfection treatment, as there is a higher risk of generation of disinfection by-products such as chloramines or trihalomethanes (THMs). Chlorine decay may be higher and risk of bacteria "re-growth" also higher.

13.3. Disinfection System and Agents

Disinfection agents shall be in line with the last international regulations and recommendations published by WHO (e.g. water safety in distribution systems; Bromide in drinking-water) and AWWA C651. Sodium hypochlorite is preferred as it has less impact on water turbidity. However, other systems are preferred for pre-commissioning disinfection and can be used, as well, in distribution systems to reduce the CAPEX, OPEX and hauling risk from production site to injection point (small tanks).

The chlorine production and storage shall be evaluated based on a cost-benefits analysis for each project, taking into account the technical requirements for each technology. The general guidelines are presented below.

13.3.1. Bulk Sodium Hypochlorite

Sodium hypochlorite, often referred to as liquid bleach, is the most widely used of the hypochlorite in potable water treatment.

- Free chlorine rate: around 15%;
- Commercial liquid bleach is usually delivered at a concentration of 149 to 158 g Cl_2/L.
- Bromide content shall not exceed internationally permissible limit by ANSI

13.3.2. Calcium Hypochlorite

Calcium hypochlorite is available on the market under different trade names, either in powder, granular or agglomerated form. Its main characteristics are as follow:

- Free chlorine rate: up to 70%
- Purchased in 25 kg or 50 kg sealed containers
- Solubility: from 215 g/l at 0 °C to 234 g/l at 40 °C
- Working solution concentration generally lower than 100 g $Ca(ClO)_2$ per liter
- Deterioration in time: 2 to 2.5% over 1 year
- Bromide content shall not exceed internationally permissible limit by ANSI

13.3.3. Sodium Dichloroisocyanurate (NaDCC)

NaDCC is the most recent product releasing chlorine. It presents some benefits (high stability under high temperatures, higher free chlorine rate) compared with bleach and other products releasing free chlorine. Like Calcium hypochlorite, NaDCC is available under different forms; powder, tablets, granular or agglomerated.

- Solubility: 30 g/ 100 mL at 25 °C
- Free chlorine rate: around 90%
- Very stable and easy to use source of free chlorine
- Not toxic

13.3.4. On-Site Electrochlorination (OSEC) System

On site sodium hypochlorite generation is recommended in locations where the chlorine demand is high.

Electro-chlorination enables the production of sodium hypochlorite continuously by using salt as a basic component. The liquid sodium hypochlorite produced is less corrosive than bleach, and the low concentrations enable a finer dosing of free chlorine into the water system.

These units shall not have exaggerated storing capacities. Storing capacity shall not be more than 21 days and shall be decided as per OEM guides.

OSEC detailed process specification is found in PAM-SPC-611.

13.4. Dosing Equipment

The following are the main components for dosing equipment:

- Mixing tank – GRP;
- Mixer with controls in day tank;
- Dosing pumps (standby / duty);
- Dampers, calibration column and piping;
- HSE equipment and protective measures.

Calcium hypochlorite and NaDCC are dissolved in water by means of mechanical mixing up to the recommended working solution concentration. Bulk hypochlorite can be diluted in water if required, but it will deteriorate more rapidly.

The liquid hypochlorite disinfectant agent is injected according to the following steps (shown in Figure below); the water is stored in a bulk storage tank (1), collected by (2), the level is checked through a level sensor (3) and the solution is pumped through (4) by the dosing pump (5). The water to be chlorinated arrives through the pipe (6) and the disinfectant agent is injected with the hydro injector (7). The injection flow can be monitored and controlled automatically by a flowmeter and an on-line chlorine analyzer, connected to the SCADA system.

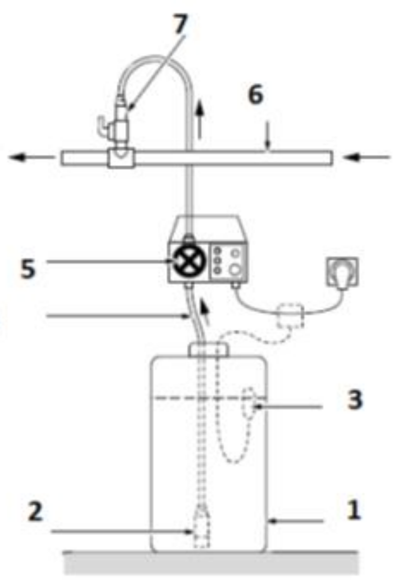

Figure 10 Schematic for liquid hypochlorite injection.

The detail design engineer shall give required specifications for the necessary equipment for safe handling with piping arrangement including flushing and cleaning.

13.5. Residual Chlorine Measurement and Monitoring

13.5.1. Regulation/control of dosage flow

Disinfection performance depends on Depending on the type of asset (well, pumping station, reservoir, etc.), the dosage flow is not controlled in the same way. Basically, the injection is controlled by:

- Strength of the solution;
- Flow of the water to disinfect;
- Residual chlorine (RCl) concentration;
- Or both (flow and RCl) for more reactivity.

The schematics below shown in Figure 19 and Figure 20 give an example of how the injection shall be controlled for wells and pumping stations or reservoirs.

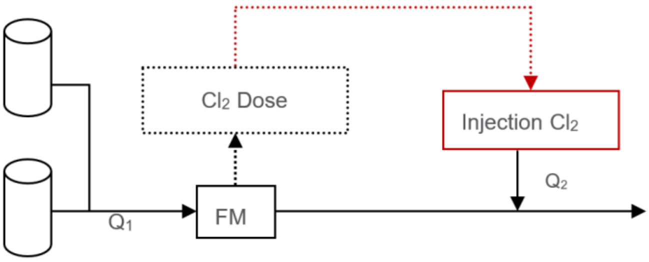

Figure 11 Schematic Chlorine dosing (wells).

13.5.1.1. Wells

The flow of the pump (Q_2) injecting chlorine in the water system is only controlled by the upstream total flow (Q_1 measured by flowmeter FM) in the pipe. The dosing pump is working only if the well pump is working.

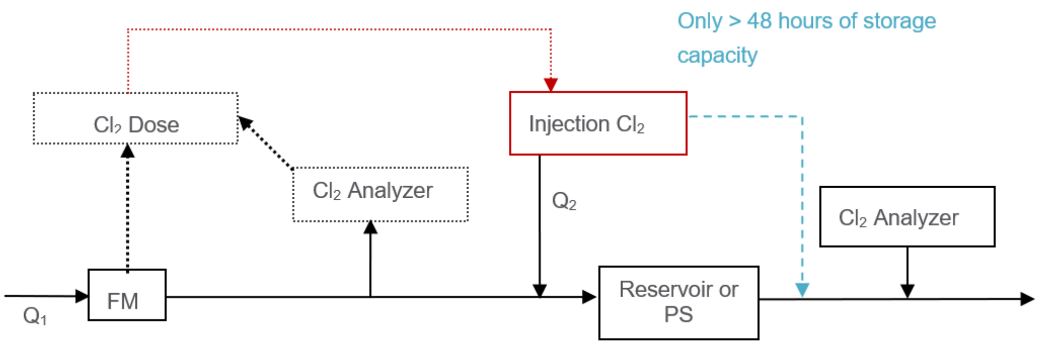

Figure 12 Schematic Chlorine dosing (Reservoir/PS).

13.5.1.2. Reservoir or Pumping Stations

The flow of the pump (Q_2) injecting chlorine in the system is monitored by both:

- The upstream flow (Q1)
- The upstream chlorine concentration

A chlorine analyzer shall be installed with chlorine downstream the reservoir/pumping station, in order to check the accuracy of the chlorine dosage.

13.5.1.3. Distribution Network

Network analysis for RCl decay against min / max flow actual & forecast for design horizon method of injection;

Min/max, RCl in the node and dosage forecast, sizing equipment.

13.5.1.4. Water sampling points

As per WHO guidelines for water sampling for RCl.

13.5.2. Type of measurement

Water Quality on-line analyzers are installed at different locations depending on the type of asset. The analyzer shall be able to measure the pH and chlorine in each site.

In the water industry, different technologies of analyses are found to titrate the different forms of chlorine. DPD colorimetric detection is a method based on N, N-Diethyl-p-Phenylenediamine reaction with active halogens. This reaction is a standard analytical approach for analysis of residual chlorine and other chlorine oxidants and is based on the formation of colored products with DPD. This method which requires a physical sample to be taken is mainly used for on-site and laboratory testing.

13.5.2.1. On-line analyzers

Water Quality can be continuously monitored through on-line analyzers, the results data being transferred to regional and national SCADA. The measuring chain for on-line analyzer which is shown on Figure 9.5 has to be followed to validate the data. The design engineer shall specify the suitable analyzers as per C&I guidelines. The designer shall specify the minimum distance between disinfectant dosing points to the analyzer so as the minimum contact time (CT) is met, which is dependent on pipe diameter, flow speed and CT required for that specific disinfectant. In order to eliminate false reading due to air entrapped in the pipeline, the detection prop of the analyzer shall be located in a position between 4 and 8 o'clock.

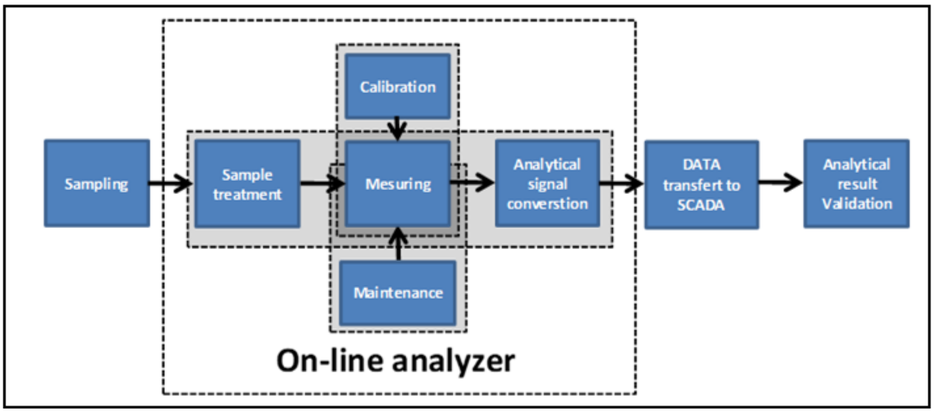

Figure 13 Chain of custody for on-line analyzers.

13.5.2.2. Water quality sampling point

A sampling point is a fixed tap located on different types of assets owned by NWS (reservoir, pumping station, TFS, water treatment plant & distribution network). At regular intervals, the samplers from the Water Quality team of NWS collect samples, and a whole batch of parameters is tested in the laboratory. Sampling and on-site water quality testing is conducted according to quality managed procedures and using standard methods for water quality testing. NWS guidelines and considering the criticality of piping loops.

The respective standard designs for sampling points are:

- PAM-STD-804 Water Quality Sampling Point for Network;
- PAM-STD-805 Water Quality Sampling Point for vertical and horizontal pipe;
- WHO guidelines for distribution systems – latest version.

13.6. Monitoring of Chlorine Levels and Alarms in SCADA

The SCADA system shall supervise, monitor and control the following parameters:

- Product tank: High, low levels

- Brine tank: High, low levels
- Production concentration
- Chlorine dosage
- Water flow
- Temperature
- Analyzer reading (treated water) - pH and Chlorine
- Other parameters recommended by the Manufacturer

All the analyzers shall be connected to the SCADA system. All this information shall be transferred to the Regional Control Room (RCR).

The Electro Chlorination system must be equipped with the OEM supplied dedicated PLC, with a panel mounted HMI with fully independent operation in AUTO mode. This PLC should be further interfaced over standard MODBUS with the Clients (NWS) PLC system, which is installed for process control at well, pumping station, reservoir, DMAs, etc. Further, these data shall be archived by the SCADA at RCR & NCC for remote Monitoring, Control, Reporting, Trending, Analyzing and MIS. Signals from local electromagnetic flowmeters may be collected either from the transmitter directly (4-20 mA) can be taken from the PLC for synchronized operation and dosing.

Prior to dispatch, a detailed FAT should be conducted at the OEMs workshop, witnessed by ICA engineers from both Consultant & Client, who shall accord and provide approval to proceed with shipment. An on-site training concerning its O&M also must be provided to the concerned engineers.

13.7. HSE Guidelines on Chlorination Units

13.7.1. HAZOP study

It is mandatory to submit an HAZOP study as a chapter of the detail design report by the design engineer for:

a. OSEC system;
b. Transporting, storing and dosing facility;
c. Sodium hypochlorite / Calcium hypochlorite tablets and dosing system.

Key information which may be required during the HAZOP should also be readily available. This could include:

- Layout drawings;
- Hazardous area drawings;
- Safety data sheets;

- Relevant codes or standards;
- Plant operating manual (for an existing plant);
- Outline operating procedures (for a new plant);
- HAZOP metrics and mitigations.

13.7.2. Risk assessment

On each new installation, in order to prevent injury, NWS requires the contractor, designer and all third parties to assess the risk on site. Hazards and risks must be reduced to prevent harm to operators, environment and property.

The following aspects shall be considered:

1. Follow guidelines of OHSE latest edition;
2. Design and build safe facilities for chlorine stores, equipment, disinfection systems and chemicals. As electro chlorination systems generate Hydrogen, ATEX directive covering the use of equipment in a potentially explosive environment applies to the design of these systems. ATEX requires that a zone is applied to any equipment and storage tanks which may contain hydrogen. Within this zone, any electrical equipment must be suitably rated for use within a potentially explosive atmosphere. In addition to these requirements, gas and explosivity detectors shall be installed (existing or new projects). Regarding the retention zone, its volume shall at least be the same as the product storage capacity. A high-level alarm shall be installed in the retention location.
3. Set up procedures that can be handed over to the operators of the plant to ensure the correct and safe systems of work are followed.
4. Design emergency plans and equipment that can deal with spills, leaks or catastrophic failure in the given area and surrounding if required
5. The designer or contractor must provide or make provision for the following for takeover by the operational owners of the asset:
    a. Servicing schedules for alarms, detectors, and other equipment designed for the security of the chlorination unit;
    b. Servicing schedules for the everyday use of equipment such as pumps, mixers, processes, etc.;
    c. Provide signage for the identification of the hazards, danger areas, prohibition, mandatory and emergency as required;
    d. Provide Information on good housekeeping, cleaning and maintenance regimes;
    e. Provide emergency response programs in line with the system installed;
    f. Provide training, information and instruction on the equipment the process used and the substances under COSHH required;
    g. Provide a list of the PPE and Specialist PPE required for the system installed;
    h. Provide HSE manual for the installation.

NWS Health, Safety and Environment department has designed policies, process safe systems of work, etc. that will accompany the end user of these systems and the contractor has to ensure all HSE documents are implemented.

They give an overview, by type of installation, of the Personal Protective Equipment and emergency equipment required on chlorination sites.

6.1. Documentation

The Consultant shall ensure that all documentation is included when a new chlorination system is installed;

- Technical Designs
- Drawings
- Functional descriptions

The Consultant shall also hand over the following documents to NWS:

- Maintenance logbook (for analyzers & chlorination units);
- Preventive maintenance schedule and List of spare parts with frequency of replacement.

14. Monitoring of Leaks

14.1. General Requirements

Leak monitoring and identification is a rapidly evolving technology field. These guidelines focus on the monitoring aspect and provide the basic requirements while allowing flexibility for technological advancement and innovation when it comes to identification.

All leak monitoring systems shall be designed with expansion capability to accommodate future technologies without significant modification to the base infrastructure.

Leak monitoring shall be classified into two primary categories: Transmission Main Monitoring and District Metered Area (DMA) Monitoring, each with specific requirements as detailed in this section.

All monitoring equipment shall comply with the NWS' SCADA requirements as specified in PAM-SPC-800.

14.2. Transmission Main Leak Monitoring System

The primary method for transmission main leak monitoring shall be mass balance analysis, comparing measured flow inputs against outputs to identify discrepancies indicative of water loss.

Mass balance calculations shall include adjustments for known operational uses and authorized consumptions.

Storage facilities (reservoirs, tanks) shall be incorporated into mass balance calculations using calibrated level measurements to determine volume changes.

14.3. Distribution leak monitoring

14.3.1. District Metered Area Approach

Distribution leak monitoring shall utilise the District Metered Area (DMA) approach as the primary methodology for identifying and quantifying water losses.

The Designer/Contractor shall present DMA designs at the earliest stage of concept designs as this will prevent the creation of cascading DMAs and aid in optimising pipeline lengths. The main objective is to eliminate cascading DMAs on all new water systems.

All aspects of DMA configuration, implementation, and operation shall comply with the requirements specified in Section 9.2.3 (Leakage and wastage control) in this document (PAM-GUD-202).

Non-Revenue Water (NRW) assessment shall be performed for each DMA using:

- Measured inflow (and outflow where applicable)
- Customer billing data within the DMA boundaries

- Minimum Night Flow (MNF) analysis
- Estimated legitimate night use based on consumer profile

14.3.2. DMA Operational Requirements

DMA integrity shall be maintained at all times to ensure accurate water balance calculations:

- Boundary valves shall remain in their designated positions (open or closed)
- Any operational changes to DMA boundaries shall be documented with location and time of modification
- Changes shall be promptly reported to the team responsible for leak monitoring

DMA watertightness verification shall be conducted:

- At minimum on an annual basis
- Following any major repairs or modifications within the DMA
- When anomalies in flow patterns are detected

Verification results shall be compiled in a formal report including status and location of all boundary valves.

Any boundary valve found in a status different from the designated DMA configuration shall be:

- Documented with location and position
- Reported immediately to the leak monitoring team
- Investigated to determine cause and duration
- Returned to proper position unless operational requirements dictate otherwise

15. Associated Document Information

The following documents that are associated, but not limited, with this Guideline is given in **Section 1.2.**

Up Stream documents		
Document Title	*Document Number*	*Responsibility*
Design Guidelines		
General Design Guidelines	PAM-GUD-201	PAM
Potable Water & Treated Effluent (TSE) Design Guidelines	PAM-GUD-202	PAM
Wastewater Design Guidelines	PAM-GUD-203	PAM
Technical Specifications		
Pipes & Fittings	PAM-SPC-2xx	PAM
Valves & Accessories	PAM-SPC-3xx	PAM
Civil Works	PAM-SPC-4xx	PAM
Metal Works	PAM-SPC-5xx	PAM
Mechanical	PAM-SPC-6xx	PAM
Electrical	PAM-SPC-7xx	PAM
ICA & OT	PAM-SPC-8xx	PAM
Standard Drawings		
General	PAM-STD-1xx	PAM
Pipes	PAM-STD-2xx	PAM
Valves & Accessories	PAM-STD-3xx	PAM
Civil Works	PAM-STD-4xx	PAM
Metal Works	PAM-STD-5xx	PAM
Mechanical	PAM-STD-6xx	PAM
Electrical	PAM-STD-7xx	PAM
Instrumentation	PAM-STD-8xx	PAM
Indicative STP, Process & Building	PAM-STD-9xx	PAM
Procedures		
Managing Asset Change requirements	PAM-PRO-201	PAM
Asset Failure Investigation Procedure	PAM-PRO-202	PAM
Technical Audit	PAM-PRO-203	PAM
Lesson Learned Procedure	PAM-PRO-204	PAM
Commissioning Procedure	PAM-PRO-205	PAM
Asset Handover Procedure	PAM-PRO-206	PAM
Decommissioning Procedure	PAM-PRO-207	PAM
Records		
Document Title	*Document Number*	*Responsibility*
Nil	Nil	Nil